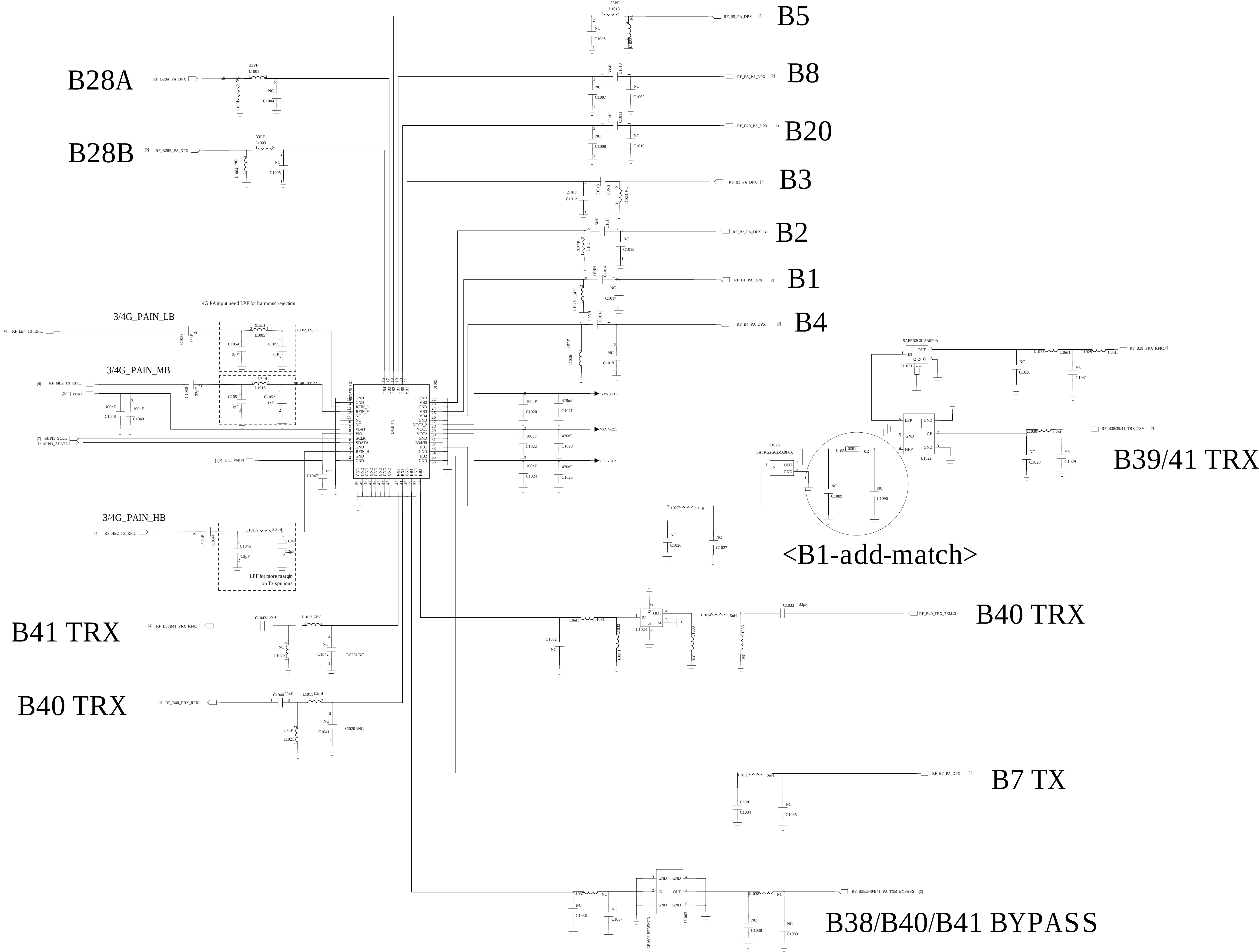
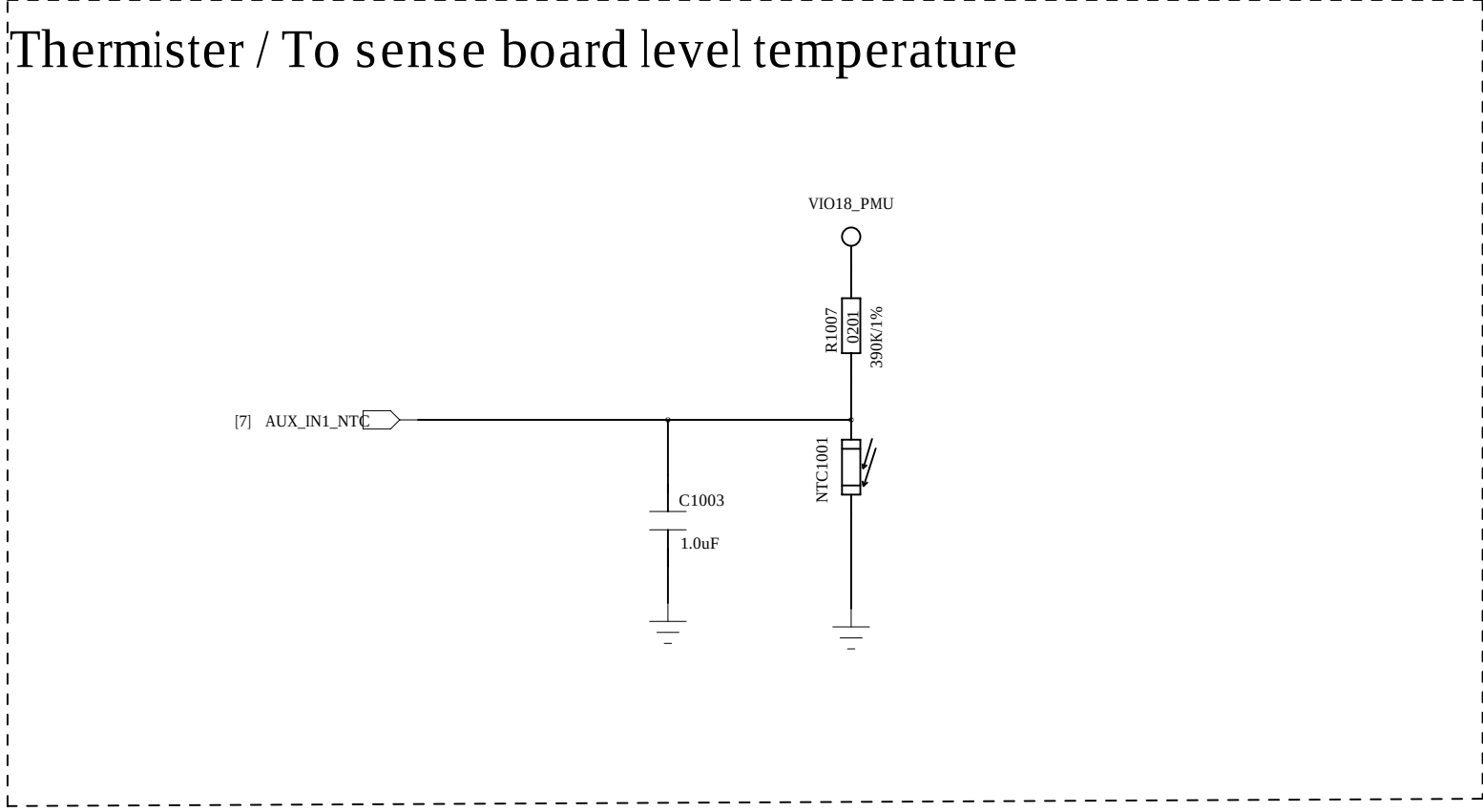
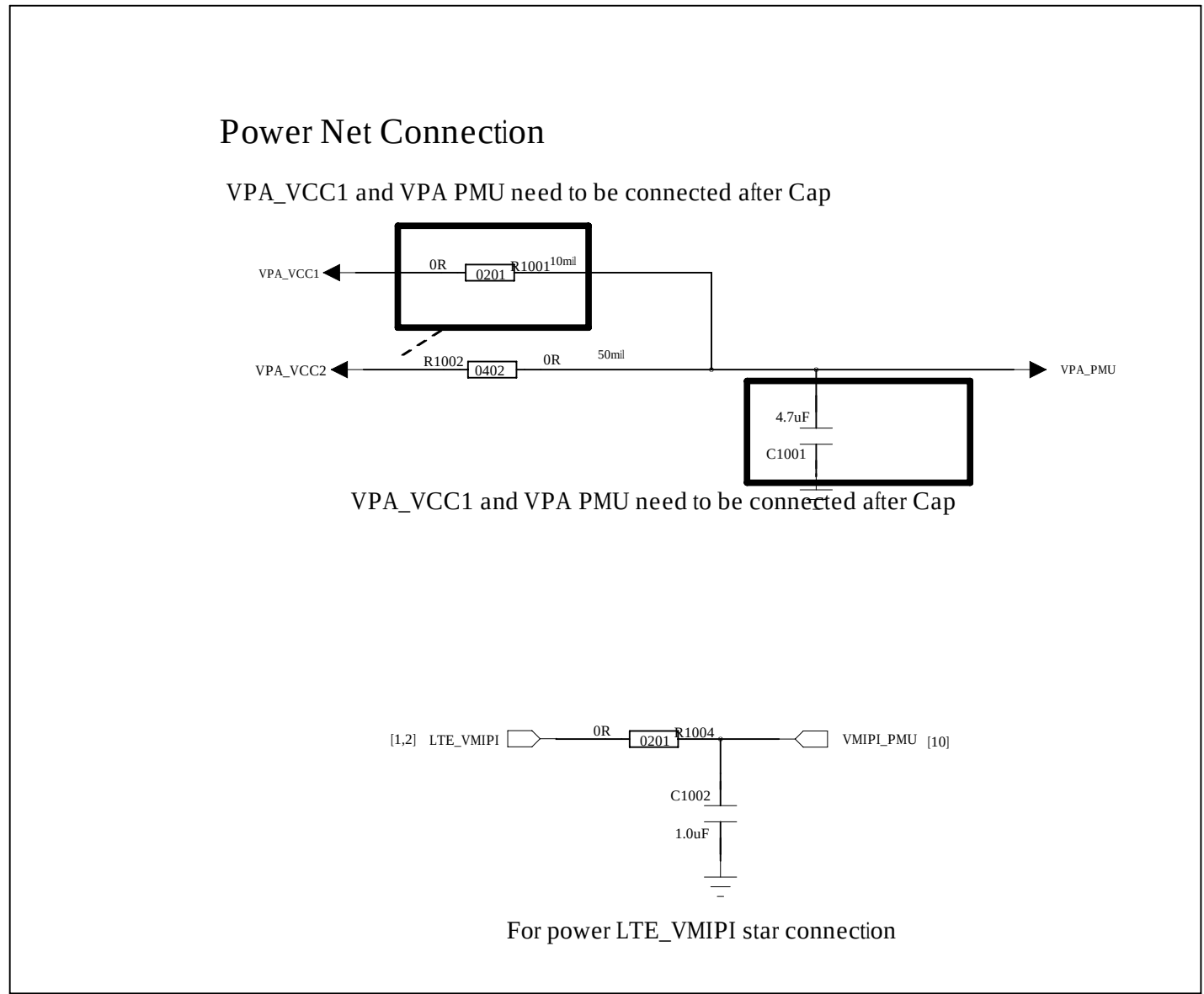


B1/B2/B3/B5/B8/B28/ B7 / B34 / B39 / B38 / B40/ B41PA

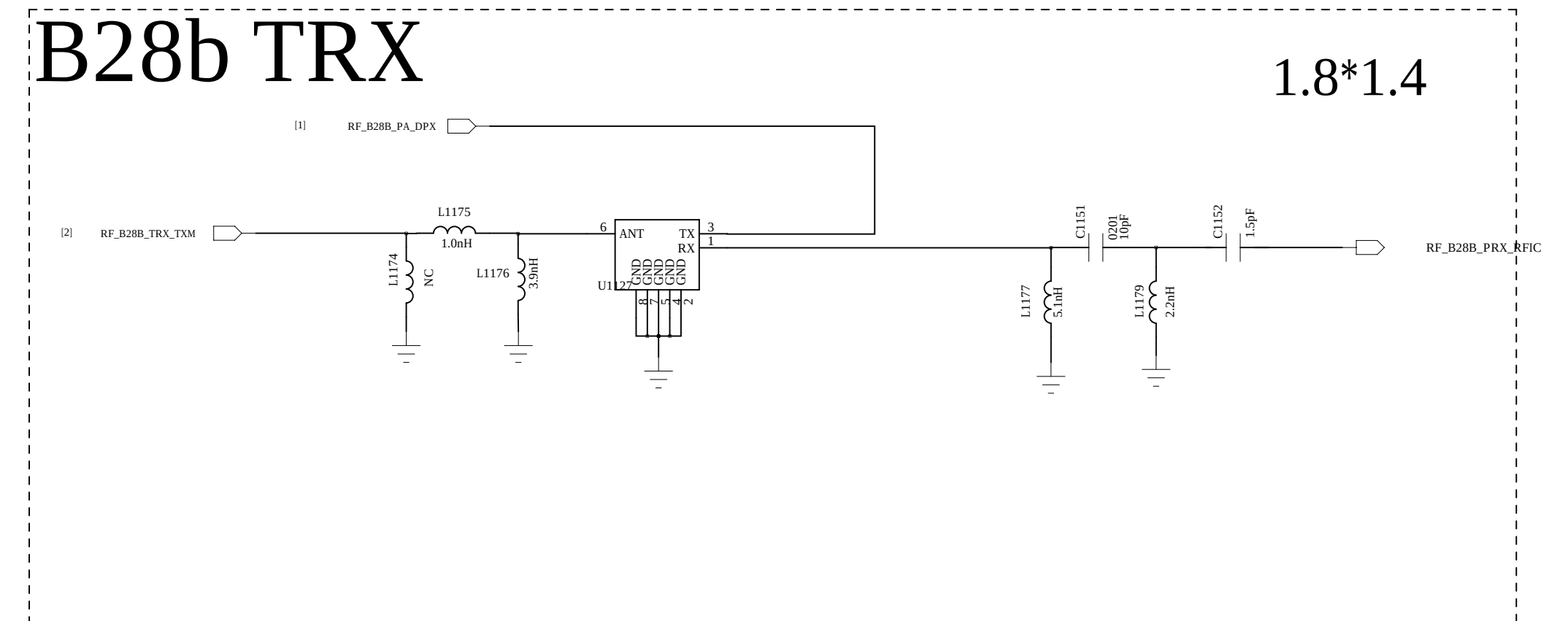
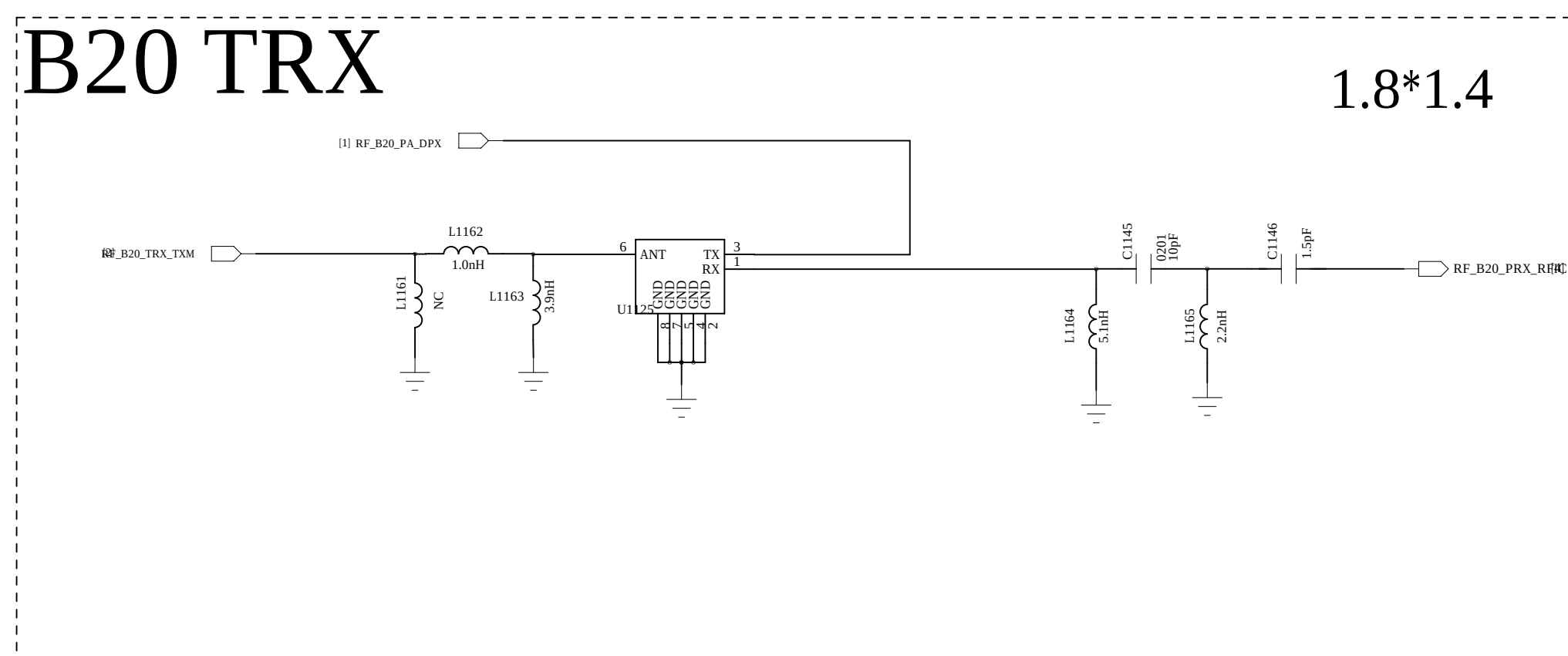
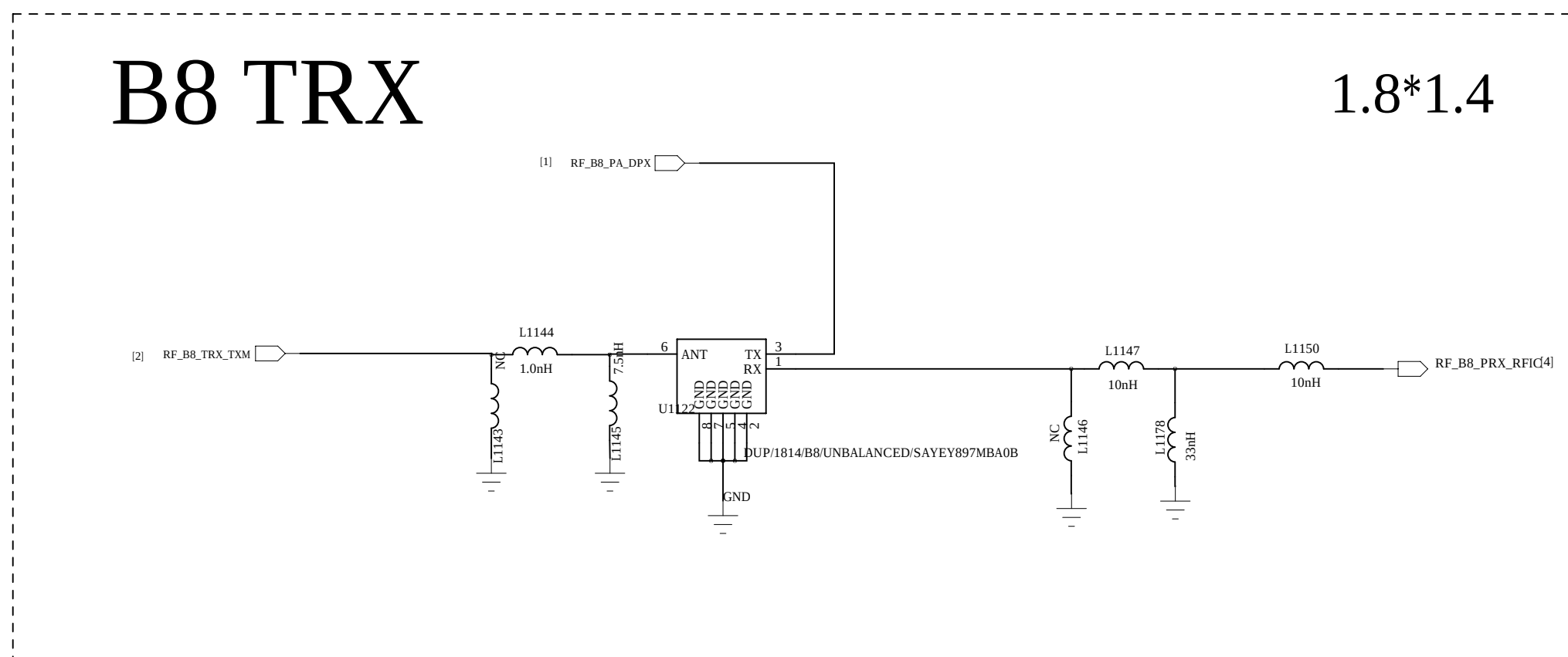
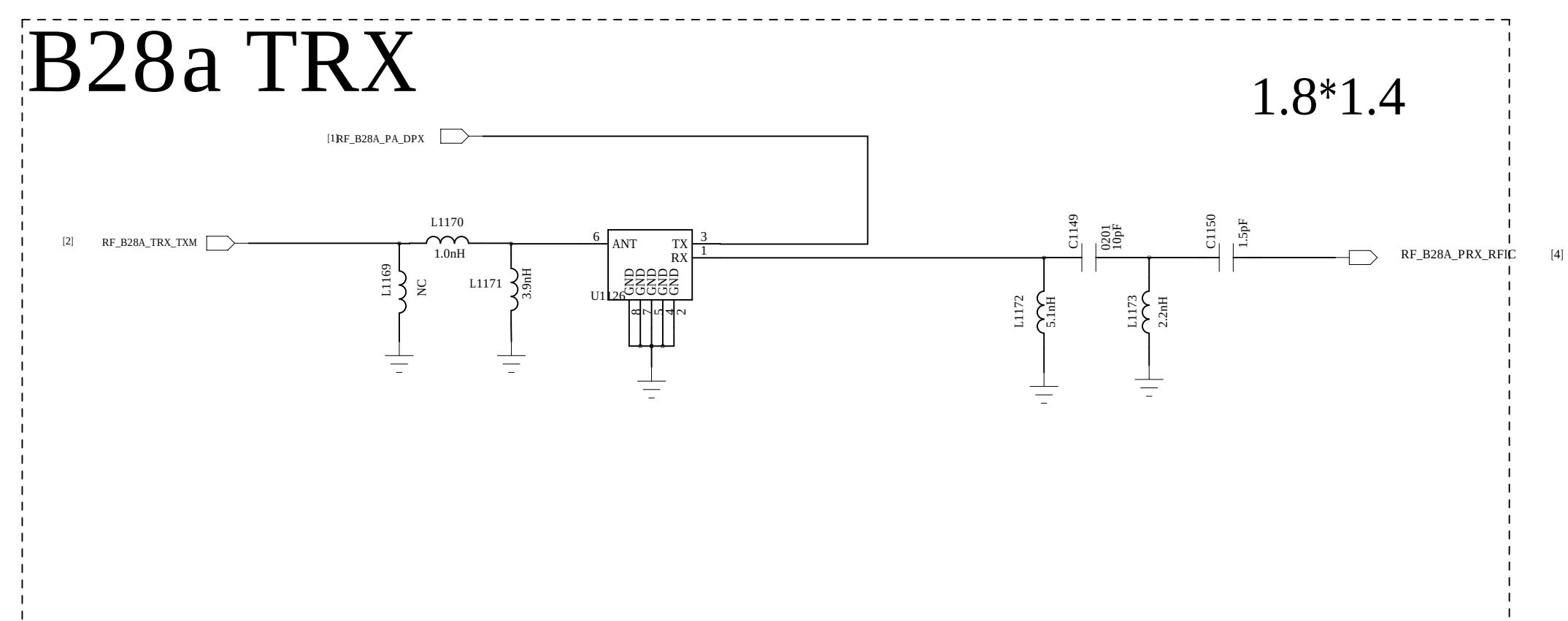
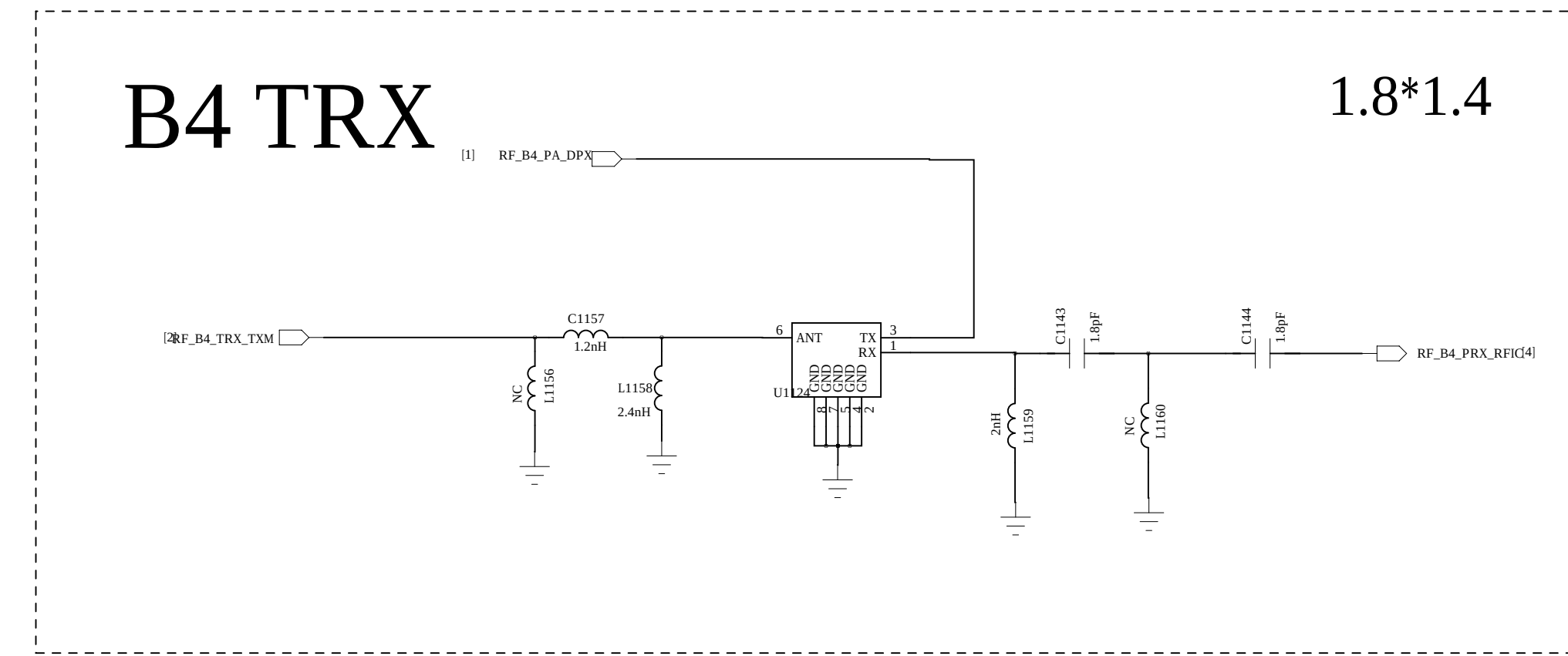
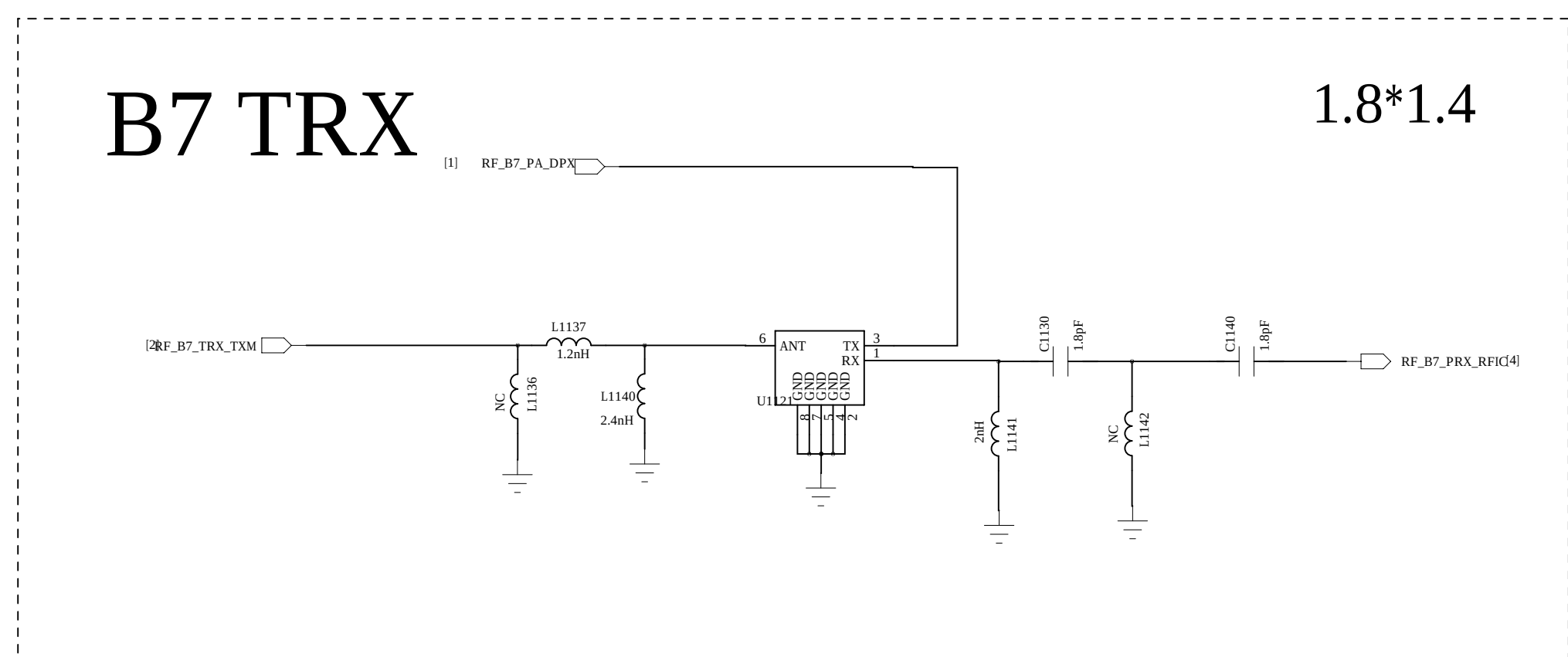
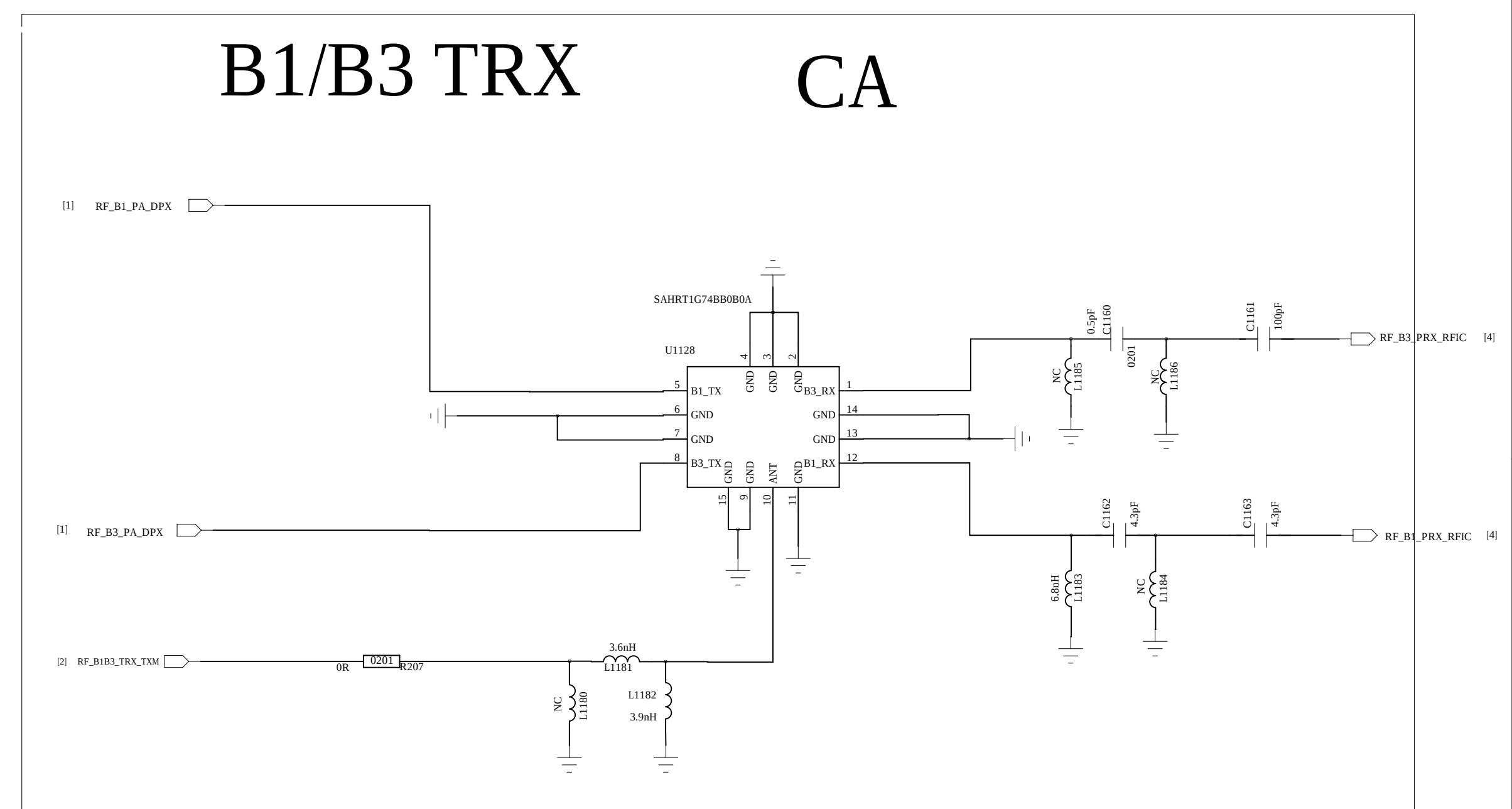
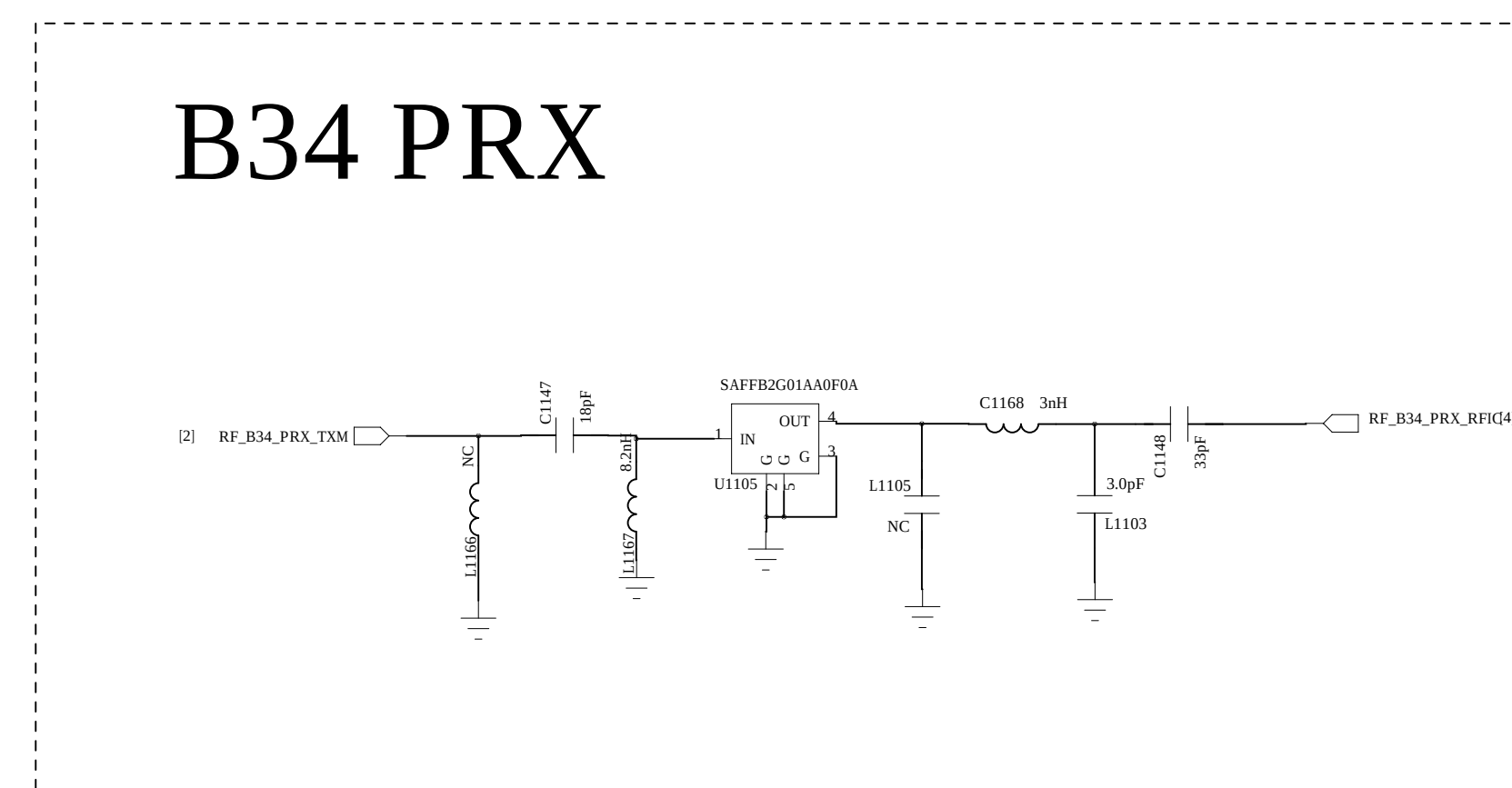
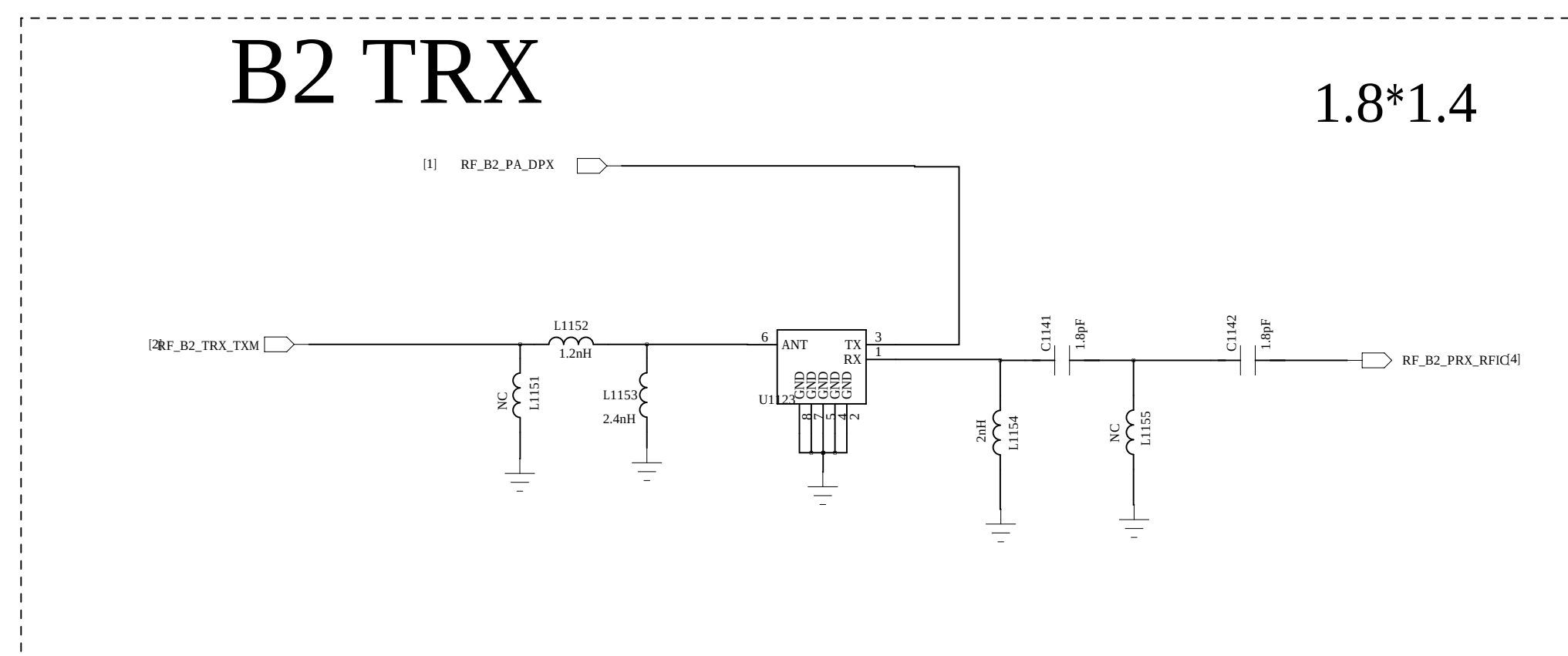
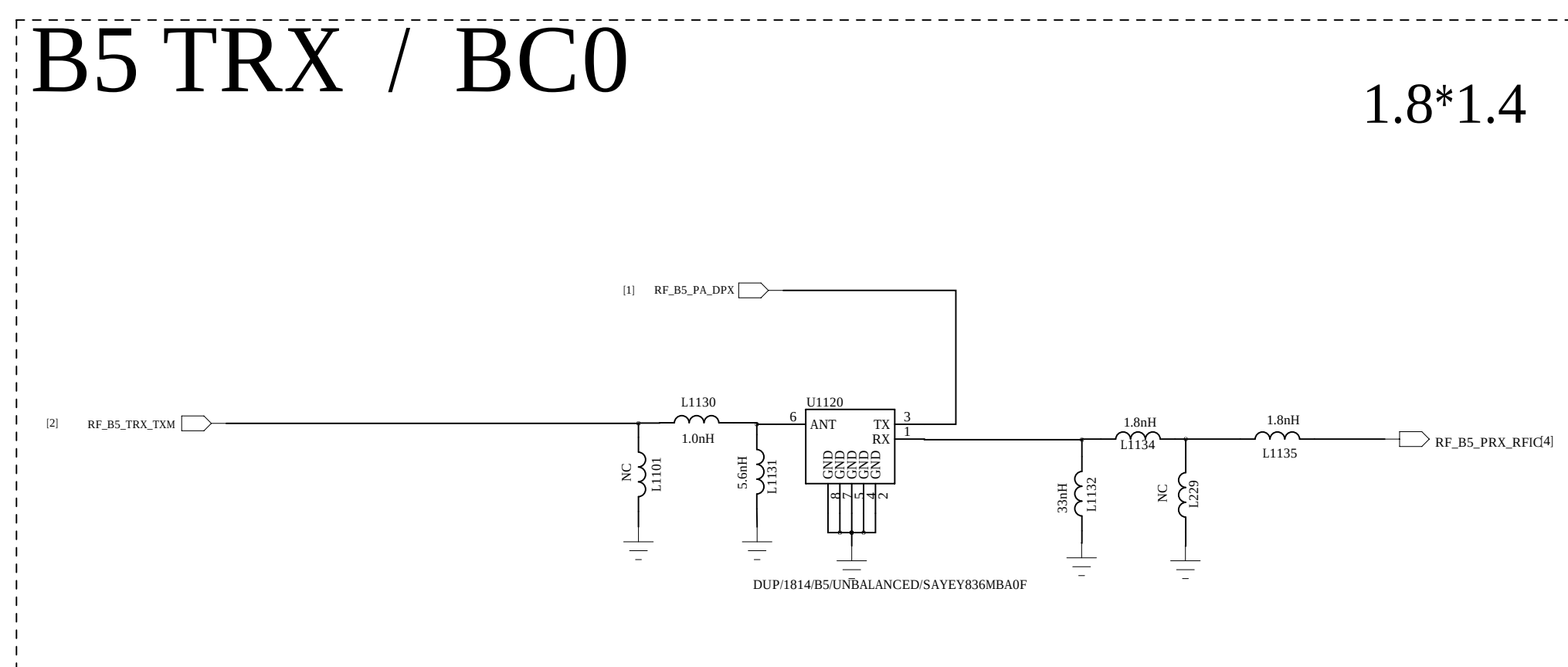
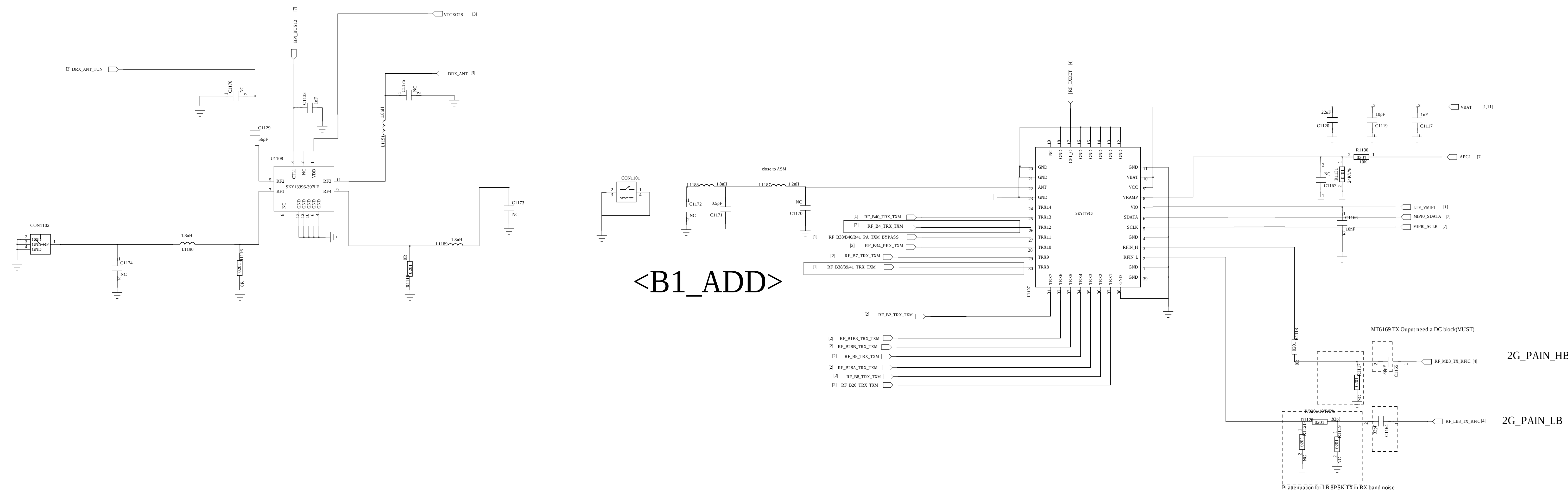


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LTB	ECOM	APPROVED	DATE

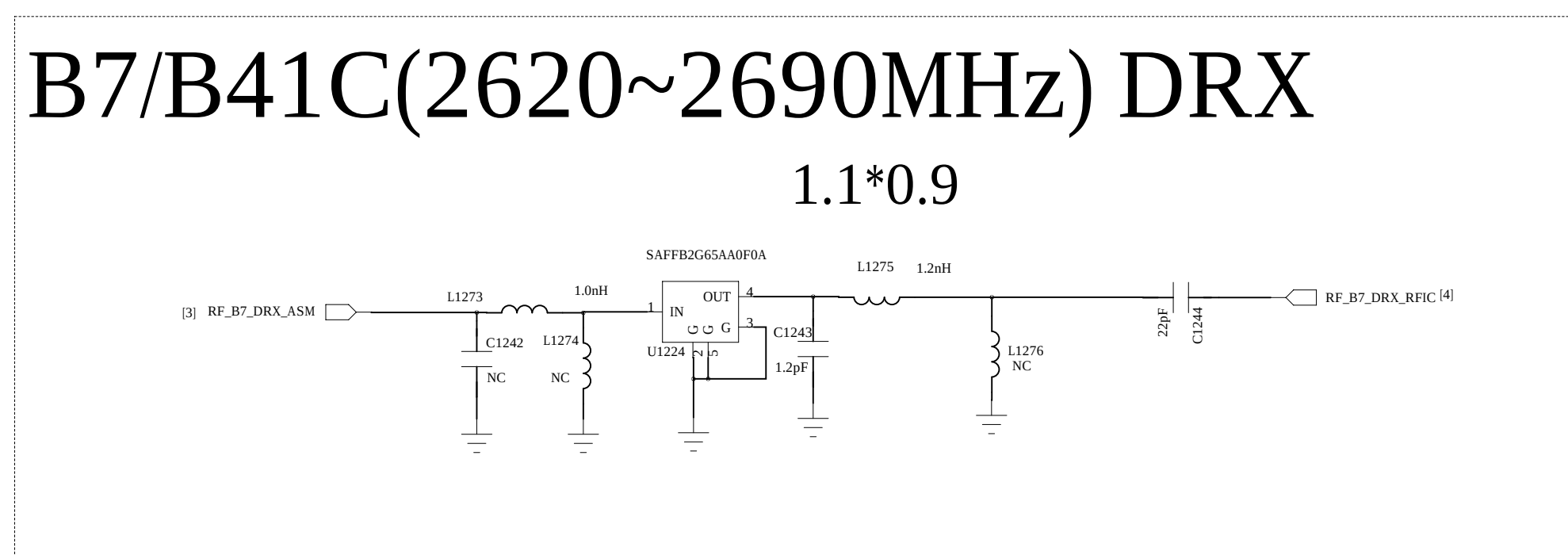
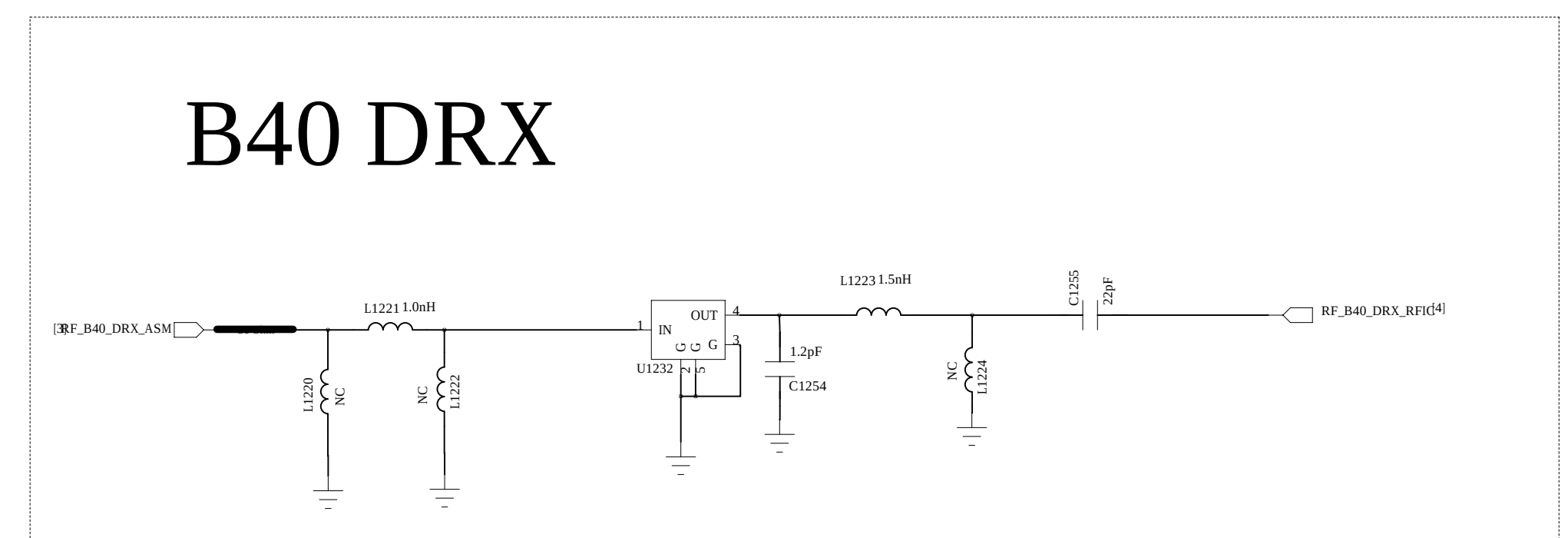
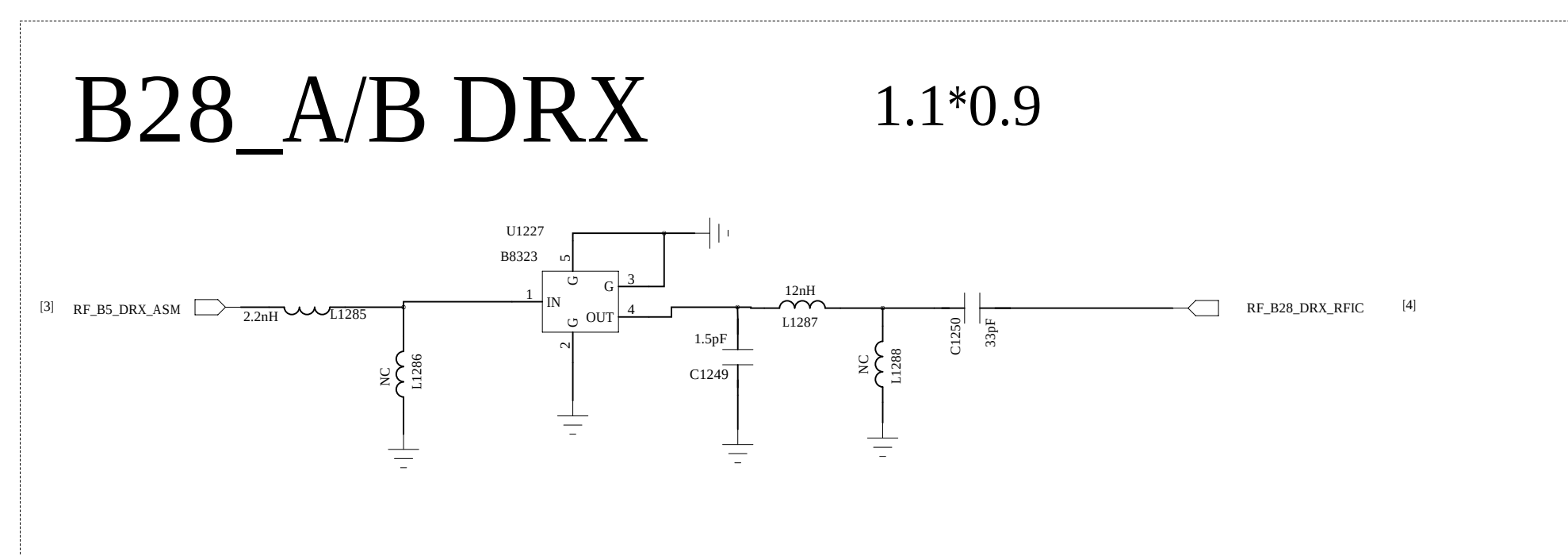
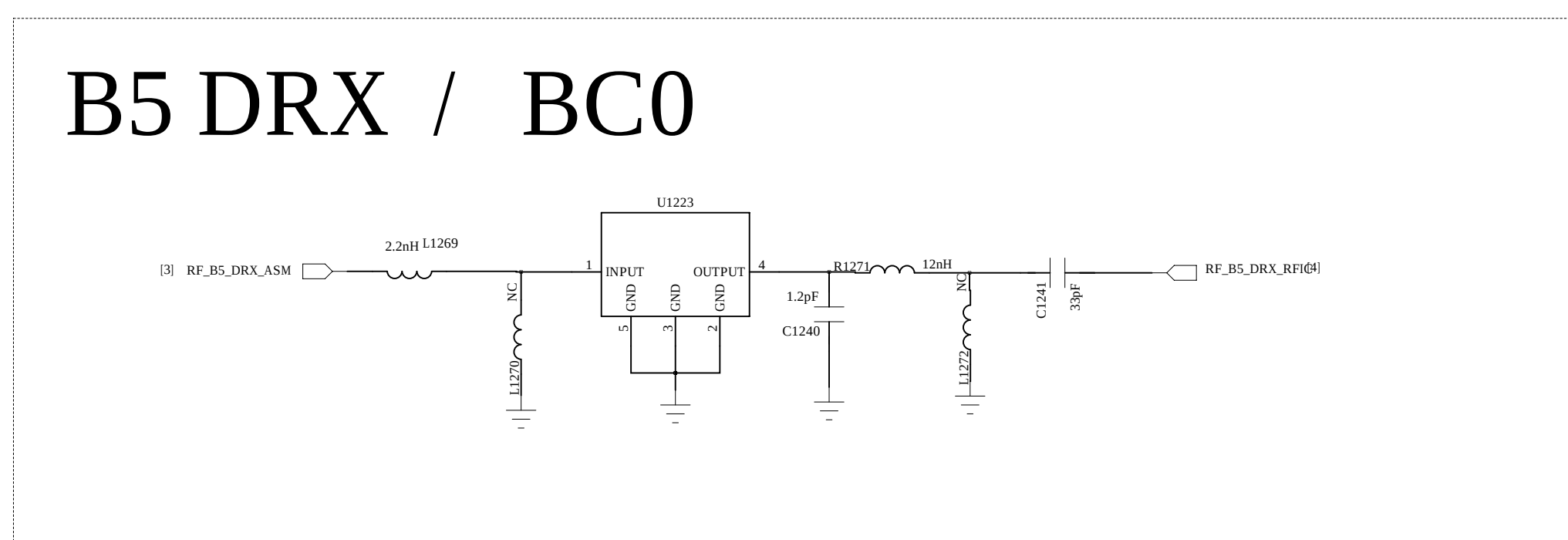
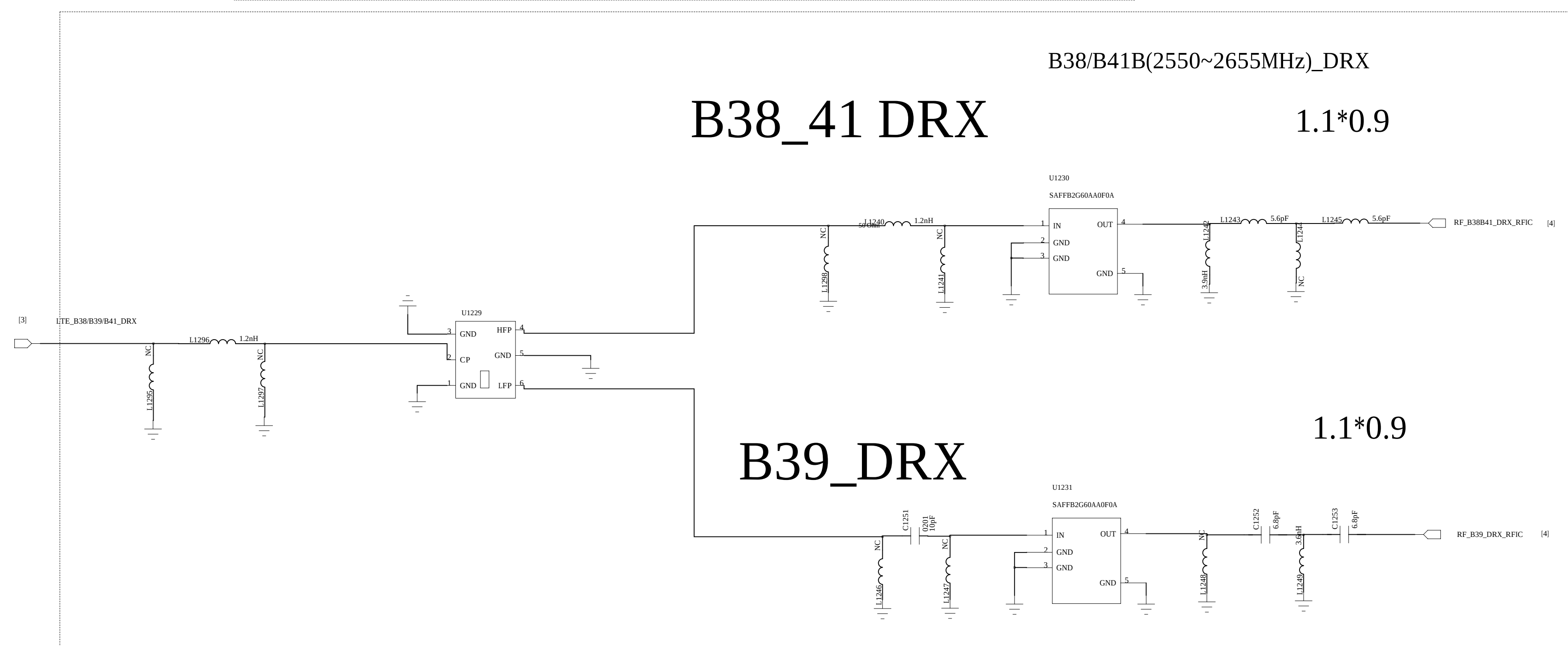
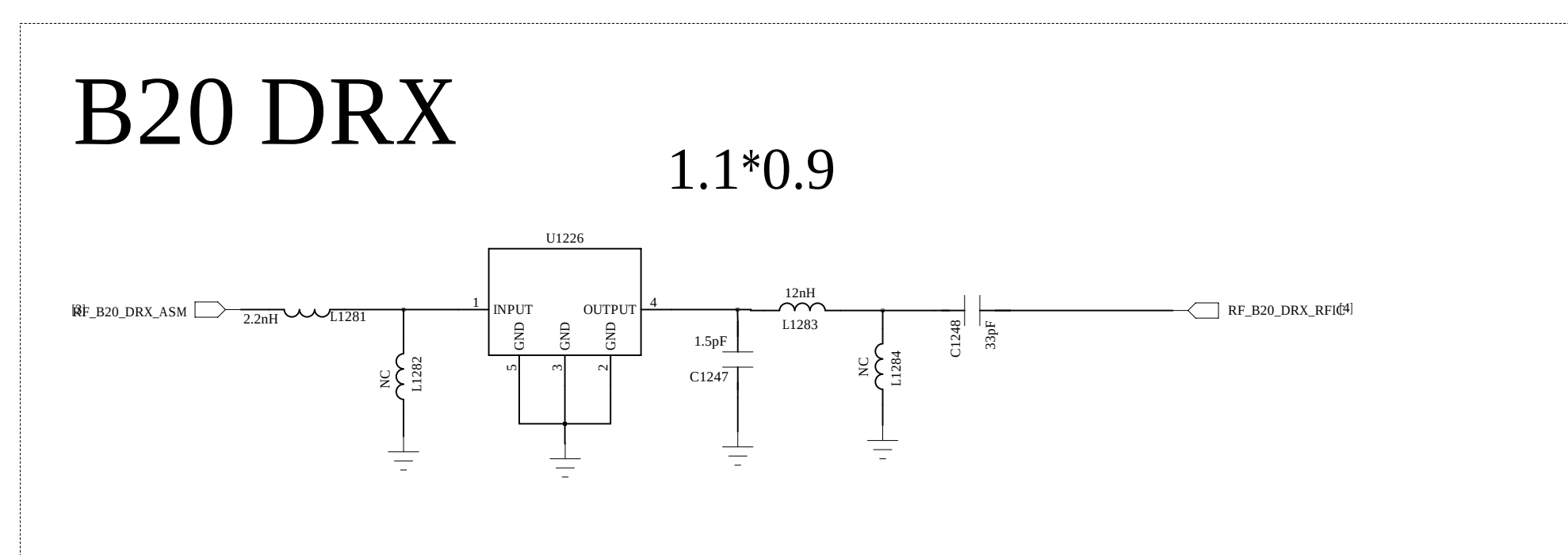
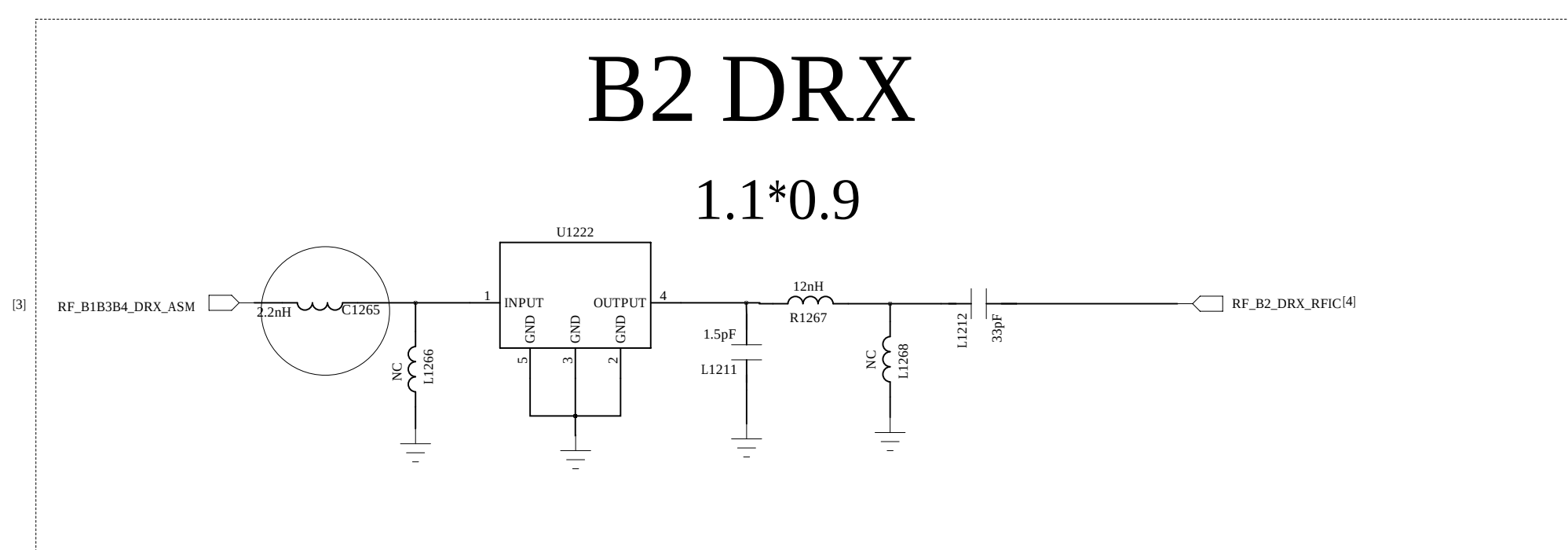
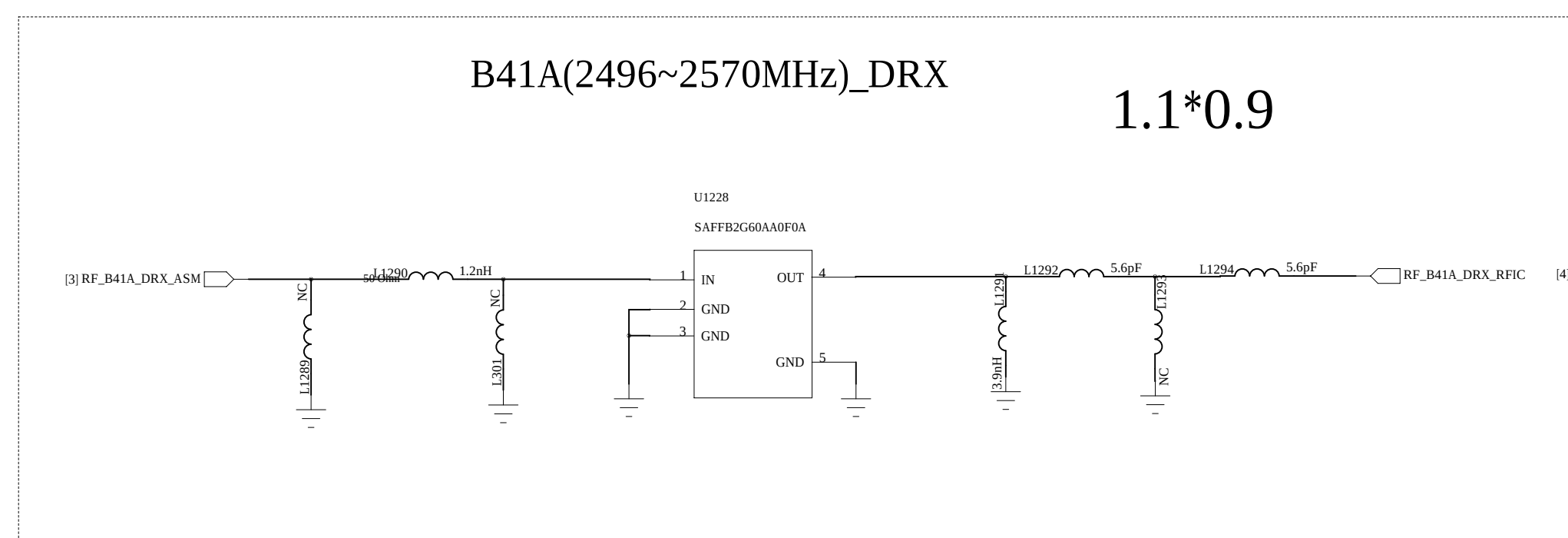
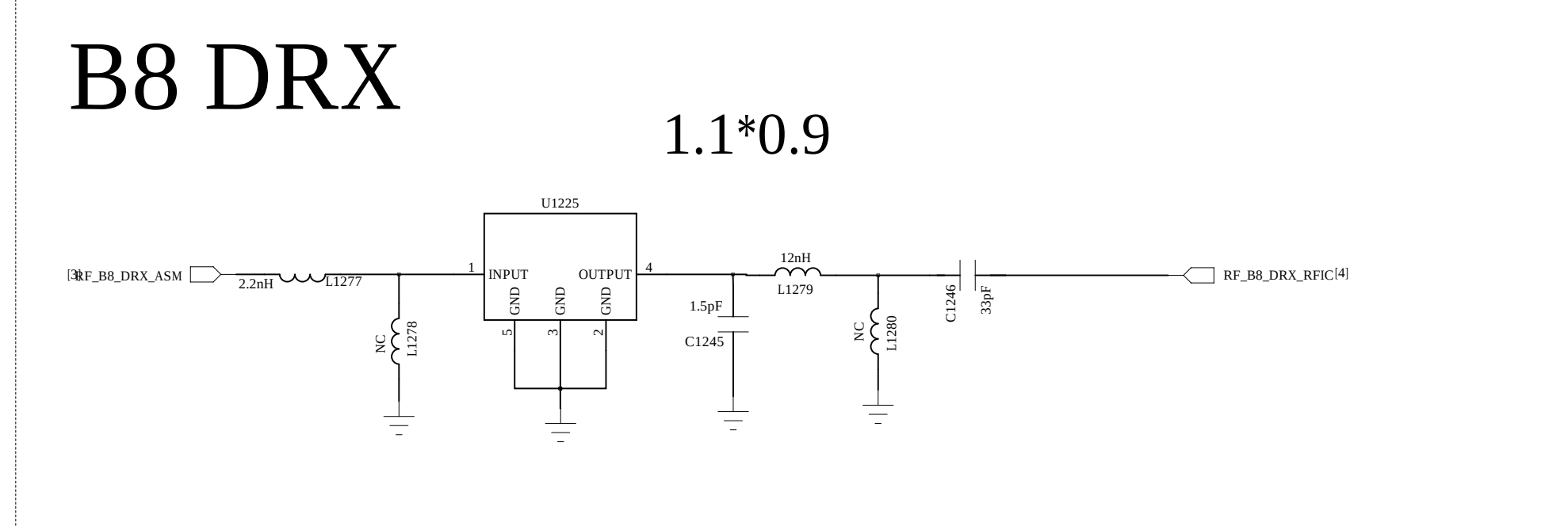
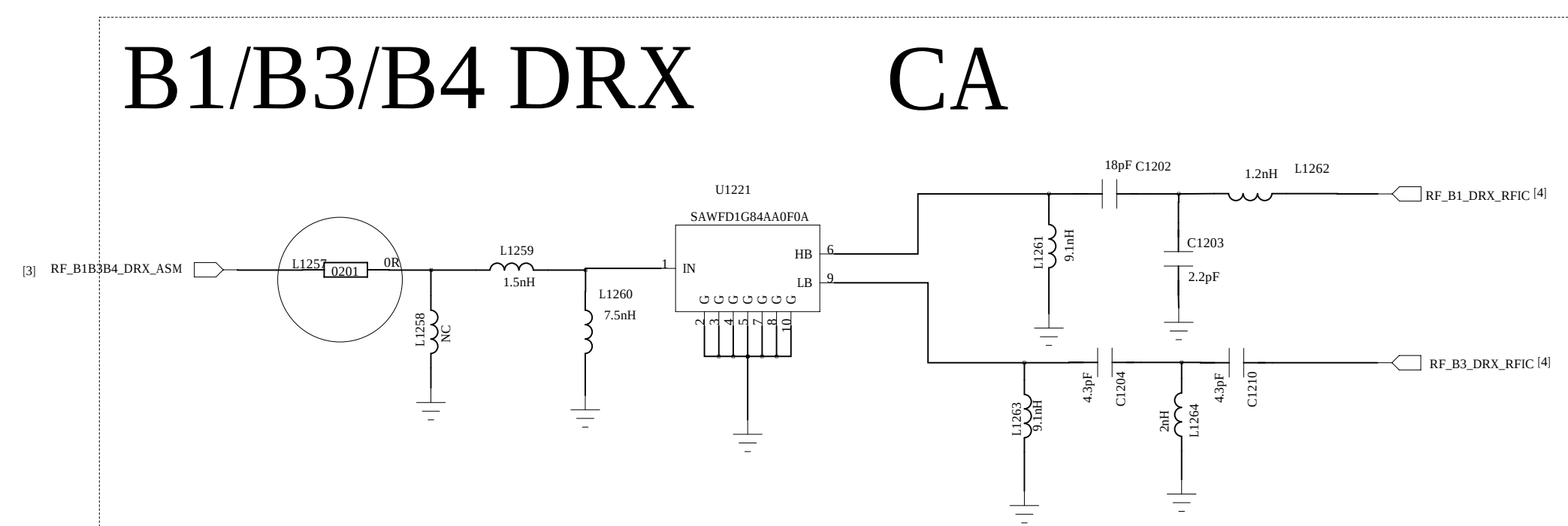
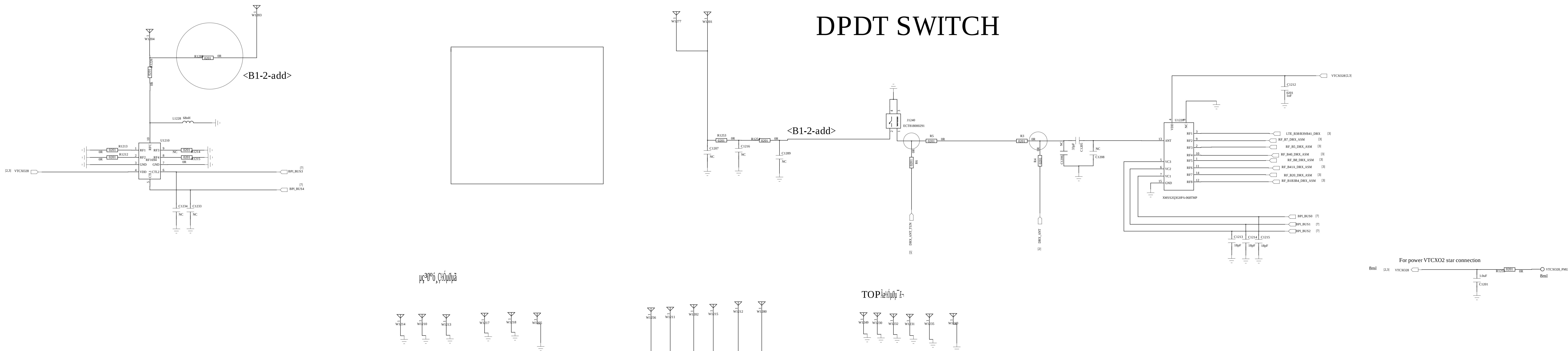
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DPDT SWITCH

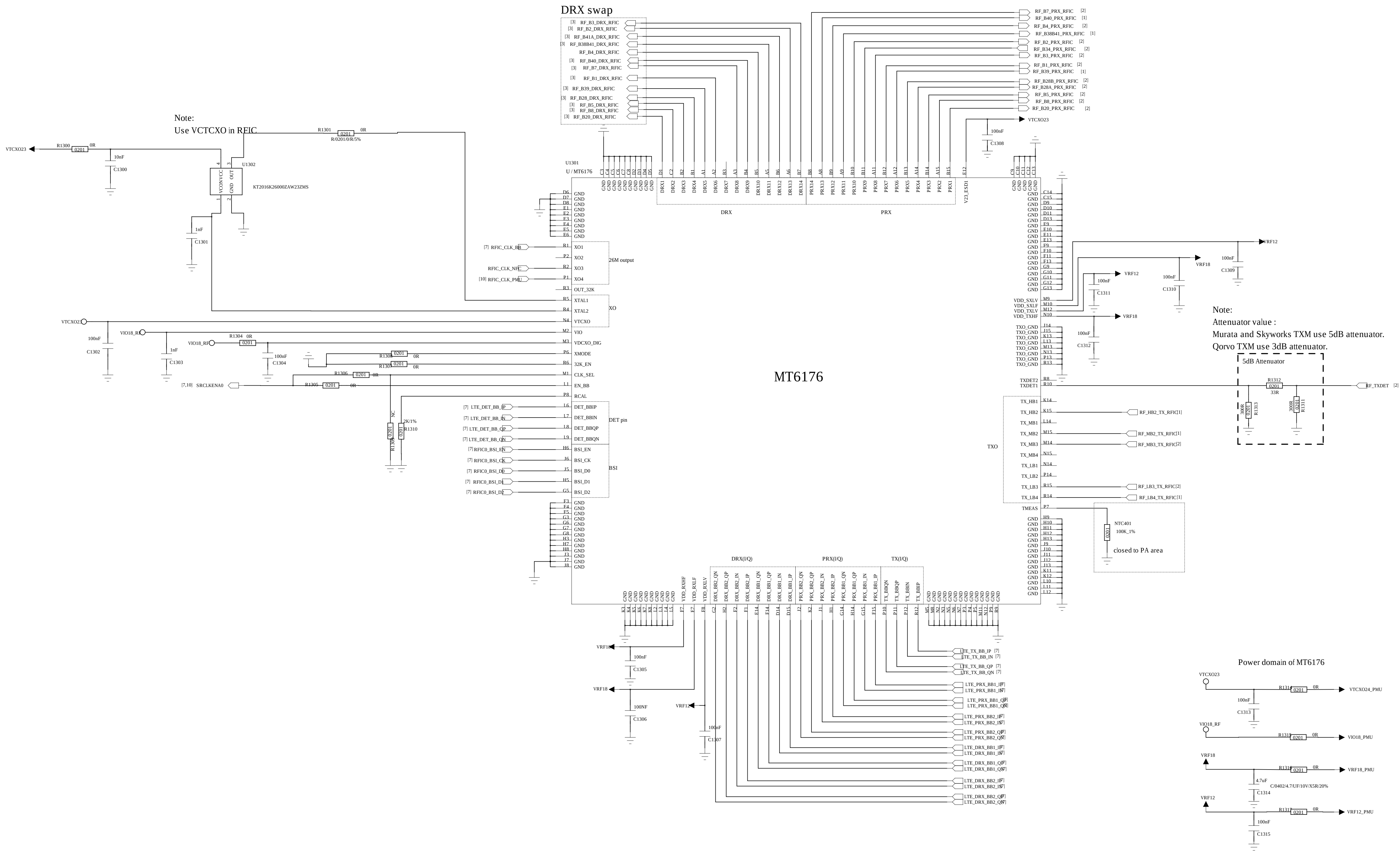
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LTR	ECD NO:	APPROVED:	DATE:



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CHECKED: ◁Checked By▷	DATED: ◁Checked Date▷	<Code>	A0	CM212	A
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REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE

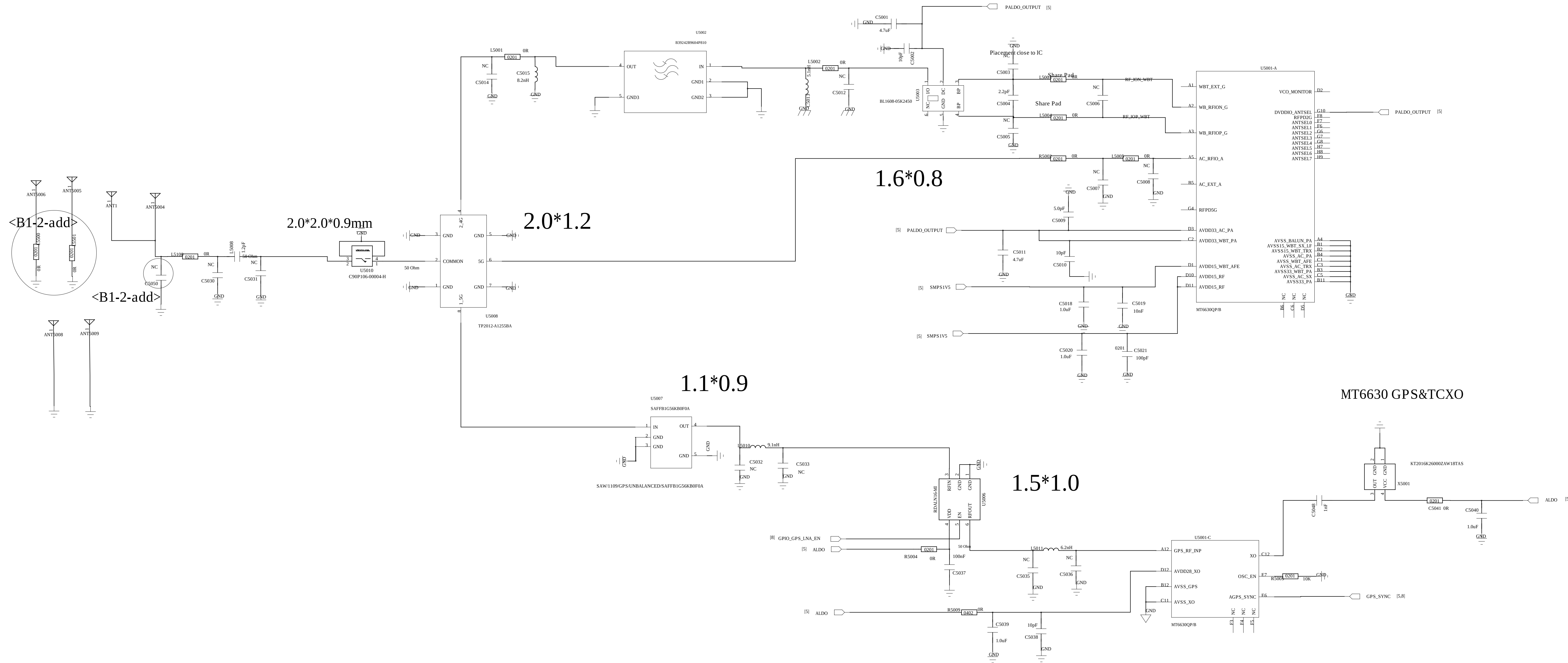


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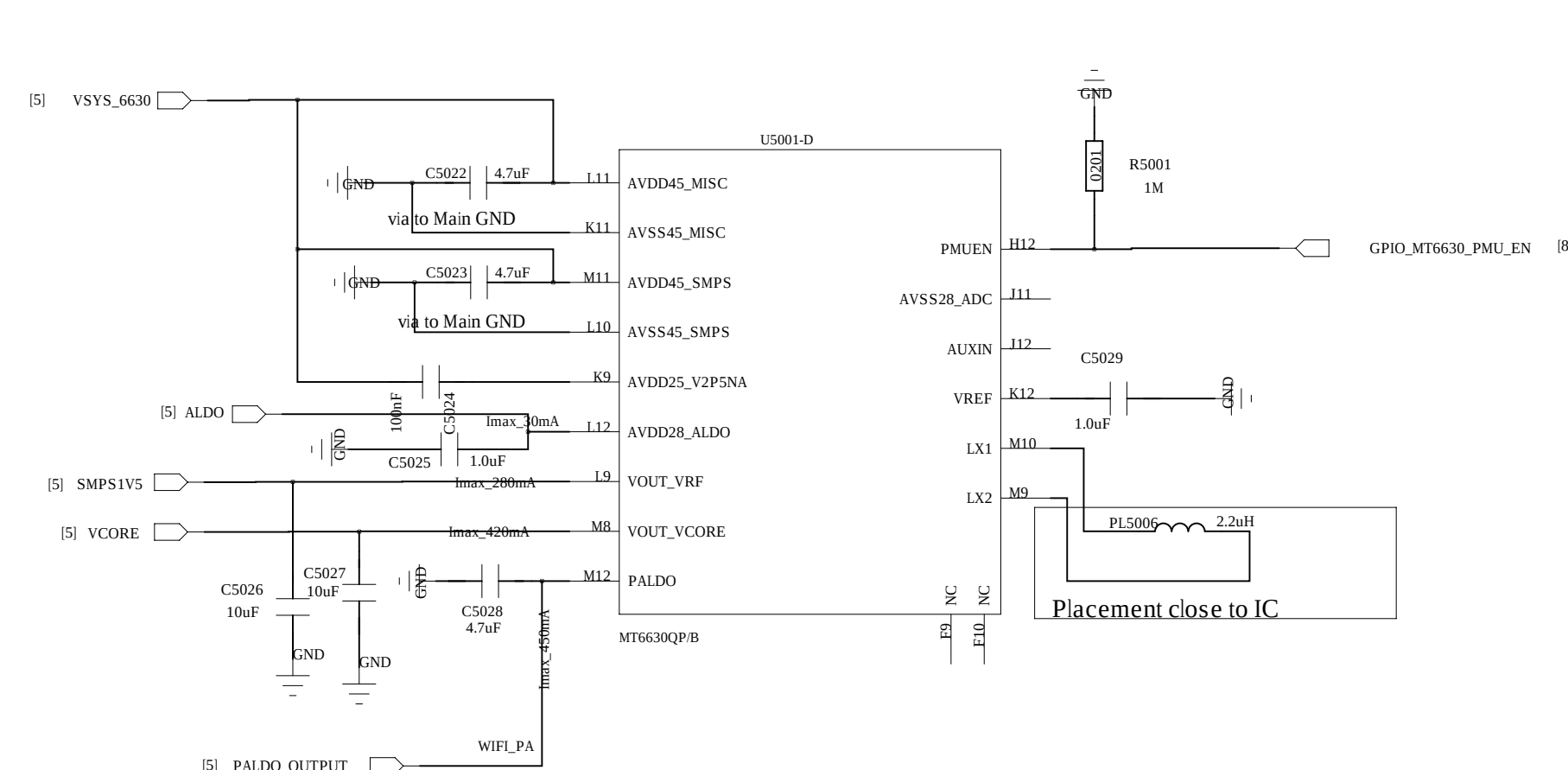
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RF_TX			
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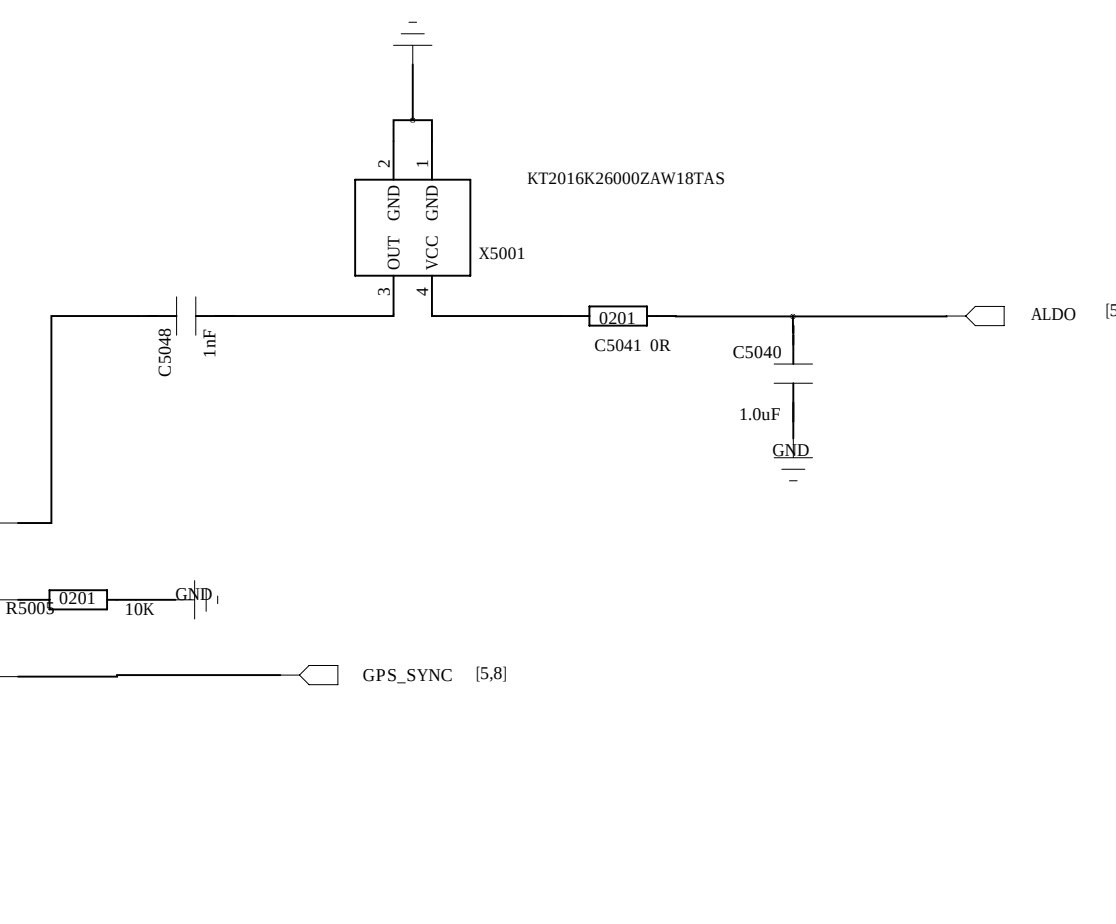
MT6630 WiFi_2.4G_5G (11ac) & BT



MT6630 PMU

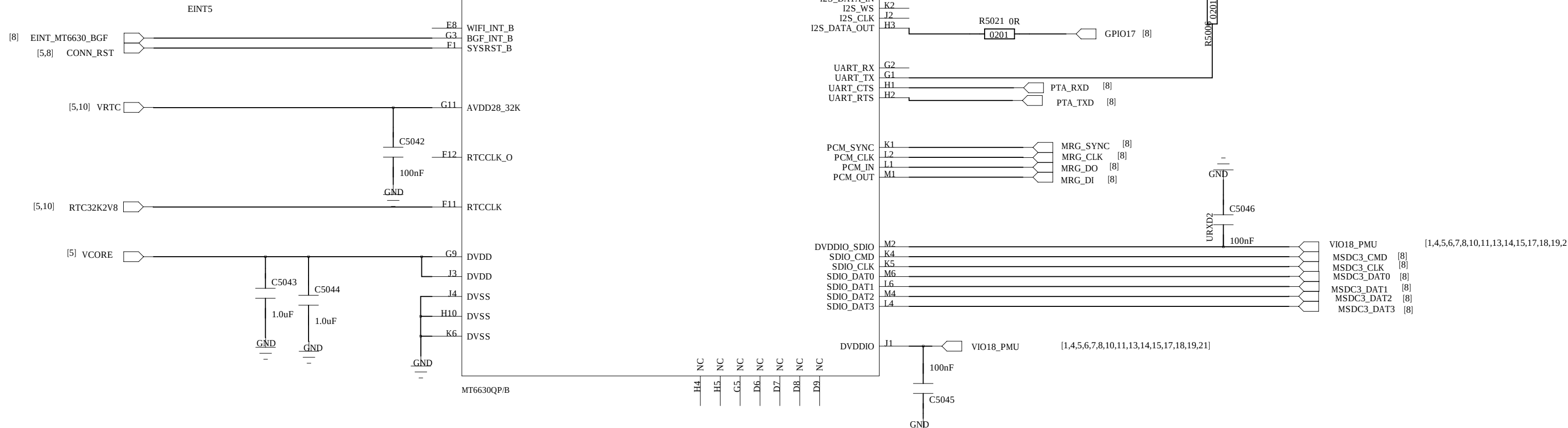


MT6630 GPS&TCXO

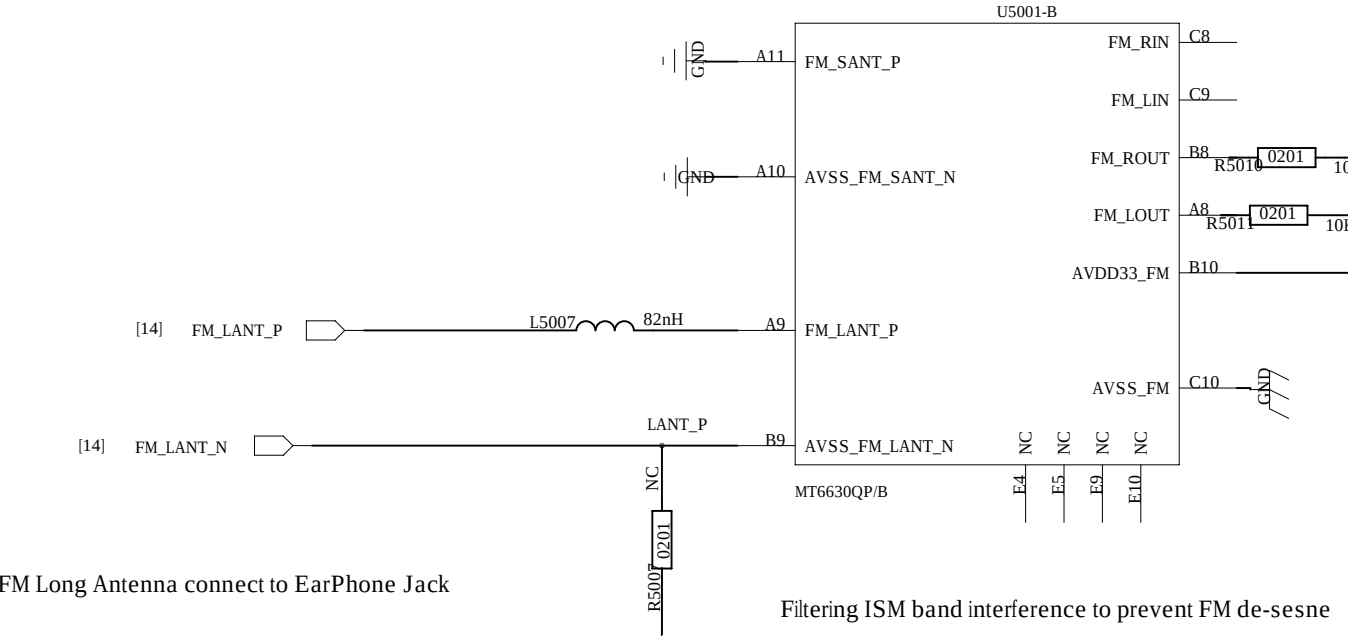
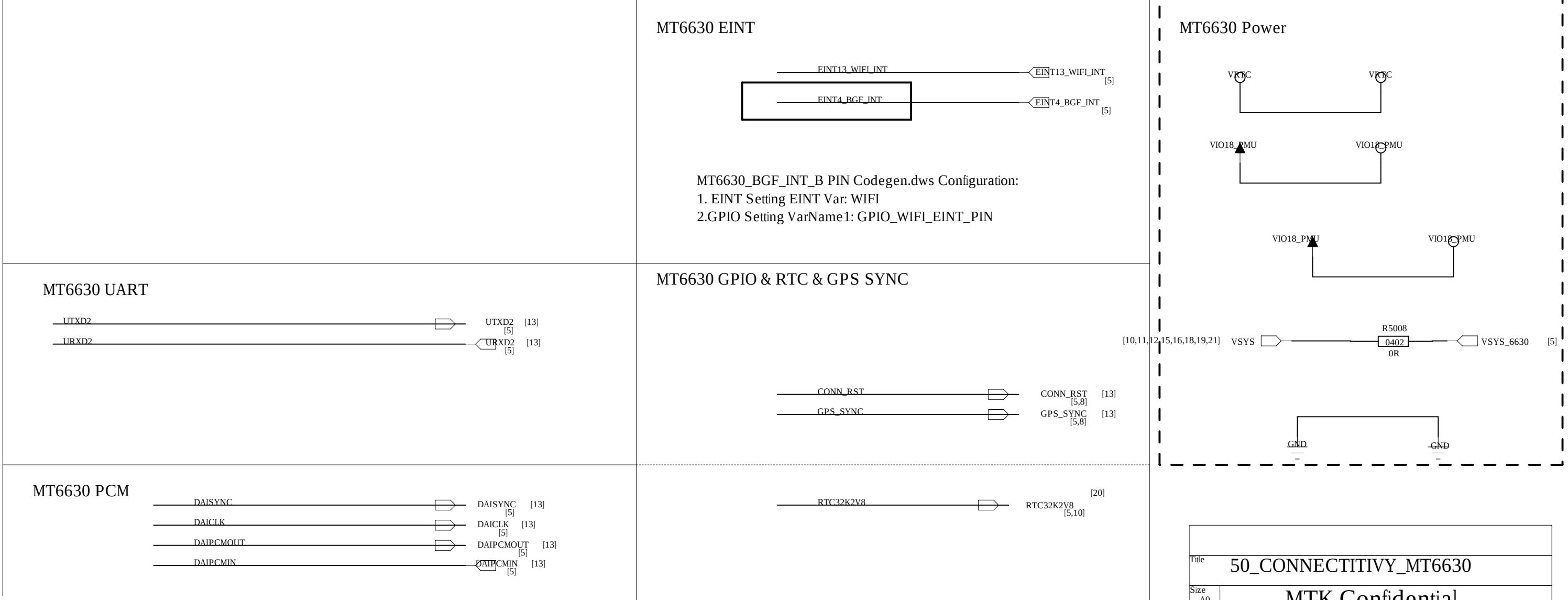


????????????

MT6630 Host Interface



OFF-PAGE_MT6630 Host Interface

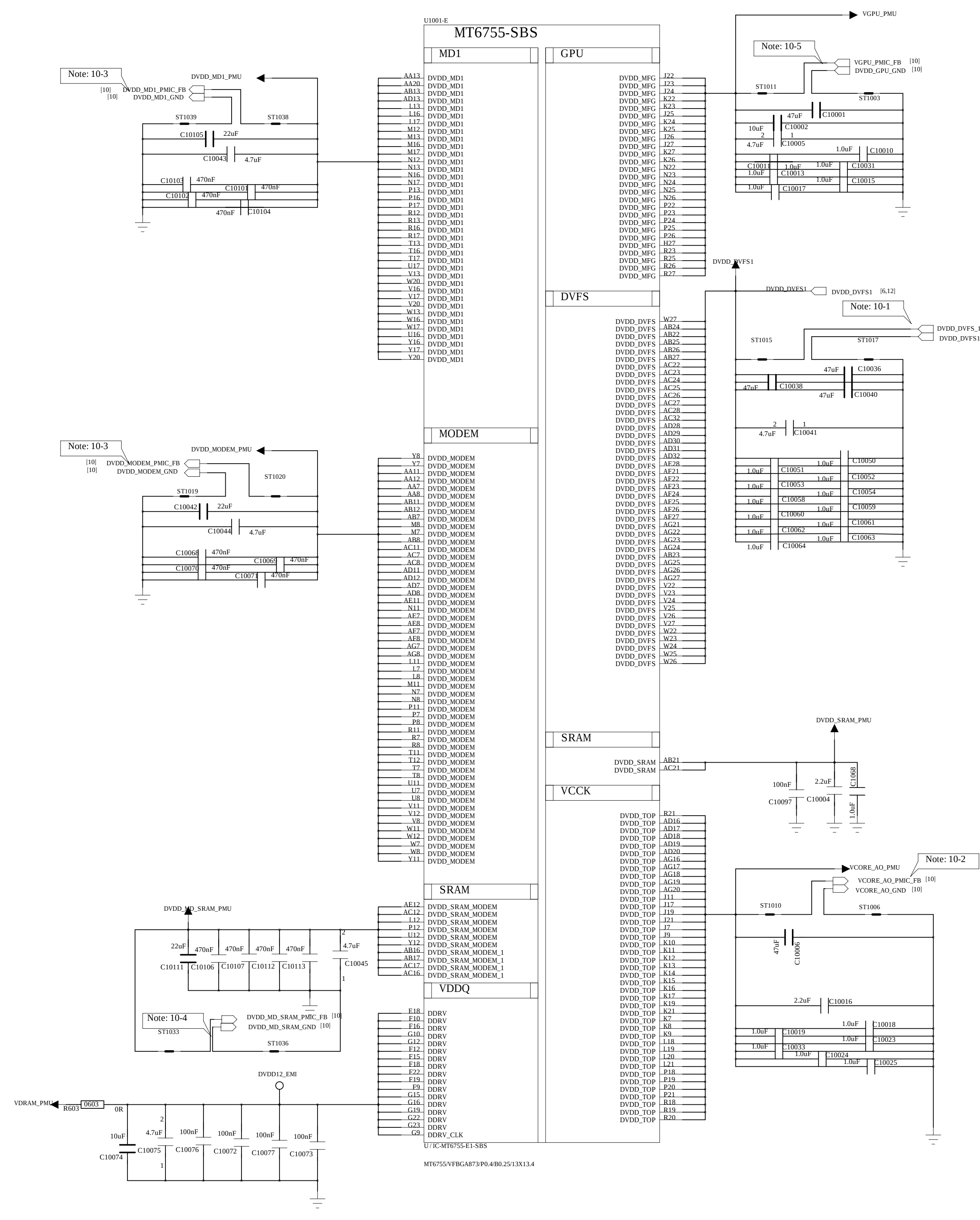


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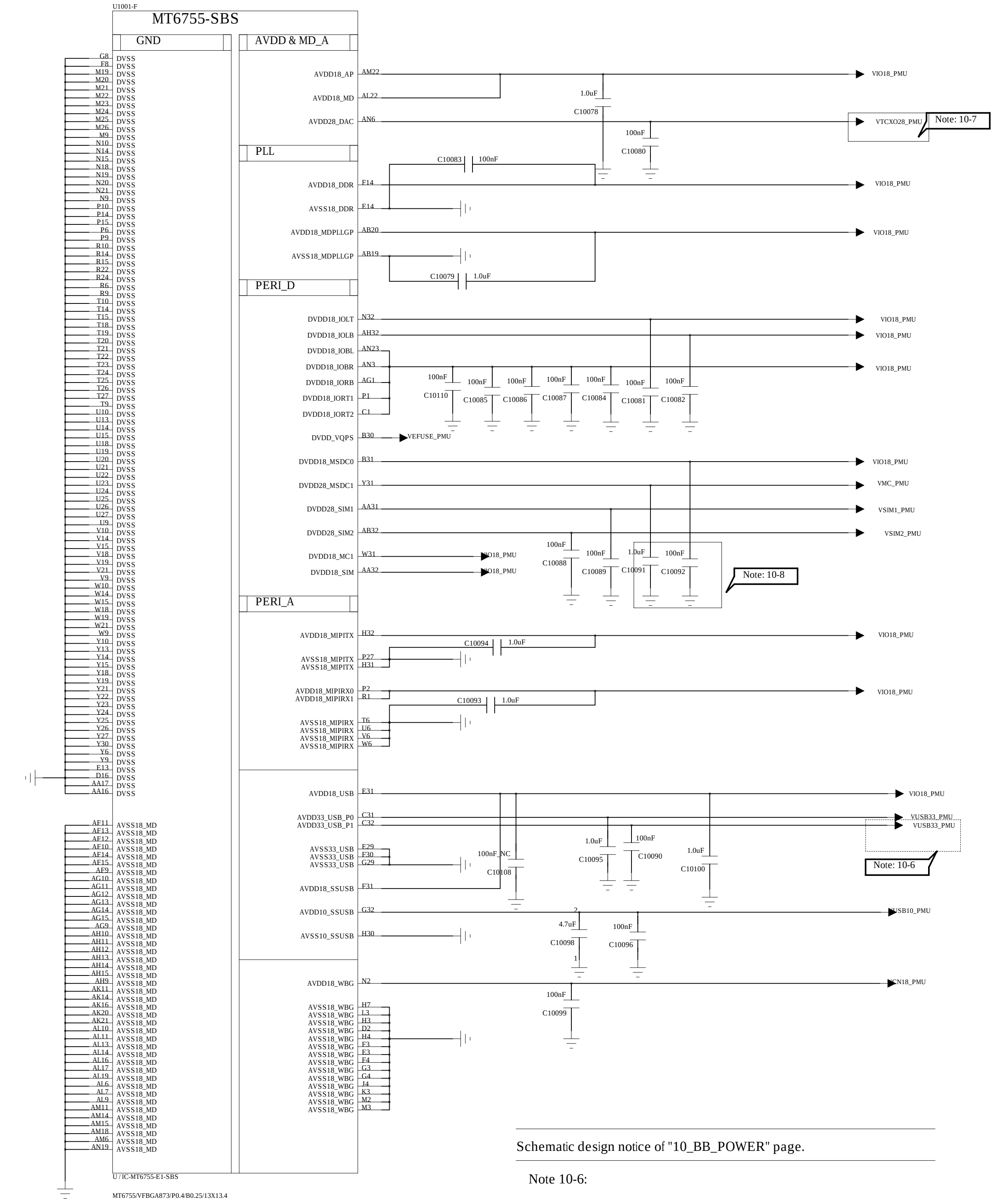
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50_CONNECTIVITY_MT6630
MTK Confidential

REVISION RECORD			
VER	ECO NO	APPROVED	DATE



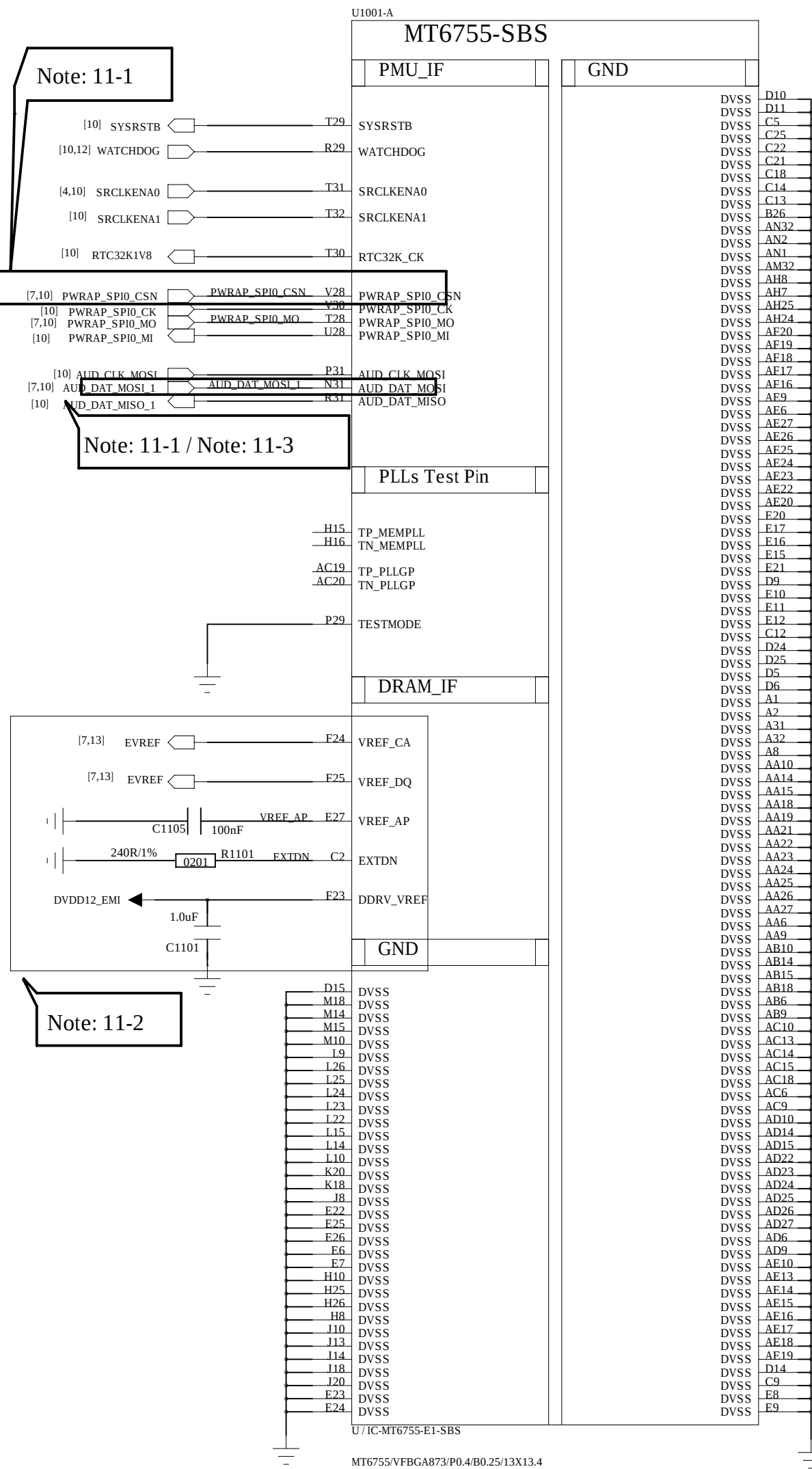
- Schematic design notice of "10_BB_POWER" page.
- Note 10-1: 4 mil GND trace with good shielding to PMIC (Differential)
Differential pair of DVFS1 remote sense must be close to BB's ball.
- Note 10-2: VCORE_AO remote sense must be close to BB's ball.
Remote sense trace with GNDshielding to PMIC (Differential)
- Note 10-3: DVDD_MD1 and DVDD_MODEM remote sense must be close to BB's ball.
Remote sense trace with GNDshielding to PMIC (Differential)
- Note 10-4: VDRAM remote sense must be close to BB's ball.
- Note 10-5: Differential pair of GPU remote sense must be close to BB's ball.
- Note 10-7: AVDD28_DAC (AN6 ball) must be powered by "VTCX028_PMU".
- Note 10-8: C10091 closed DVDD28_MSDC1 150mH
C10092 closed DVDD18_MSDC0 150mH



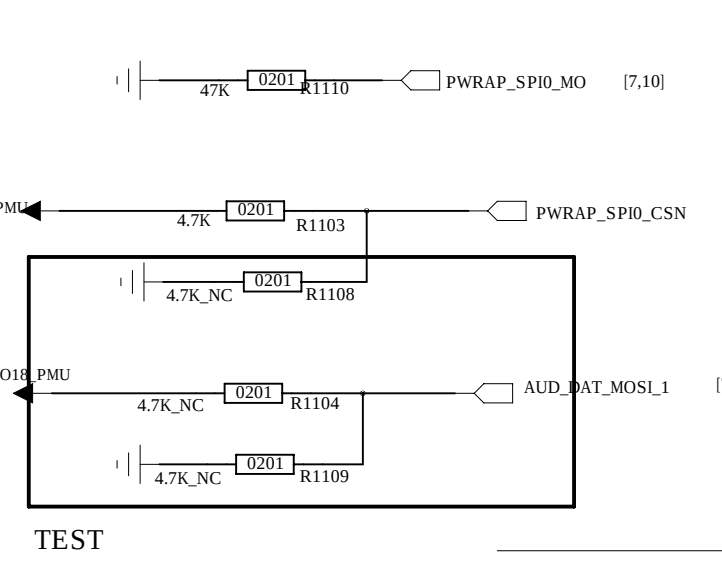
- Schematic design notice of "10_BB_POWER" page.
- Note 10-6: Connect AVDD33_USB_P1 (C32 bal) to "VSIM1_PMU" for IC-USB / Samrt card application.
Connect AVDD33_USB_P1 (C32 bal) to "VUSB33_PMU" for USB application or not used.

COMPANY: Longcheer			
TITLE: <15_AUDIO>			
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CHECKED: <Checked By>	DATED: <Checked Date>	CODE: <Code>	SIZE: A0
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		SHEET: 6F	23

REVISION RECORD			
VER	ECO NO	APPROVED	DATE



PWRAP_SPI0_CSN	AUD_DAT_MOSI	JTAG
HI	LO	N/A
HI	HI	SPI_CS/SPI_CLK/SPI_M0/SPI_M1
LO	LO	CAM_PDN0/CAM_PDN1/CAM_RST0/CAM_RST1
LO	HI	MSDC1_CLK/MSDC1_CMD/MSDC1_DAT0/MSDC1_DAT1

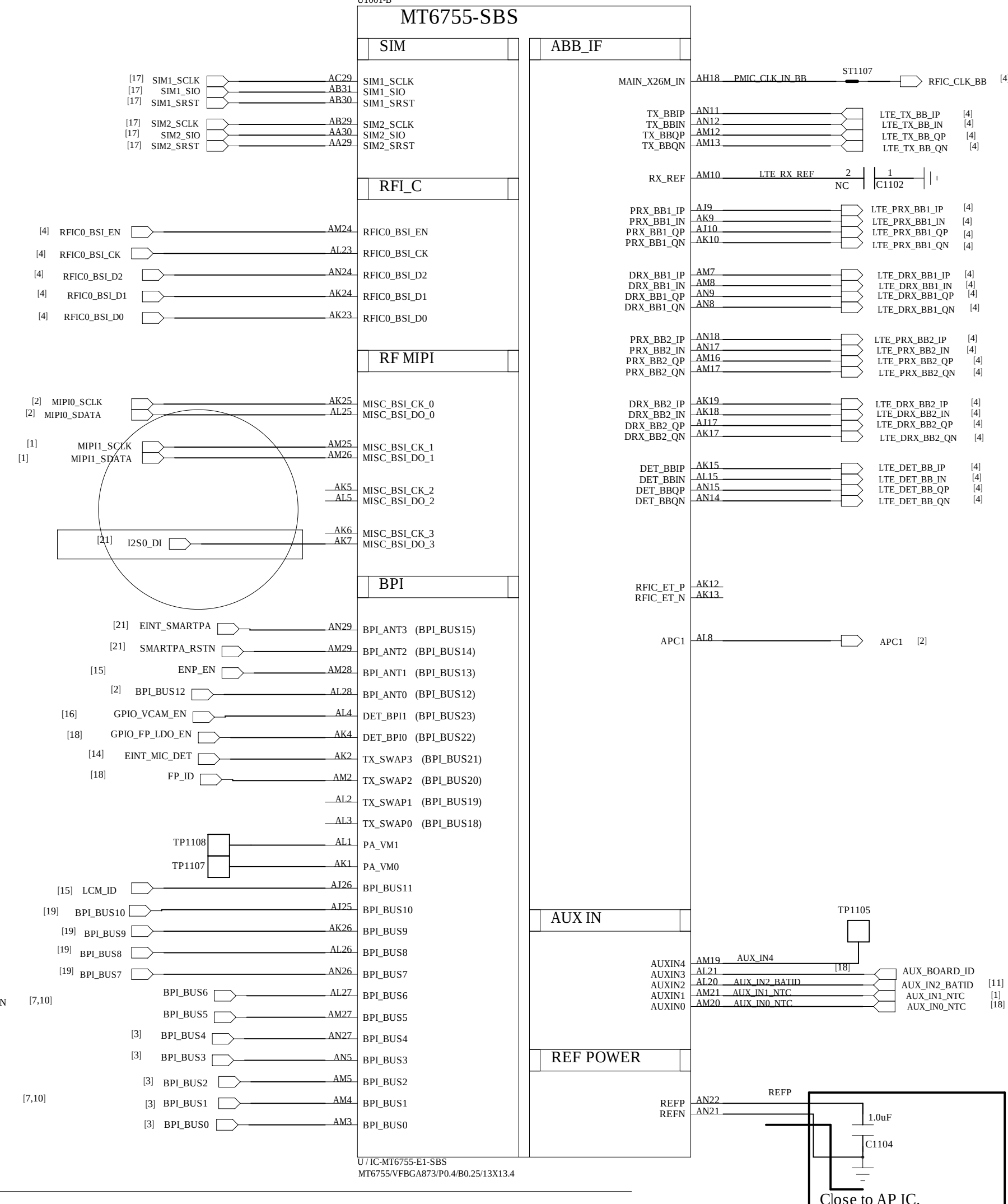


Schematic design notice of "11_BB_1" page.

Note 11-1: PWRAP_SPI0_CSN and AUD_DAT_MOSI are JTAG feature in bootstrap.

Note 11-2: The de-coupling cap. of DRAM VREF have to be placed as close to BB as possible.

Note 11-3: AUD_CLK_MOSI/AUD_DAT_MOSI/AUD_DAT_MISO should surround shield with GND
AUD_CLK_MOSI/AUD_DAT_MOSI/AUD_DAT_MISO layout trace should be same length and cannot over 1350mils



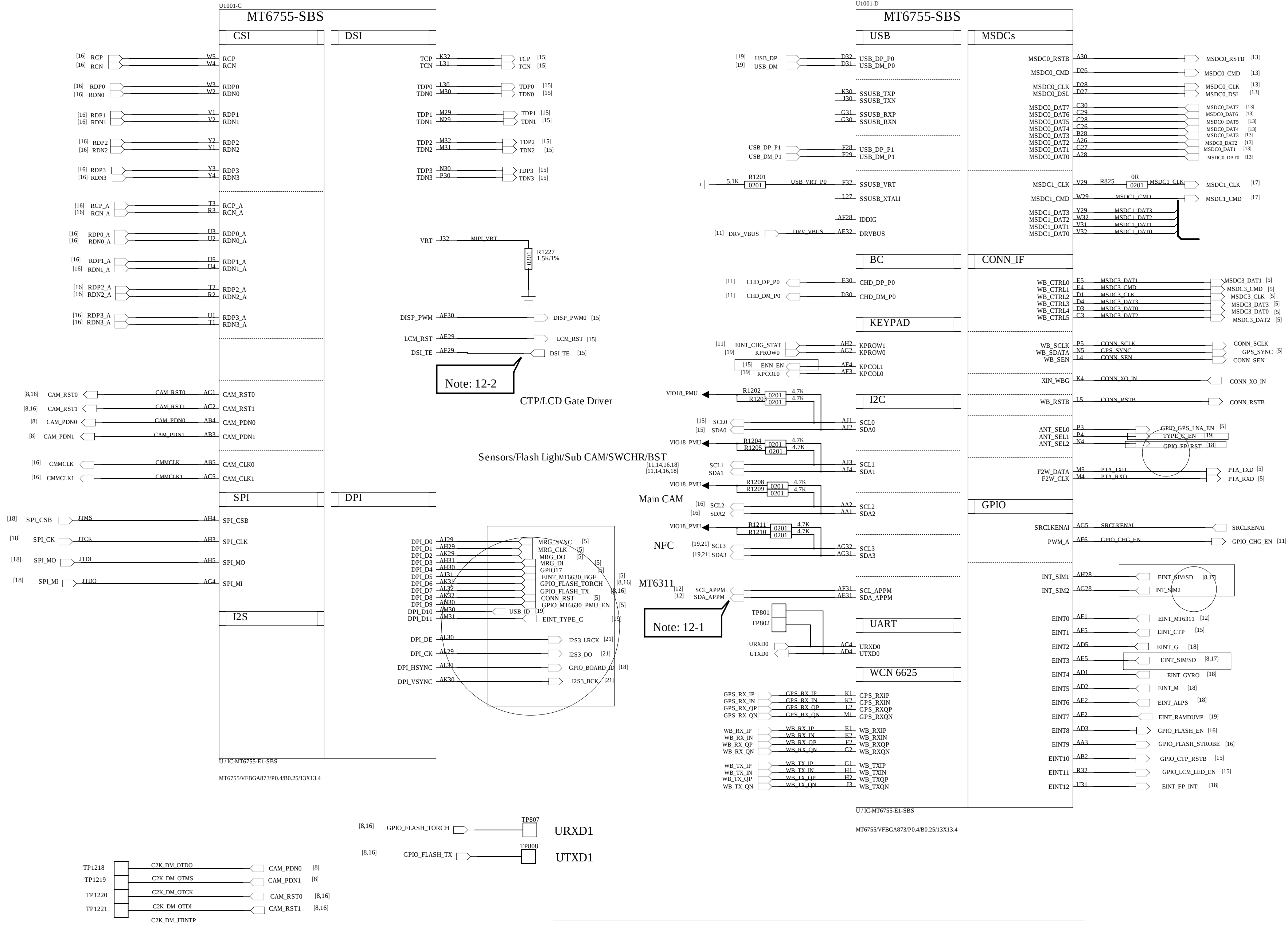
COMPANY: Longcheer

TITLE: <15_AUDIO>

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RELEASED	<Released By>	DATED	<Release Date>				

<Code>	A0	CM212	A
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REVISION RECORD			
LTB	ECO NO	APPROVED	DATE



Schematic design notice of "12_BB_2" page.

Note 12-1: I2C APM is dedicated for MT6311 2-phase buck control.

Note 12-2: DSI_TE must connect to LCM's TE pin

COMPANY: Longcheer			
TITLE: <15_AUDIO>			
DESIGN: <Drawn By>	DATE: <Drawn Date>	CODE: <Code>	SIZE: A0
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SHEET: Bf 23			

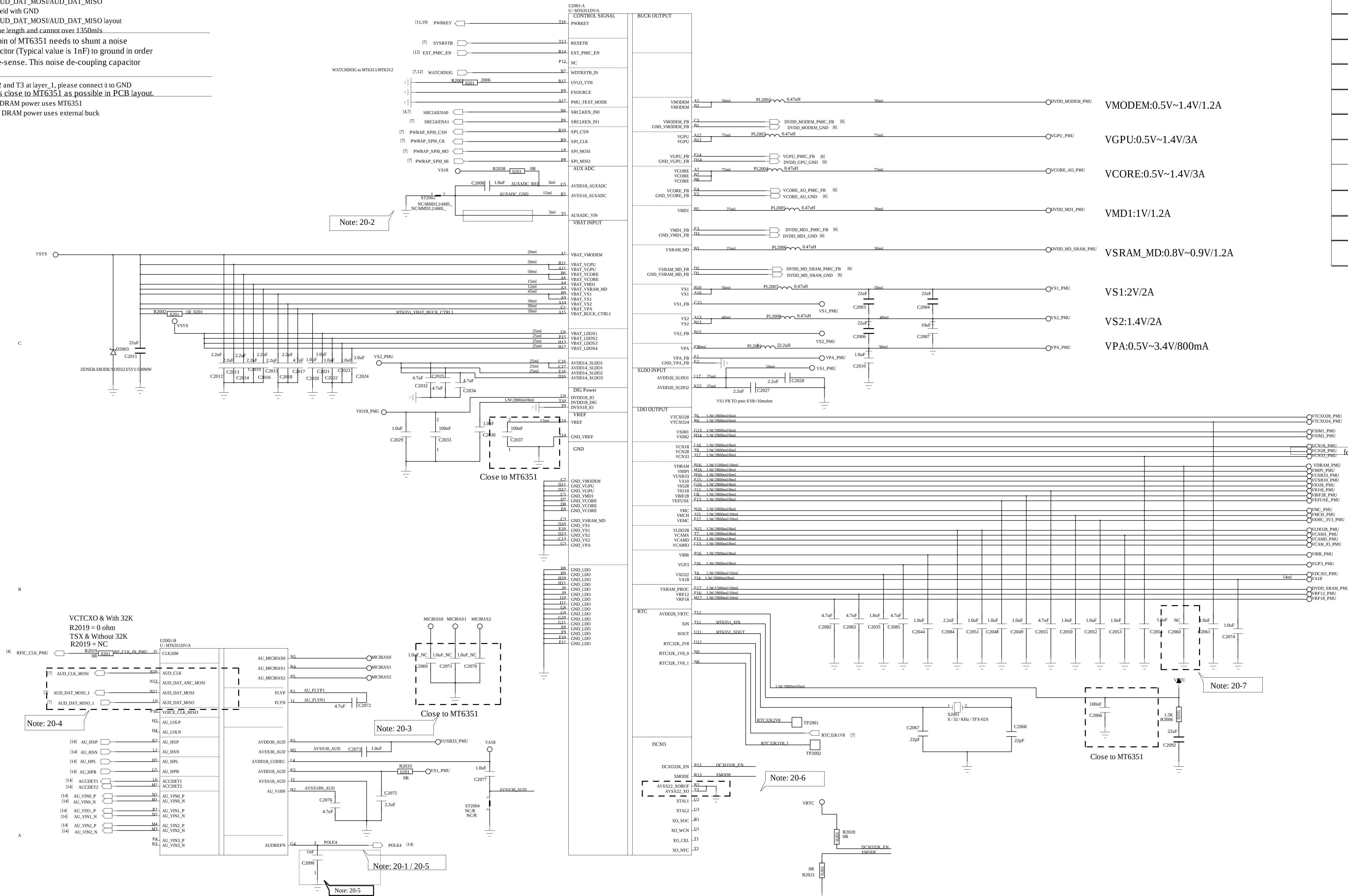
REVISION RECORD			
VER	ECO NO	APPROVED	DATE

Type	Name	Input power plan	Default On/Off	Local side cap	Application side cap	Max cap
ALDO	VTCX028	VBAT_LDOS1	Off	No need	1uF±20%	3uF±20%
ALDO	VTCX024	ISENSE or BATSNS	On	No need	1uF±20%	3uF±20%
ALDO	VX022	ISENSE or BATSNS	On	No need	1uF±20%	3uF±20%
SLDO	VRF18	AVDD20_SLDO1	Off	No need	9.4uF±20%	10uF±20%
SLDO	VRF12	AVDD14_SLDO2	Off	No need	4.7uF±20%	6.9uF±20%
DLDO	VCN33	VBAT_LDOS4	Off	1uF±20%	4.7uF±20%	9.4uF±20%
ALDO	VCN28	VBAT_LDOS1	Off	No need	1uF±20%	3uF±20%
SLDO	VCN18	AVDD20_SLDO1	Off	No need	1uF±20%	3uF±20%
ALDO	VCAMA	VBAT_LDOS1	Off	1uF±20%	2.2uF±20%	6.9uF±20%
SLDO	VCAMIO	AVDD20_SLDO1	Off	No need	1uF±20%	3uF±20%
SLDO	VCAMD	AVDD14_SLDO2	Off	No need	10uF±20%	30uF±20%
SLDO	VA10	AVDD14_SLDO2	On	No need	4.7uF±20%	6.9uF±20%
ALDO	VA18	VBAT_LDOS3	On	No need	1uF±20%	3uF±20%
SLDO	VSRAM_PROC	AVDD14_SLDO3	On	No need	2.2uF±20%	3.3uF±20%
SLDO	VDRAM	AVDD14_SLDO1	On	No need	22uF±20%	30uF±20%
DLDO	VSIM1	VBAT_LDOS3	Off	No need	1uF±20%	3uF±20%

Type	Name	Input power plan	Default On/Off	Local side cap	Application side cap	Max cap
DLDO	VSIM2	VBAT_LDOS3	Off	No need	1uF±20%	3uF±20%
SLDO	VMIP1	AVDD20_SLDO1	Off	No need	1uF±20%	3uF±20%
DLDO	VLD028	VBAT_LDOS2	Off	1uF±20%	4.7uF±20%	7uF±20%
SLDO	VGF3	AVDD20_SLDO2	Off	No need	4.7uF±20%	6.9uF±20%
DLDO	VEFUSE	VBAT_LDOS3	Off	No need	1uF±20%	3uF±20%
DLDO	VIO28	VBAT_LDOS4	On	No need	2.2uF±20%	6.6uF±20%
DLDO	VMC	VBAT_LDOS2	On	No need	1uF±20%	3uF±20%
DLDO	VMCH	VBAT_LDOS3	On	1uF±20%	4.7uF±20%	7uF±20%
DLDO	YEMC	VBAT_LDOS4	On	1uF±20%	4.7uF±20%	7uF±20%
DLDO	VIBR	VBAT_LDOS2	Off	No need	1uF±20%	3uF±20%
ALDO	VUSB33	VBAT_LDOS4	On	No need	1uF±20%	3uF±20%
ALDO	VBIF28	VBAT_LDOS1	Off	No need	1uF±20%	1.2uF±20%
SLDO	VIO18	AVDD20_SLDO2	On	No need	8uF±20%	20uF±20%
VRTC	VRTC	ISENSE or BATSNS	On	0.1uF±20%	R1.5k + super cap%	1000uF±20%
VDIG18	DVDD18_DIG	ISENSE or BATSNS	On	No need	1uF±20%	1uF±20%

Schematic design notice of "20_POWER_MT6351" page.

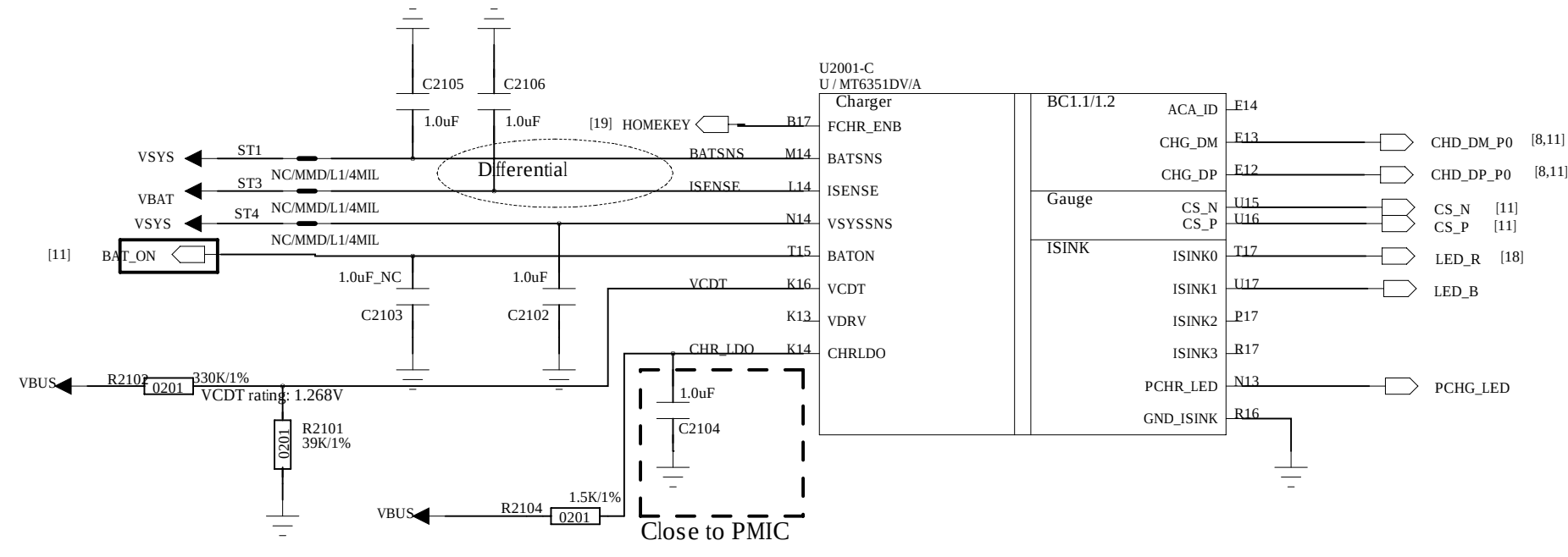
- Note 20-1: Layout trace from MT6351 ball G4 AUDREFN to Audio jack GND must surround shield with GND
- Note 20-2: Route AUXADC_REF/AUXADC_TSX as differential trace (3 mil each) both w/ GND shielding and route AUXADC_GND with 15mil trace width under AUXADC_TSX/AUXADC_REF trace to provide return current path.
- Note 20-3: For MEMS Mic could not use the 100nF near Mic. For ECM mic could use 1uF, if very care mic input noise Detail Capacitor support range please refer Design Note
- Note 20-4: AUD_CLK_MOSIAUD_DAT_MOSIAUD_DAT_MISO should surround shield with GND AUD_CLK_MOSIAUD_DAT_MOSIAUD_DAT_MISO layout trace should be same length and cannot over 1350mils
- Note 20-5: The AUDREFN pin of MT6351 needs to shunt a noise de-coupling capacitor (Typical value is 1nF) to ground in order to minimize FM de-sense. This noise de-coupling capacitor by ground via
- Note 20-6: Don't connect pin R2 and T3 at layer 1, please connect it to GND must be placed as close to MT6351 as possible in PCB layout.
- Note 20-7: C2060 is NC when DRAM power uses MT6351 Mount C2060 when DRAM power uses external buck



DCX032K_EN	XMODE	CLK OUT
1	0	VTCXO on MT6176 (OFF XO in MT6351)

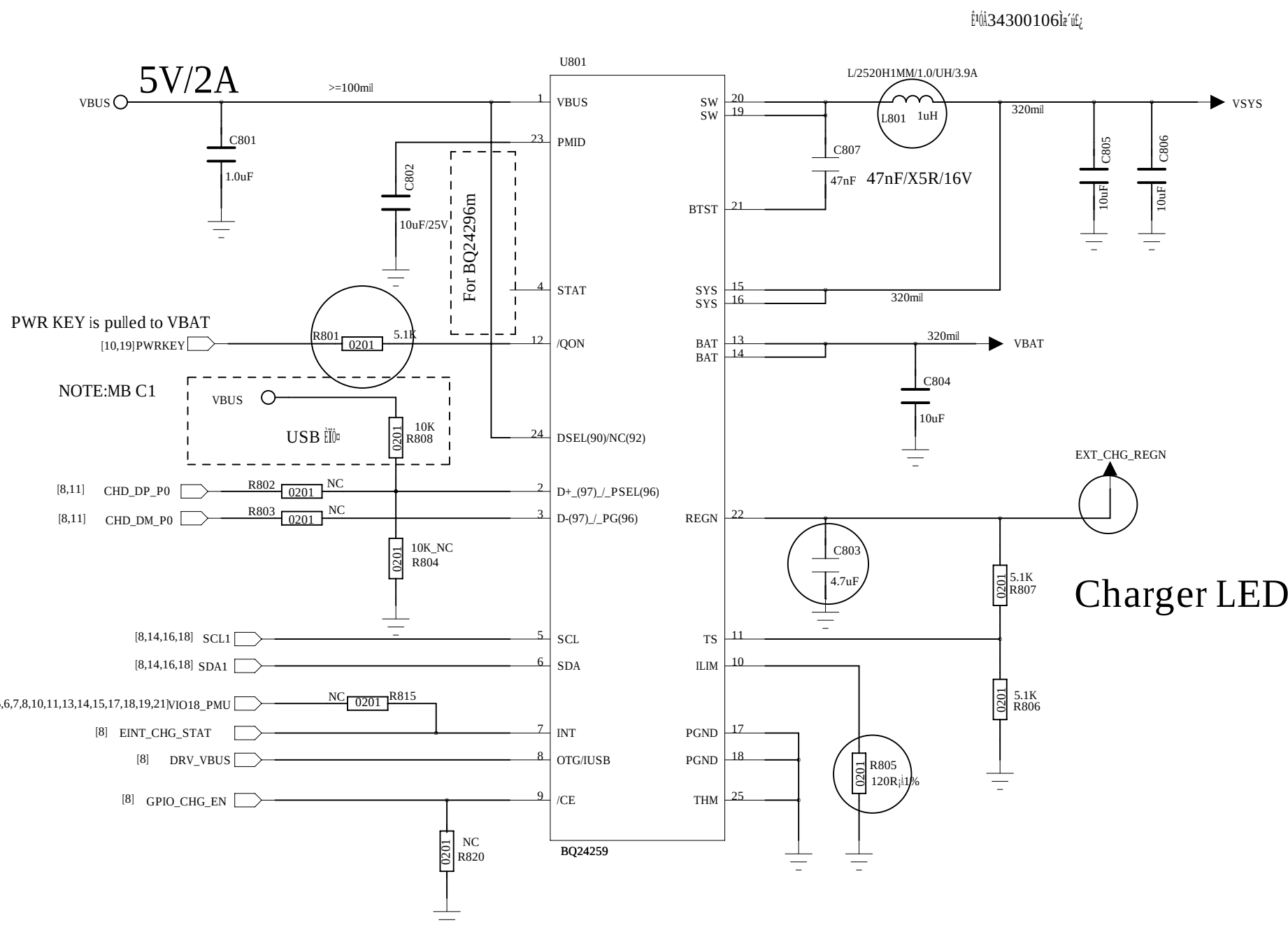
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SHEET				REV	
40				A	

REVISION RECORD			
VER	ECNO	APPROVED	DATE

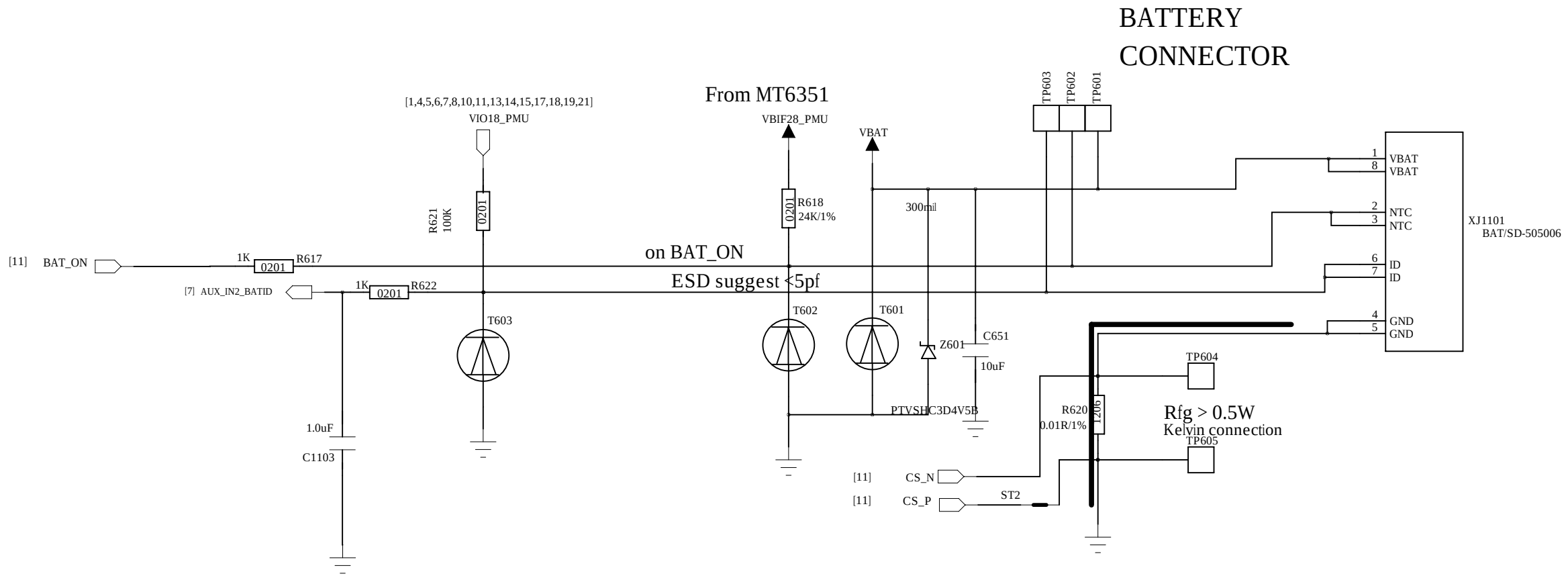


SW Charger / Power Path

bq25896 / SW Charger I2C address: 0X6B (Write:0xD6, Read:0xD7)
bq24296m / SW Charger I2C address: 0X6B (Write:0xD6, Read:0xD7)



PSEL
High: USB 500mA mode
Low: 3.25A mode

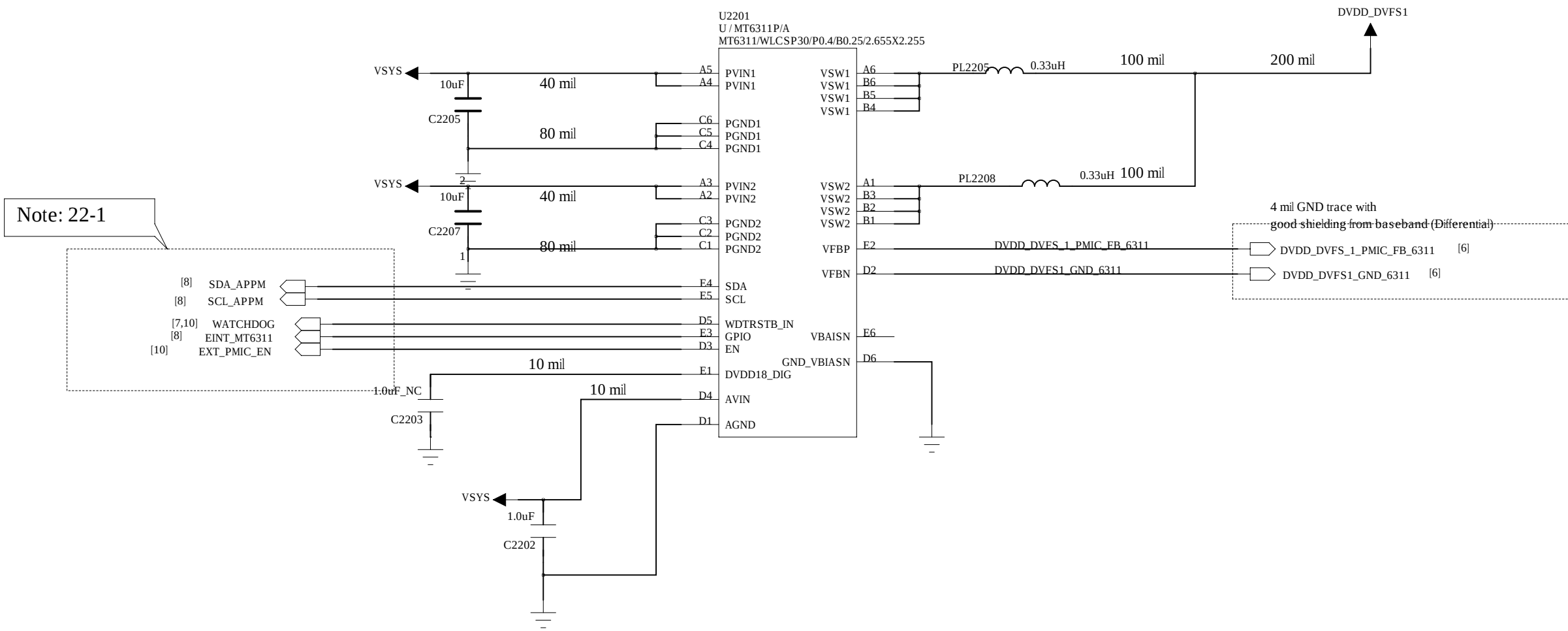


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RELEASED: <Release Date>	DATED: <Release Date>	SCALE: <Scale>	SIZE

REVISION RECORD			
LTR	ECNO	APPROVED	DATE

2-Phase Buck

MT6311 / 2-Phase Buck I2C address: 0X6B



Schematic design notice of "22_POWER_MT6311" page.

Note 22-1: MT6311 => Buck EN controlled by EXT_PMIC_EN from MT6351 and I2C.

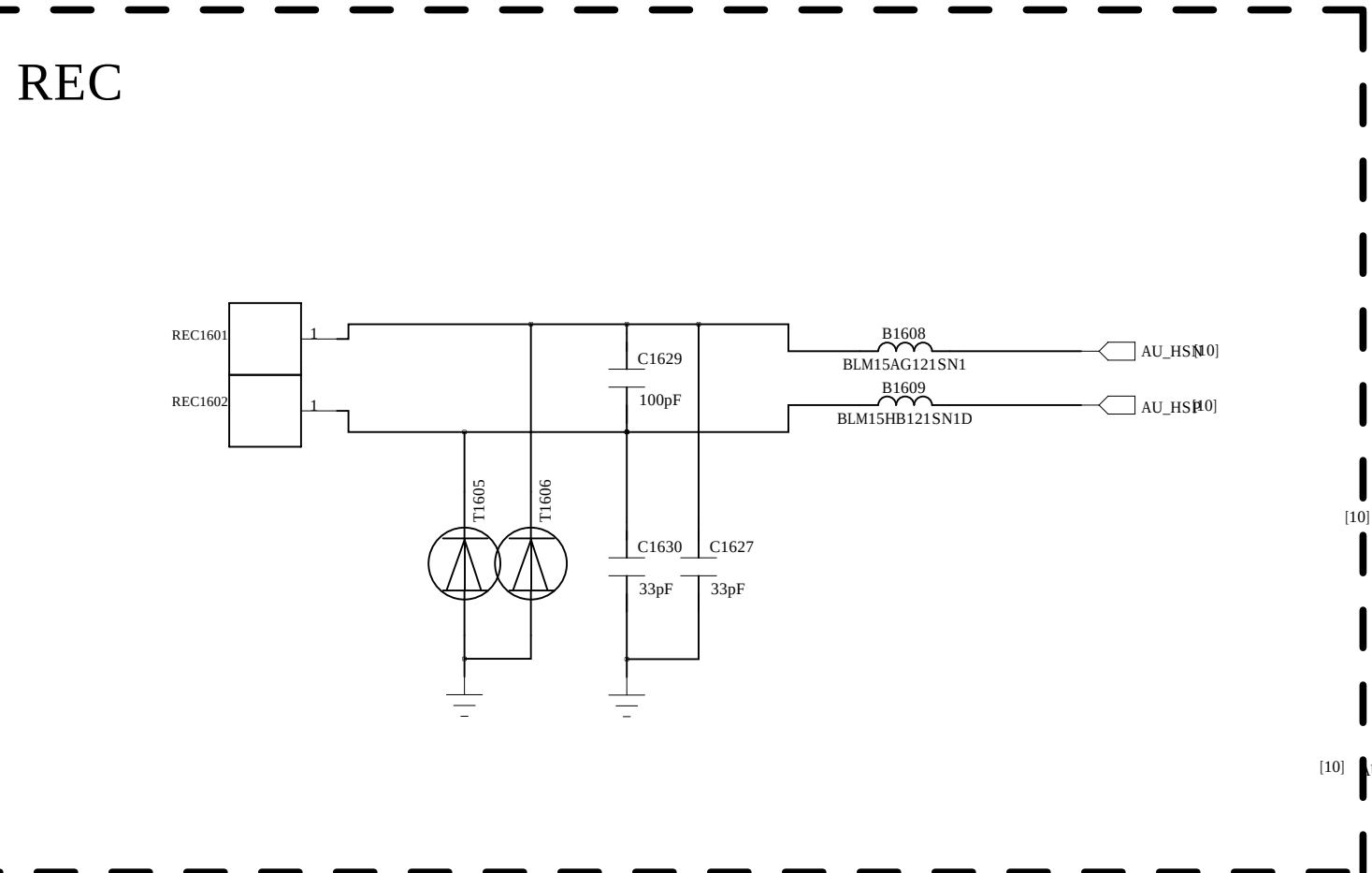
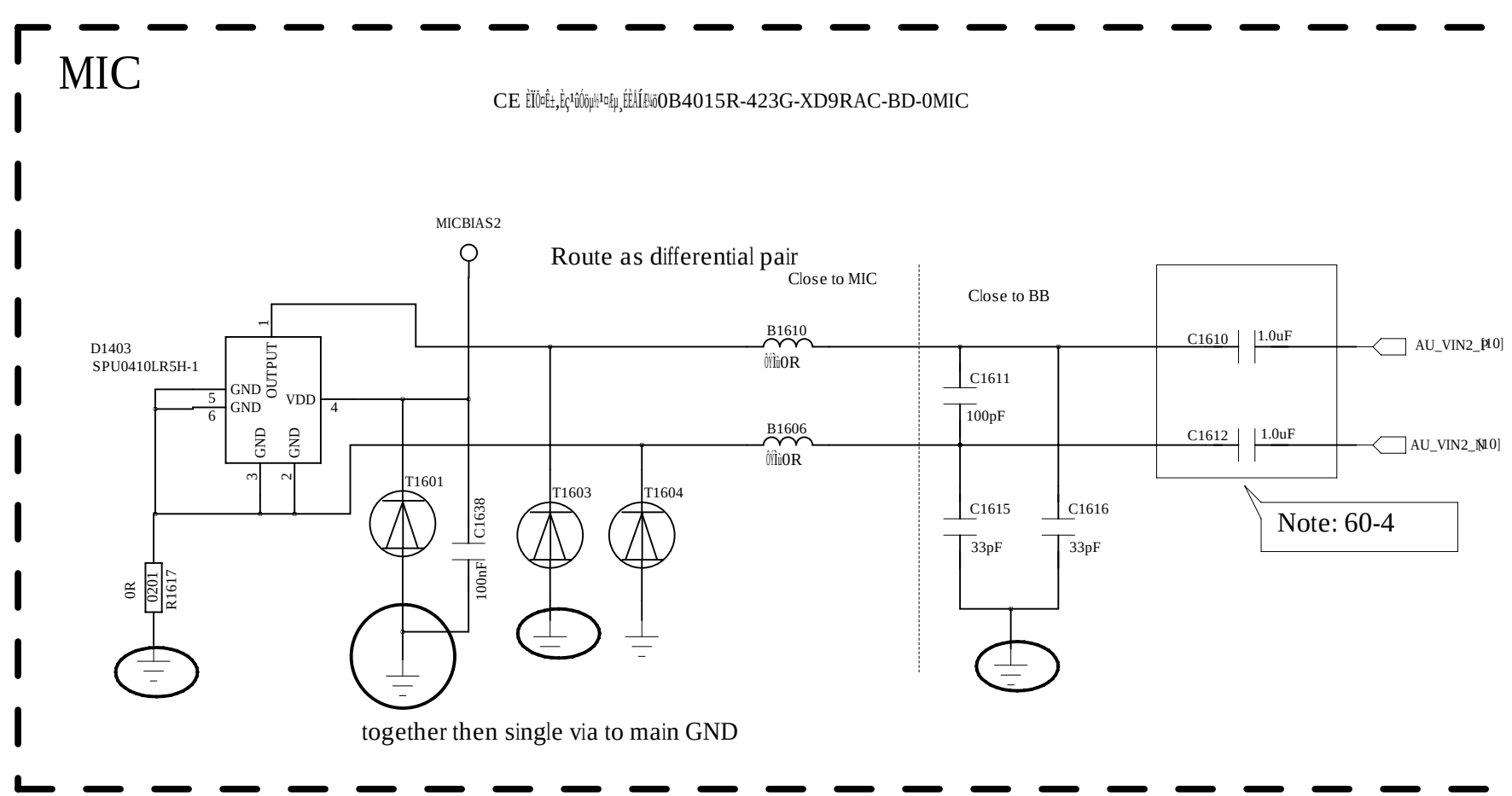
Schematic design notice of "24_POWER_THIRD-PARTY_VDRAM" page.

Note 24-1: Reserved external buck for DRAM power

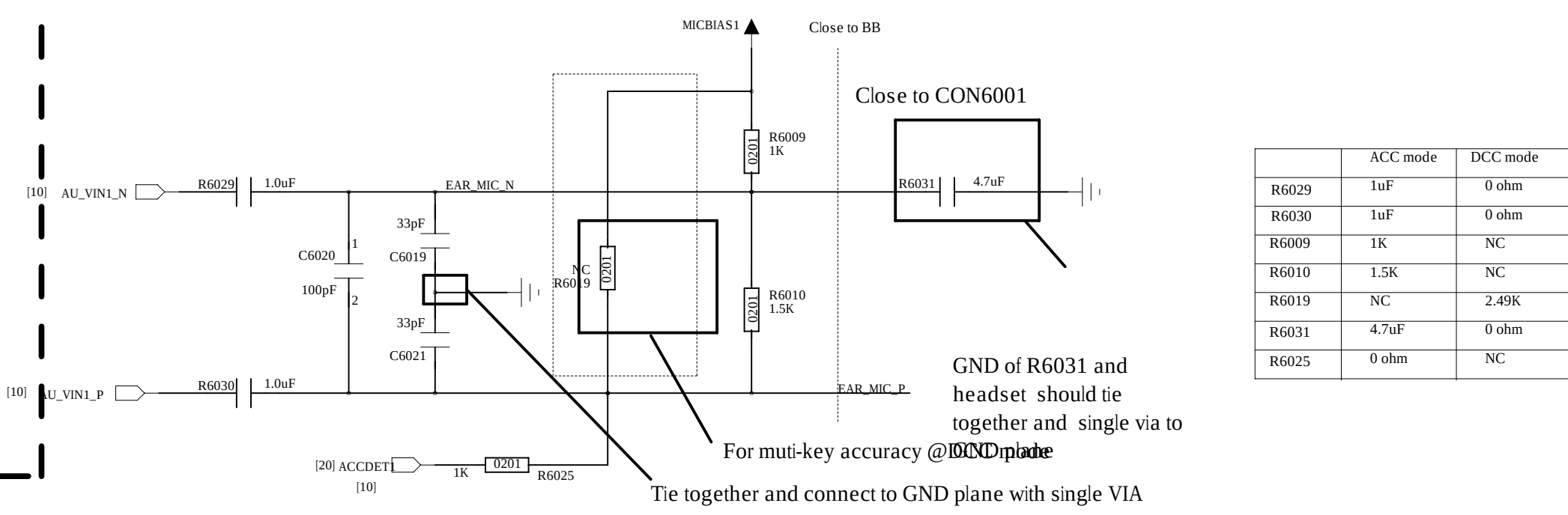
Note 24-2: Optional for low power performance especially in flight mode

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REVISION RECORD			
LTB	REV NO.	APPROVED	DATE

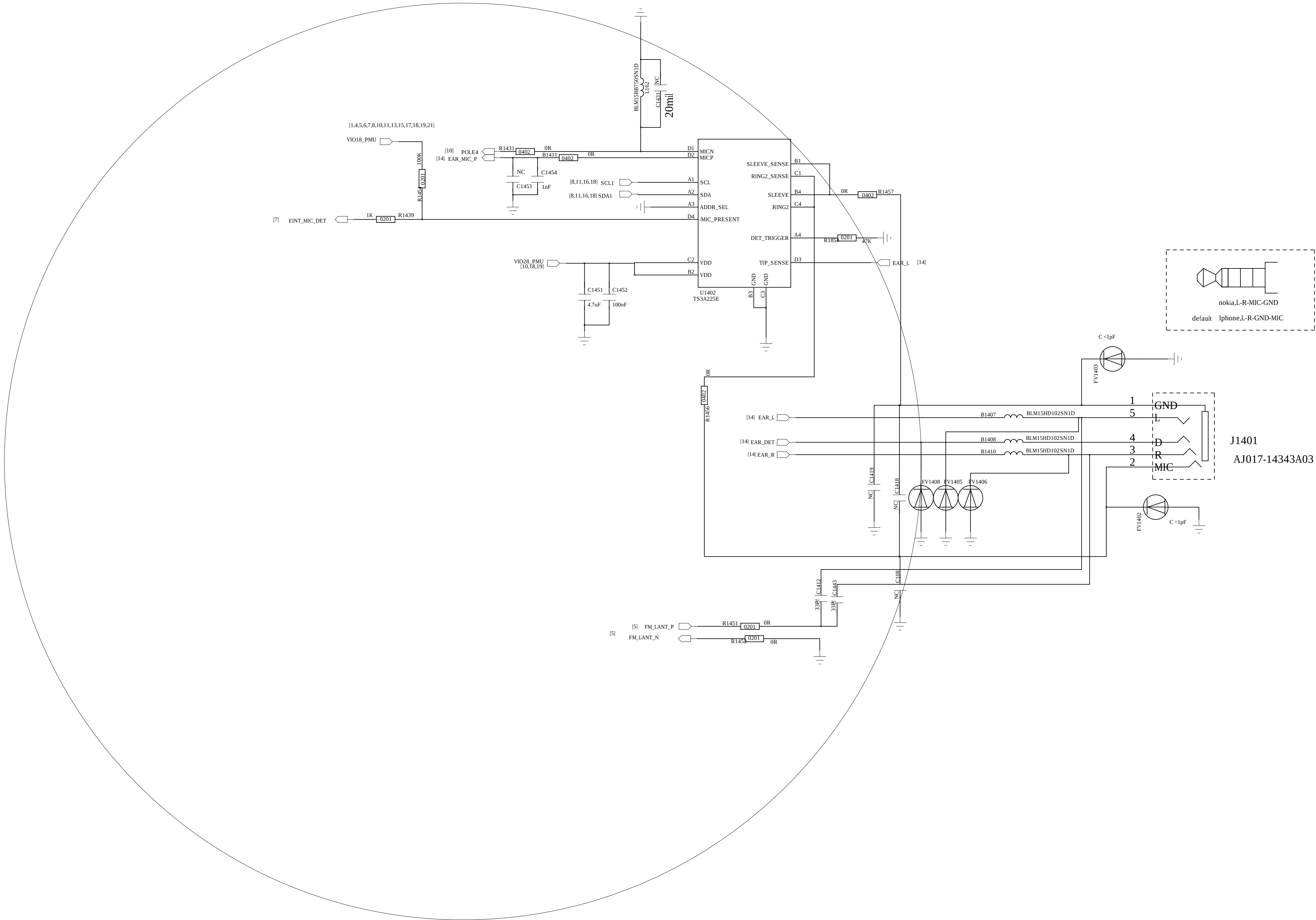
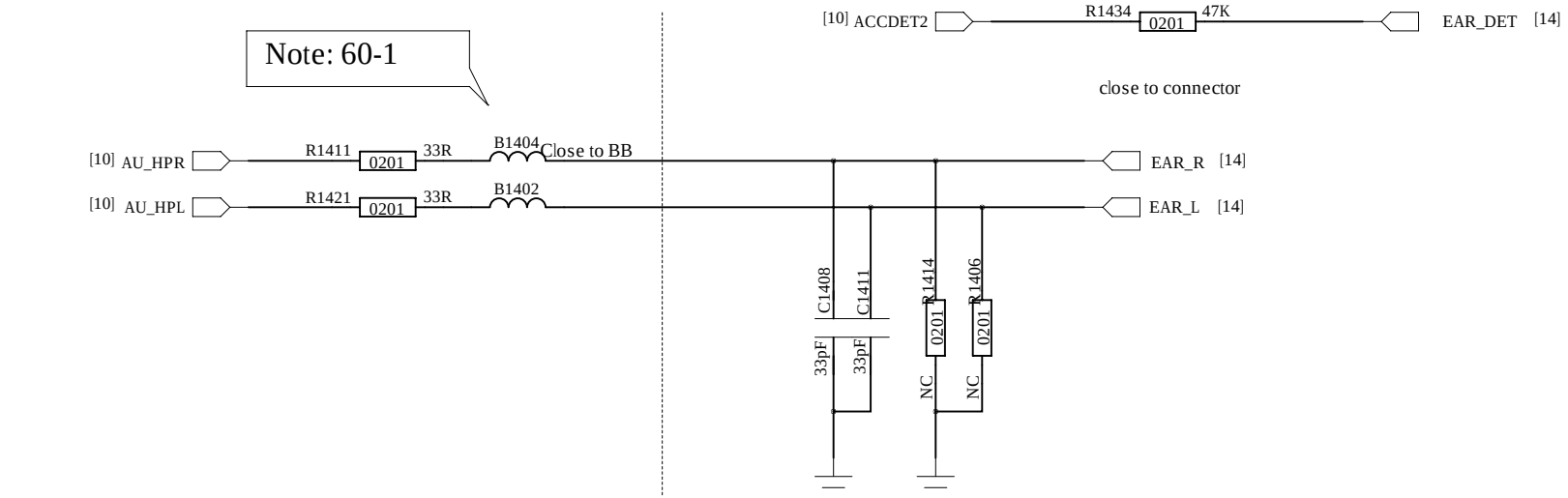
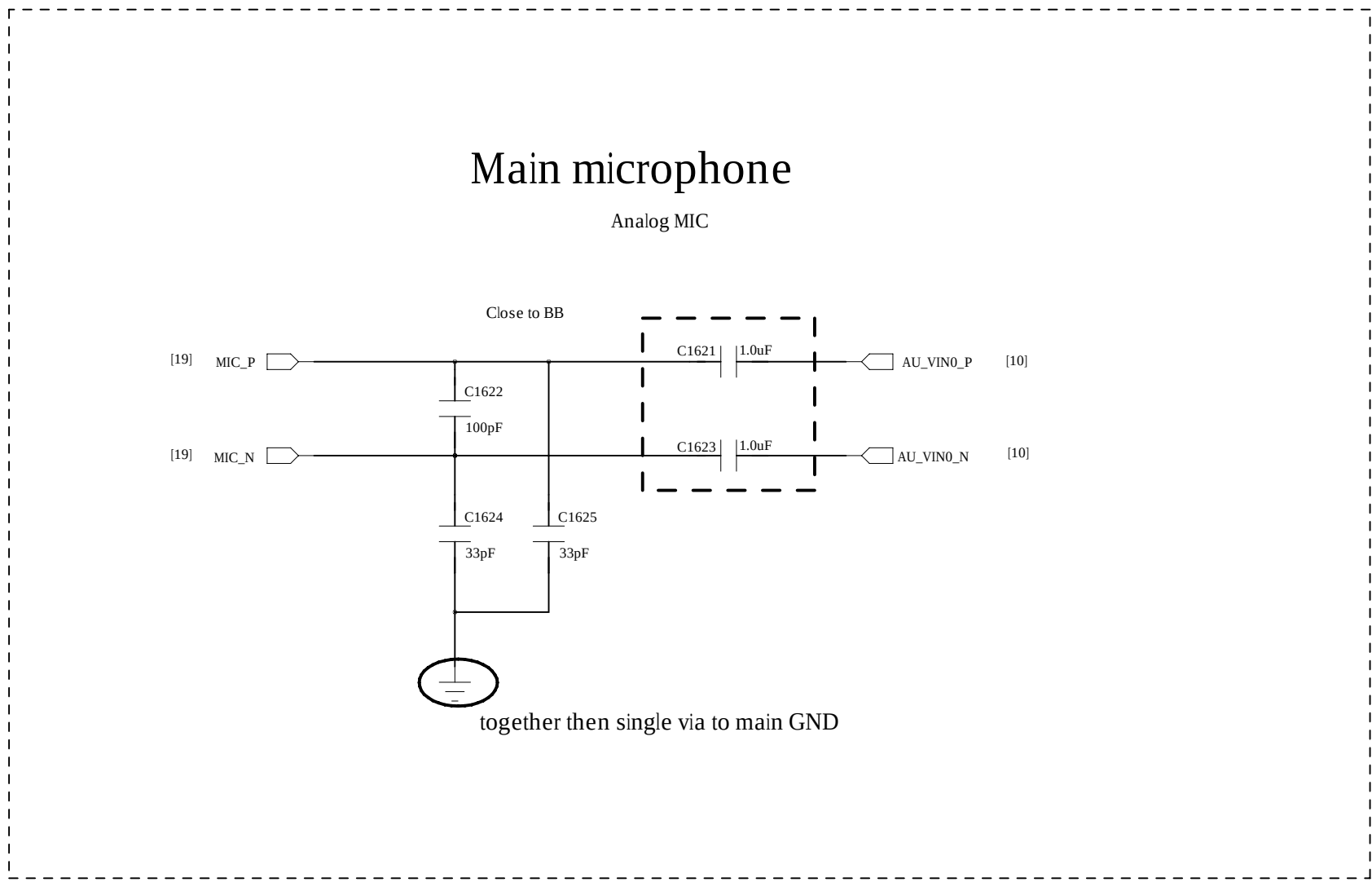


Earphone MICROPHONE



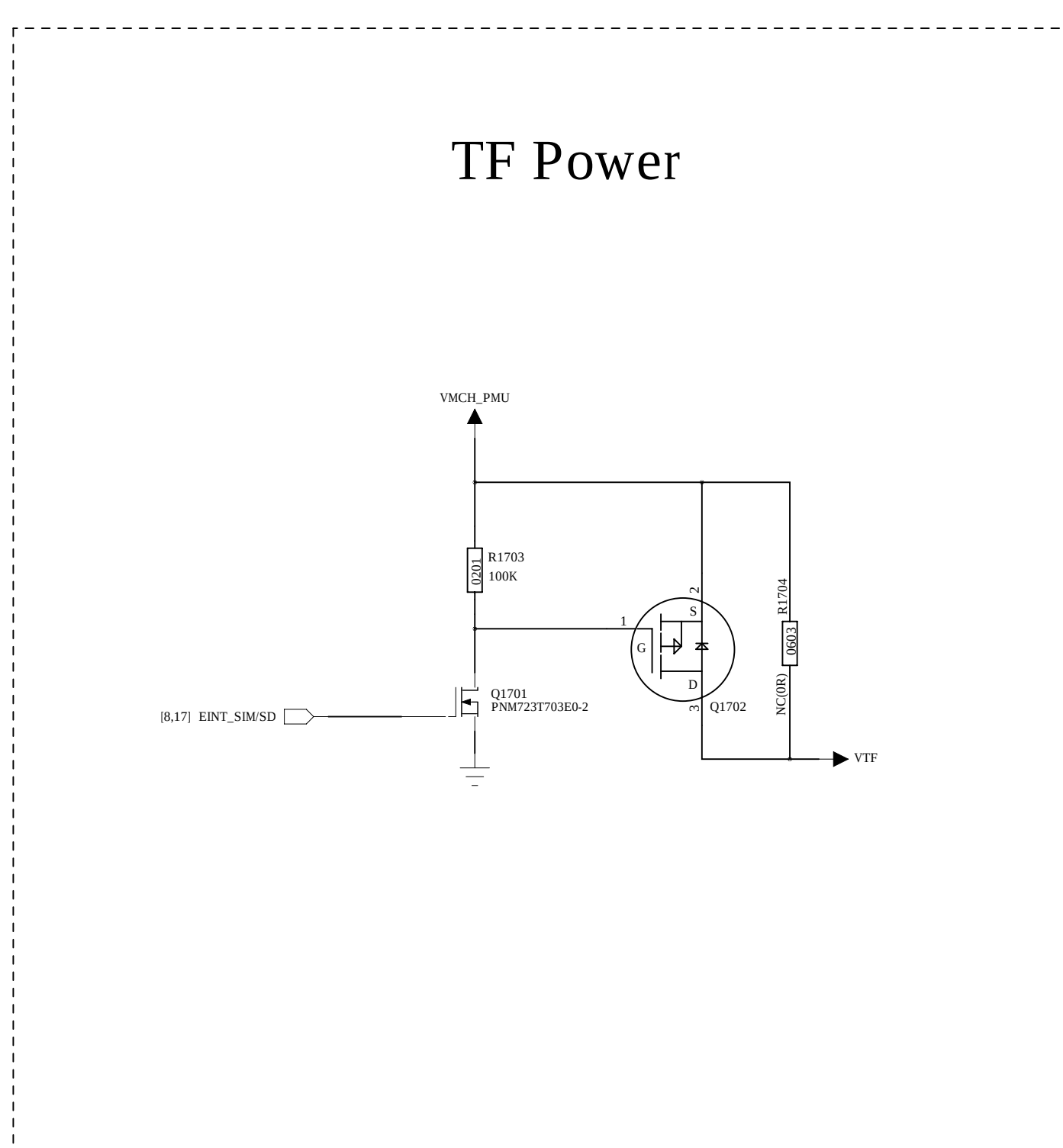
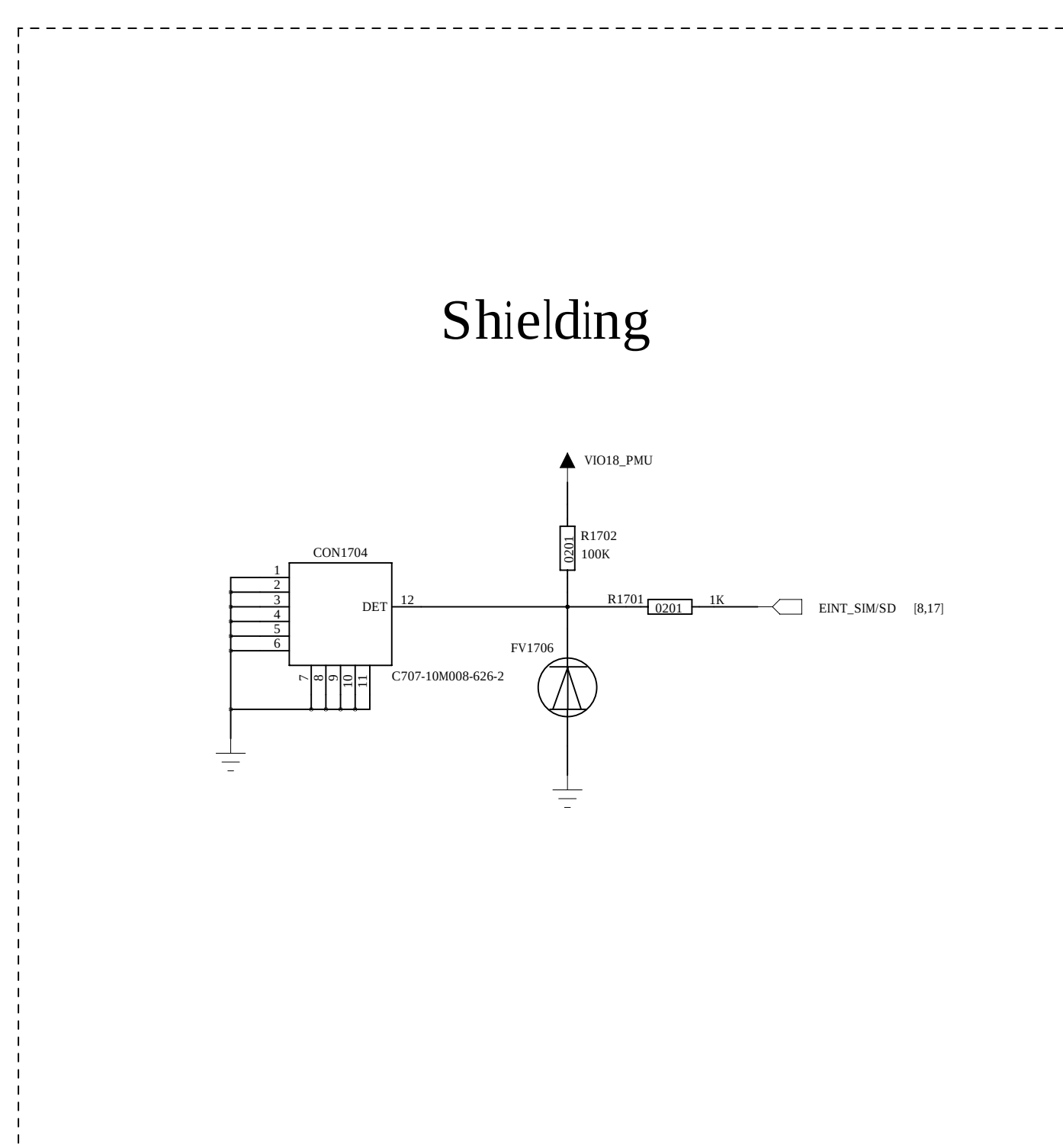
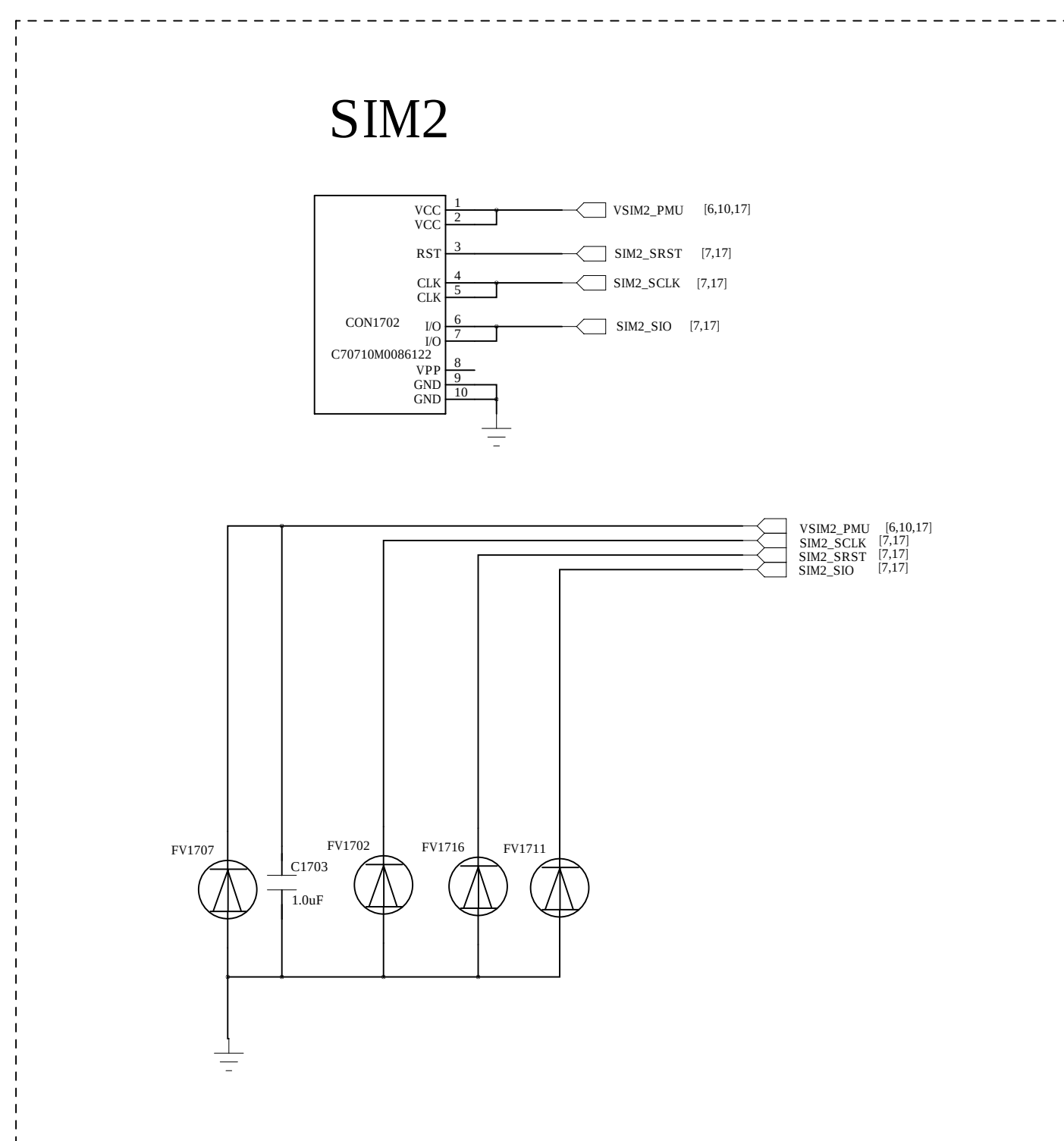
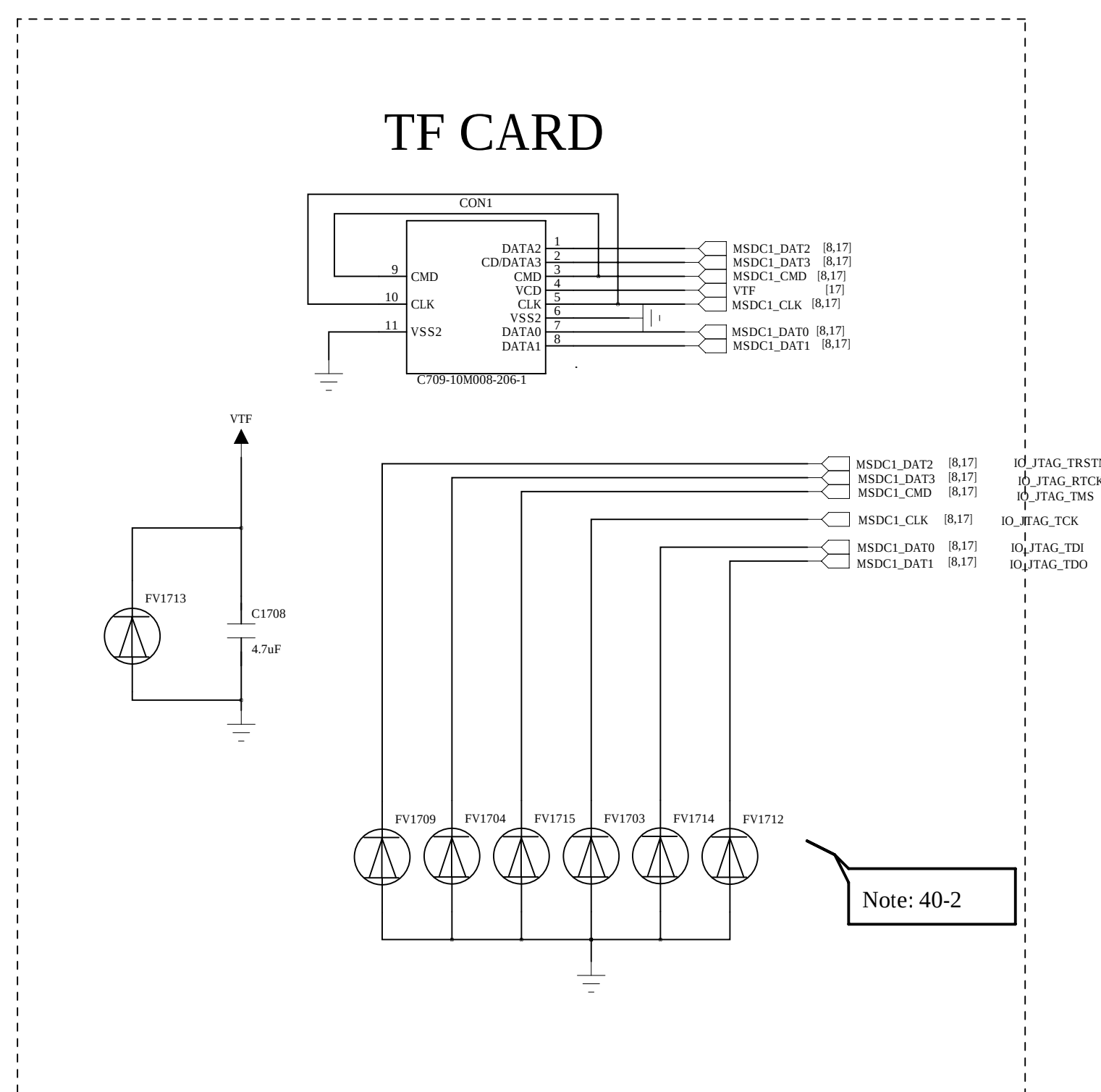
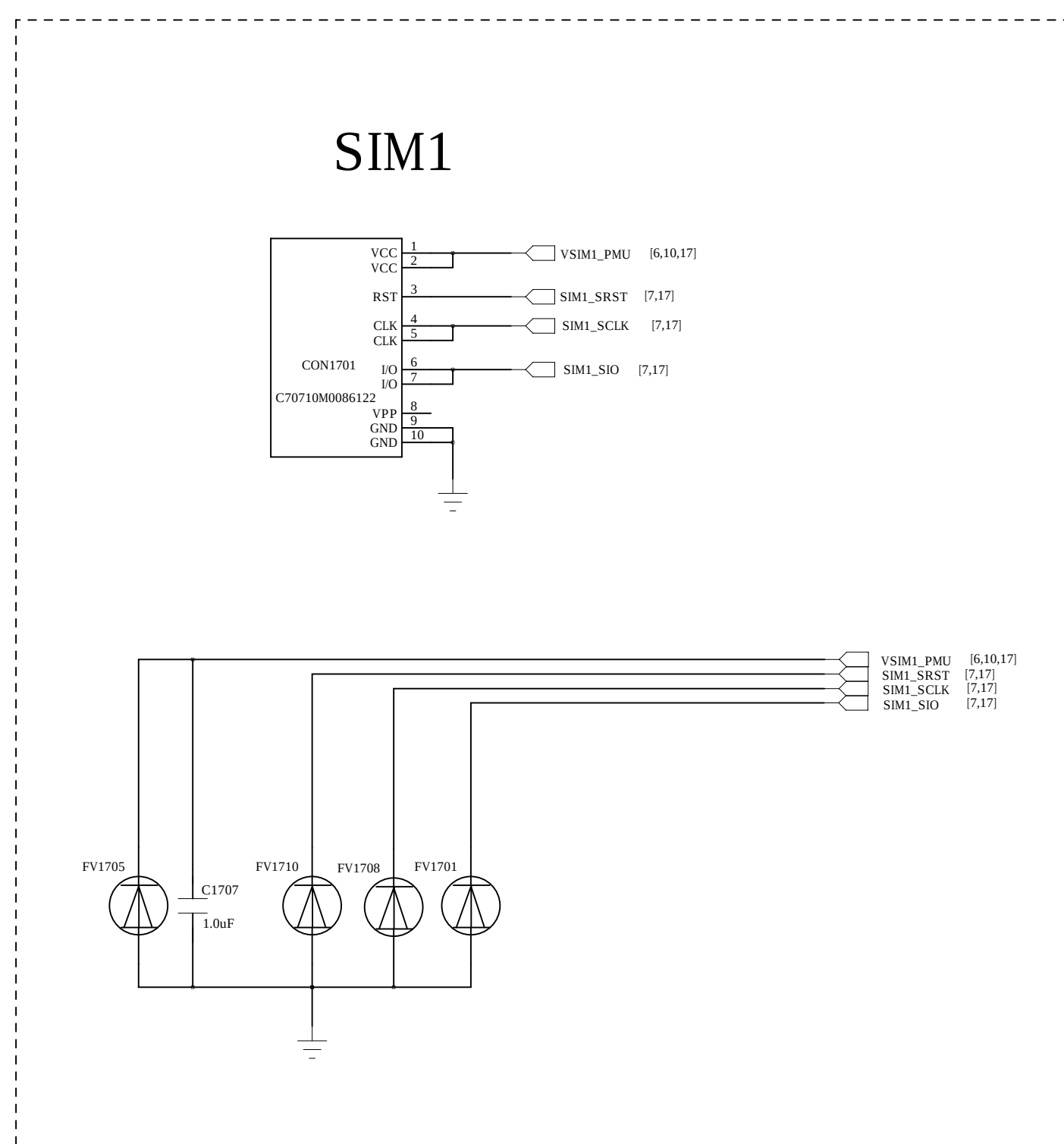
Schematic design notice of "60_PERI_AUDIO_IO" page.

- Note 60-1: Part # of B1402, B1404, B1408 and B1410 needs changed to "BLM18BD1025NT" for high THD performance (<90dB) but this BOM change will result in FM RSSI 10dB degraded .
- Note 60-2: To reserve a resistor in HPL and HPR in series connection both in order to optimize headphone pop noise. The recommended value of this resistor is 33R.
- Note 60-3: Layout trace from MT6351 bal G4 AUDREFN to Audio jack GND must surround shield with GND
- Note 60-4: 0.1/1 uF for ACC mode (1uF for WB_AMR Speech/0.1uF for NB_AMR Speech) 0 ohm for DCC mode



COMPANY: Longcheer			
TITLE: <15_AUDIO>			
DRAWN: <Drawn By>	DATE: <Draw Date>	CODE	SIZE
CHECKED: <Checked By>	DATE: <Checked Date>	CODE	SIZE
QUALITY CONTROL: <QC By>	DATE: <QC Date>	CODE	SIZE
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SCALE: <Scale>		DRWING NO: CM212	REV: A
SHEET: 44		23	

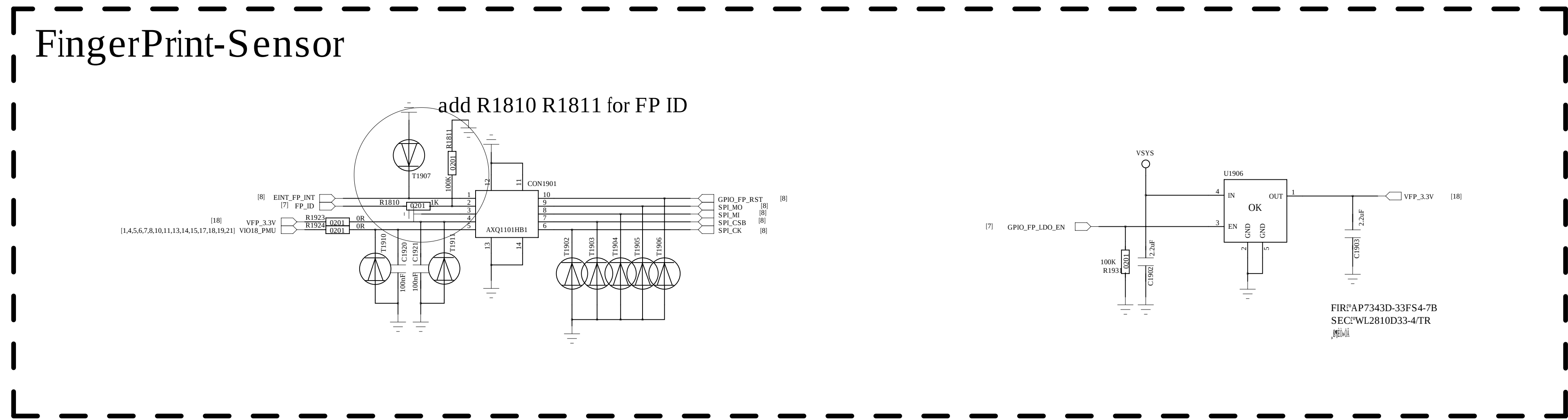
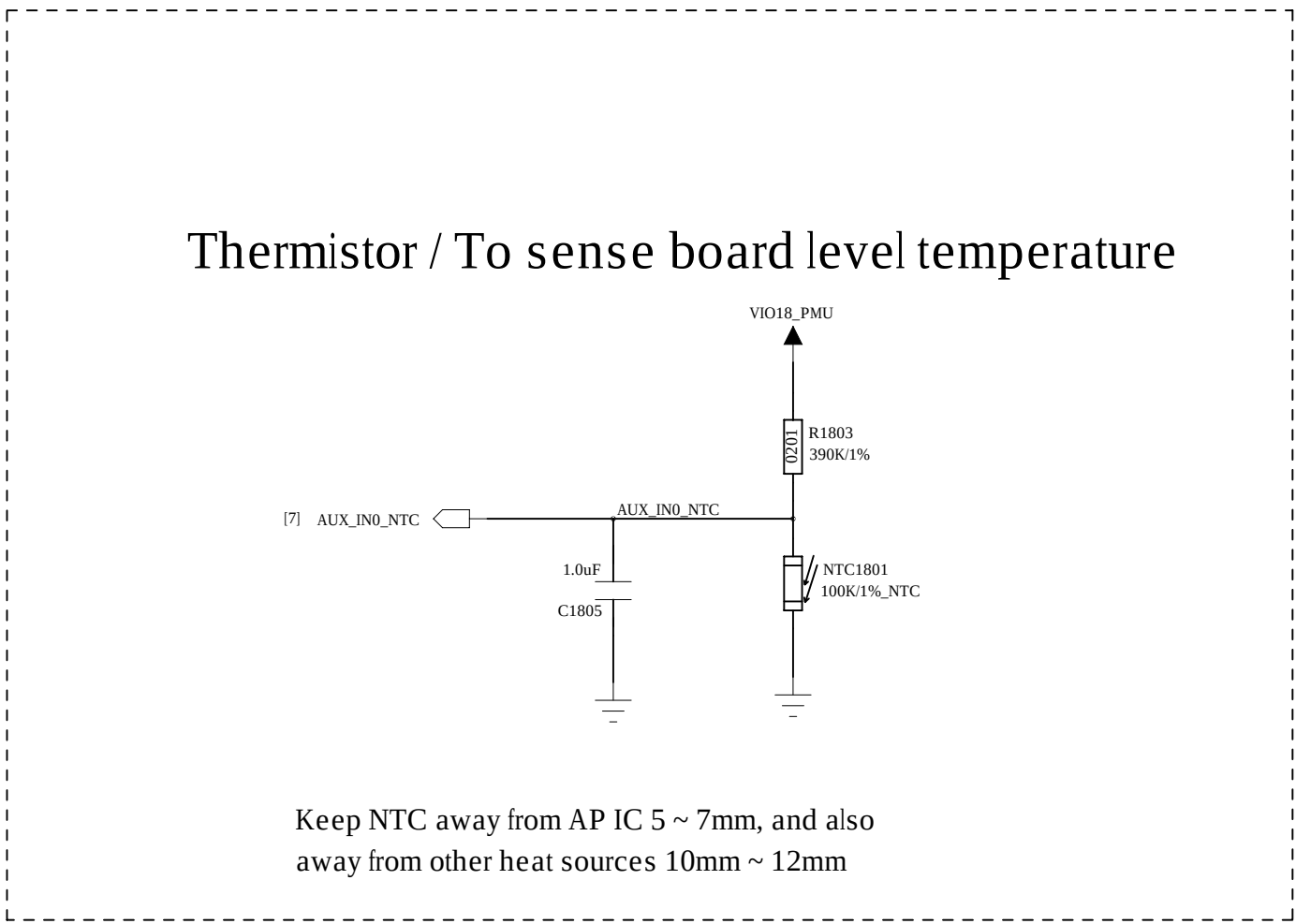
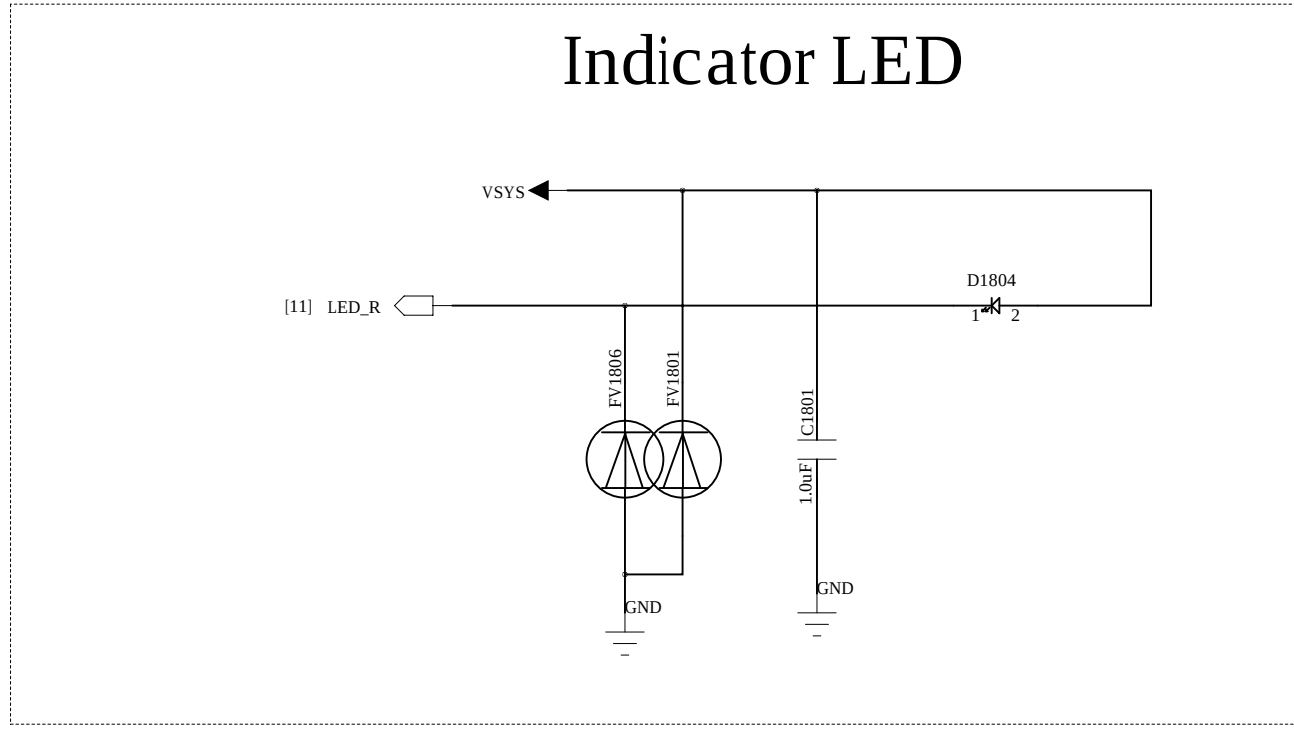
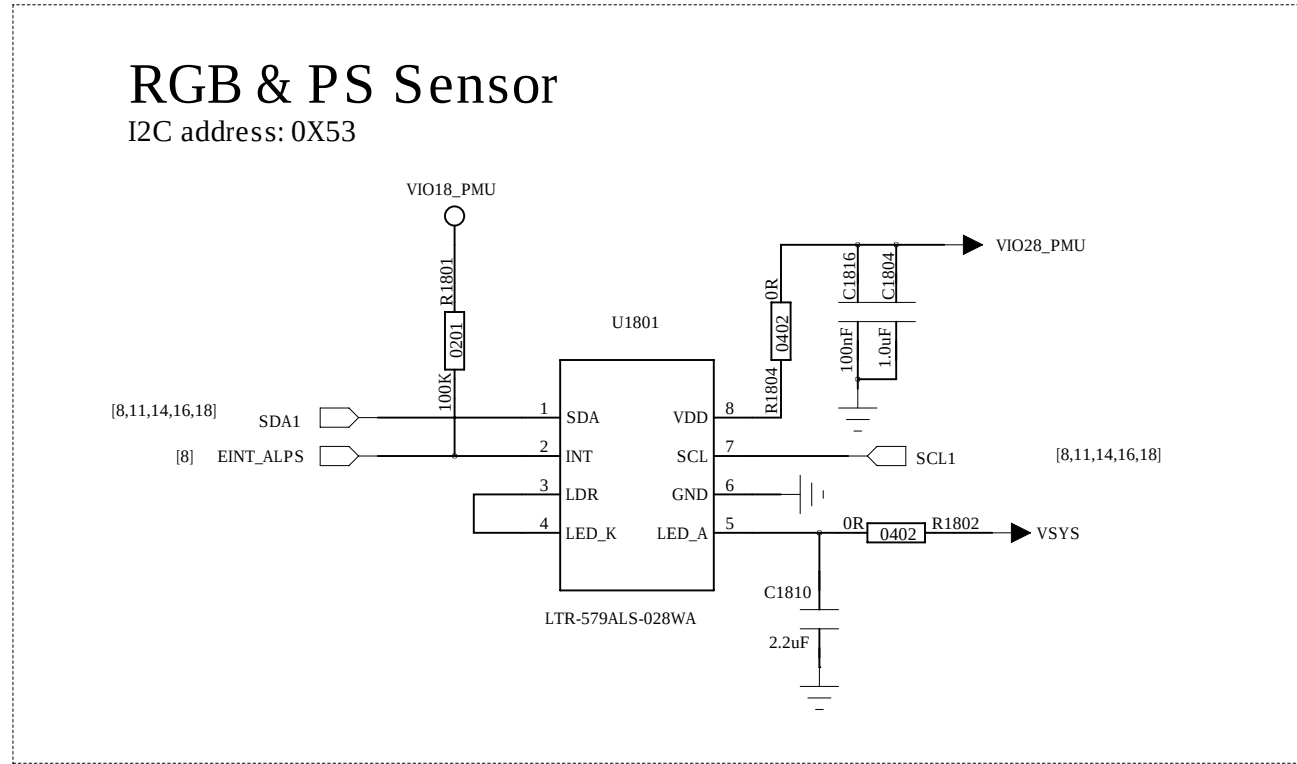
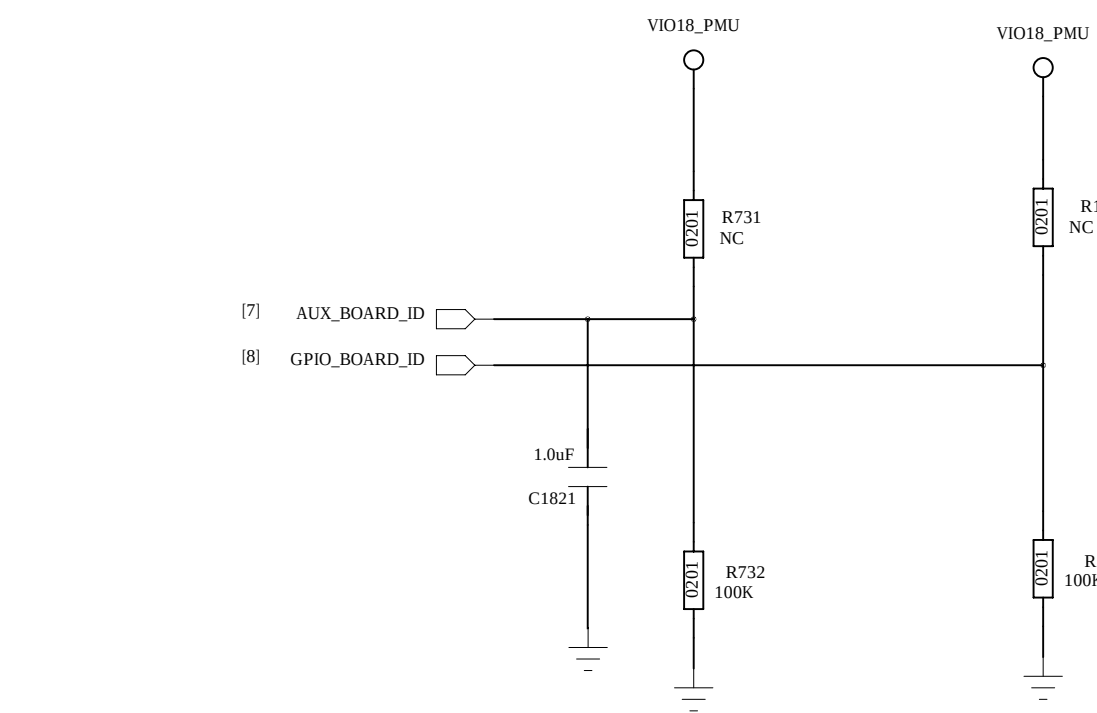
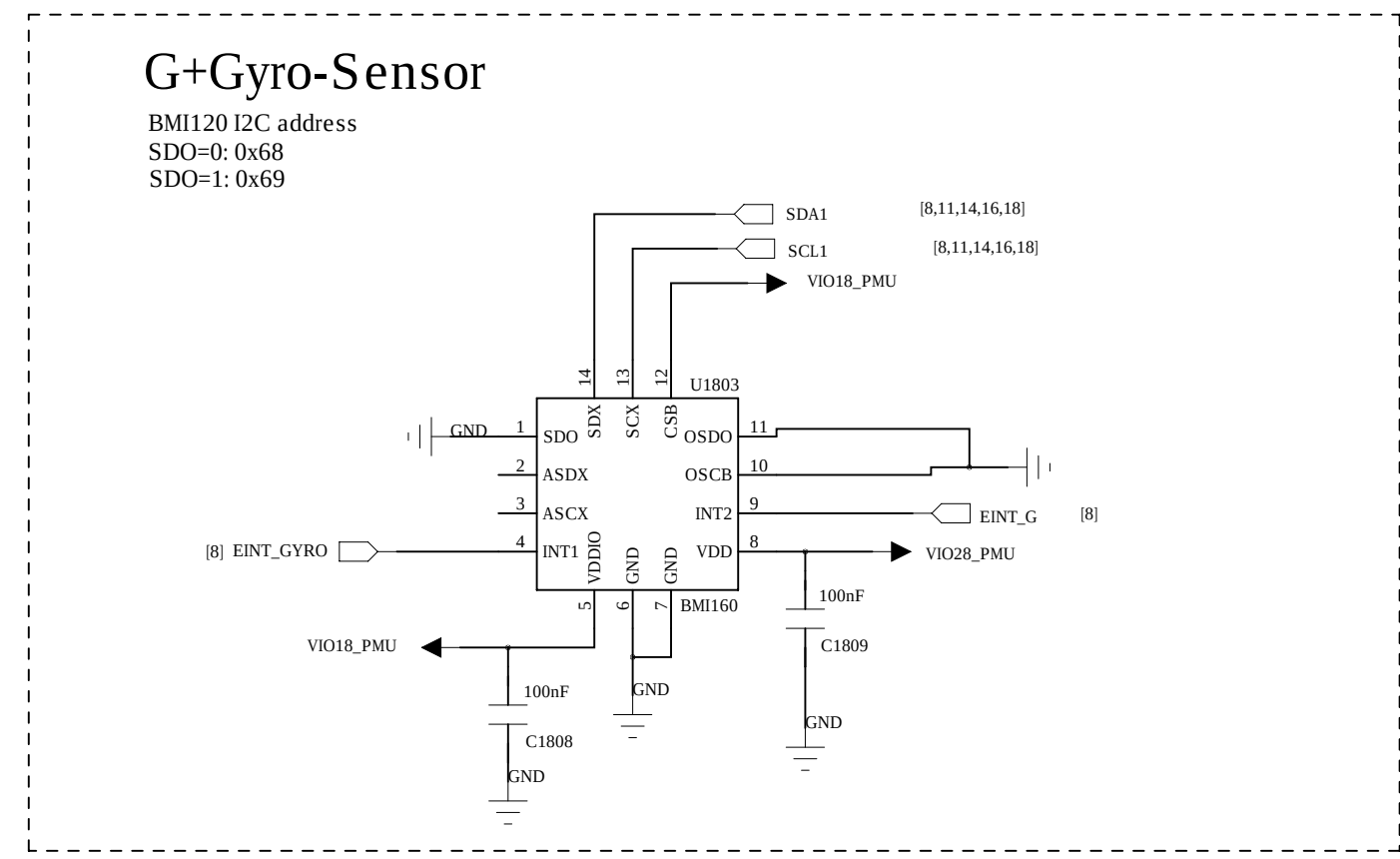
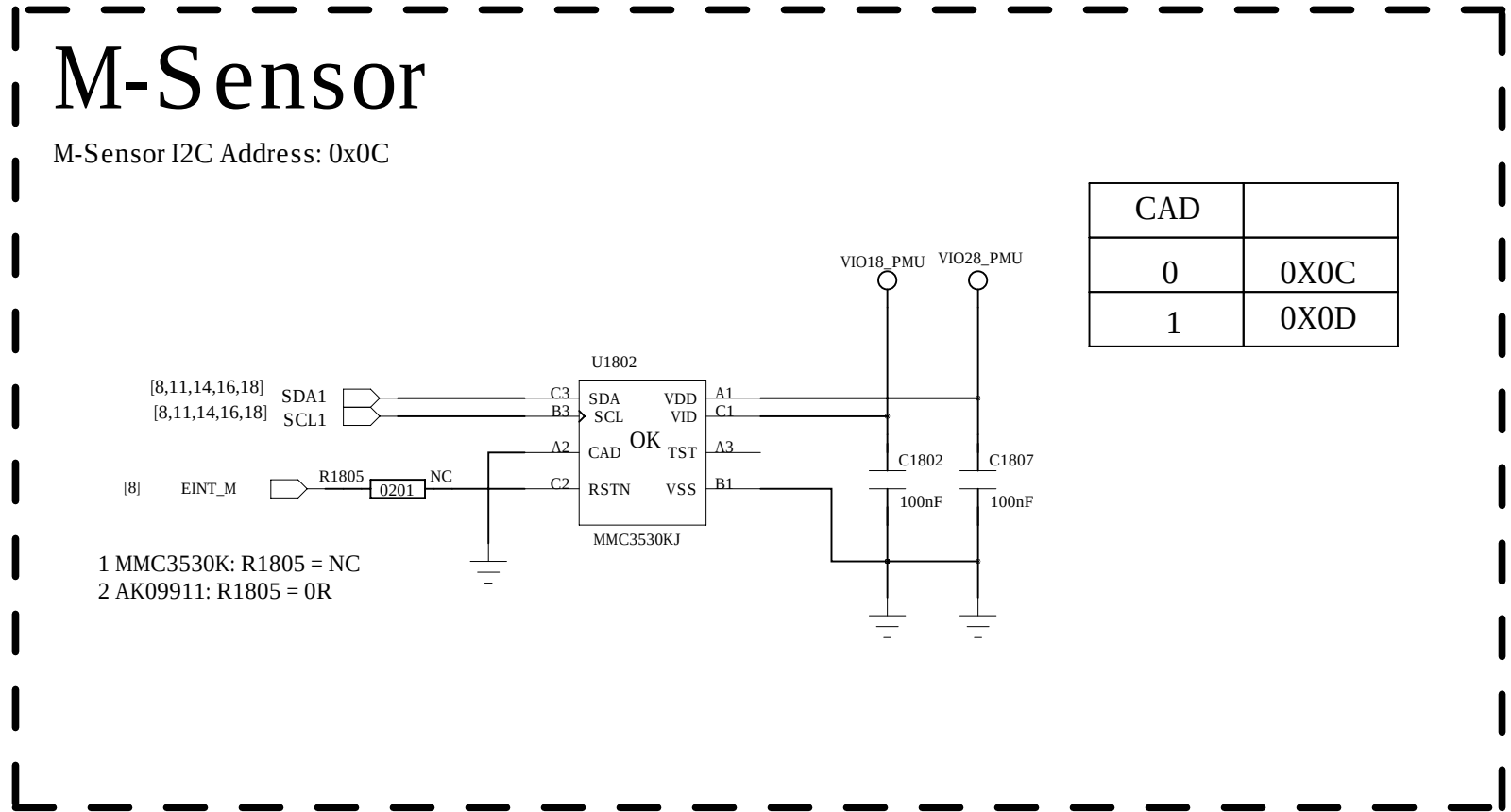
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Note 40-1:
The equivalent capacitance of ESD protection device must be $\leq 0.5\text{pF}$
-- otherwise it will result in NFC card mode function fail.

		COMPANY: Longcheer			
		TITLE: <15_AUDIO>			
DRAWN	<Drawn By>	DRAWN	<Drawn Date>		
CHECKED	<Checked By>	DATE	<Checked Date>		
QUALITY CONTROL	<QC By>	DATE	<QC Date>		
RELEASED	<Release Date>	DATE	<Release Date>		
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		<Code>	A0	CM212	A
		SCALE: <Scale>			SHEET <i>M</i> 23

REVISION RECORD			
VER	ECO NO	APPROVED	DATE



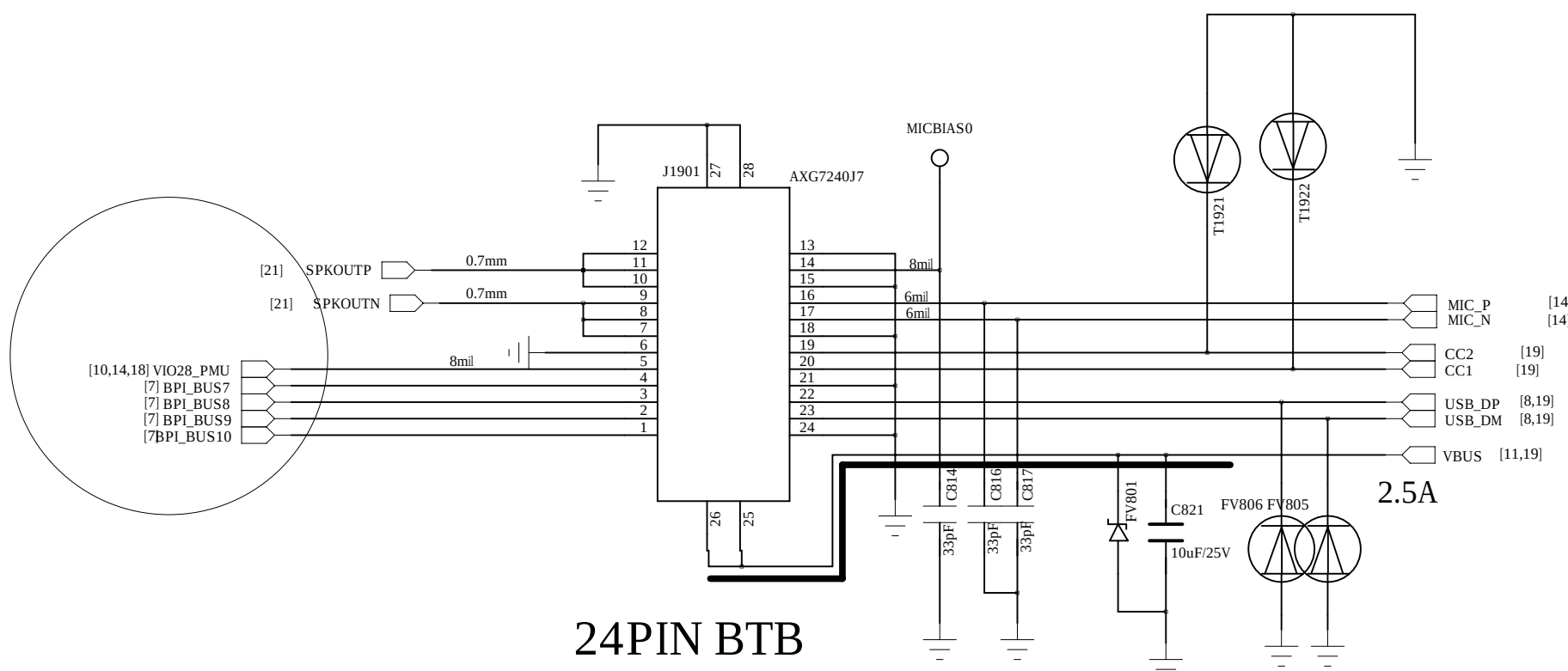
COMPANY: Longcheer

TITLE: <15_AUDIO>

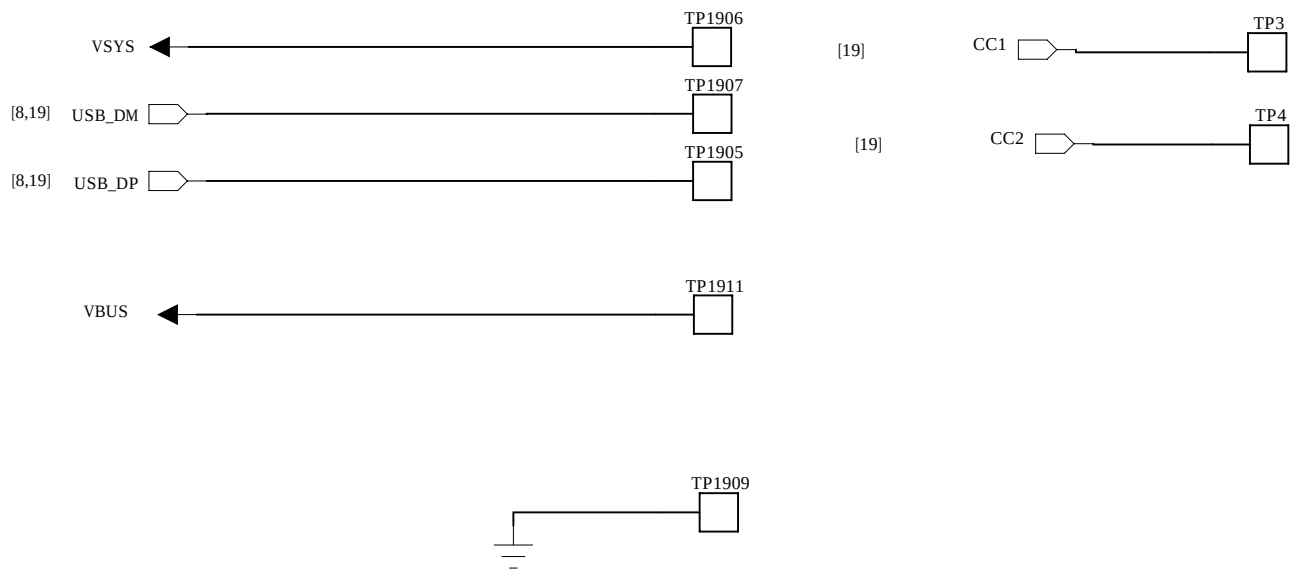
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CHECKED	DATE				
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<QC By>	<QC Date>				
RELEASED	DATE				
<Released By>	<Release Date>				

SCALE: <Scale>

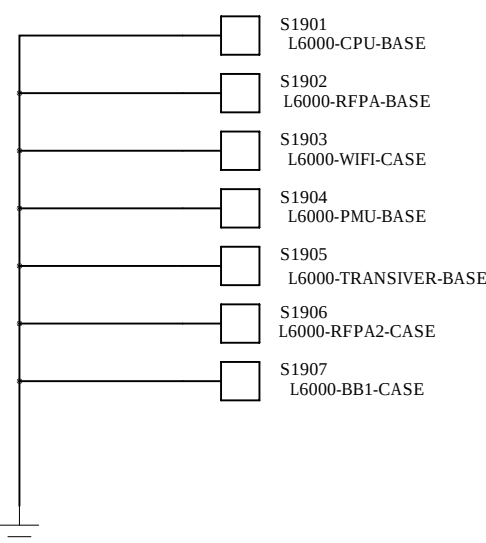
REVISION RECORD			
LTR	REVISION	APPROVED	DATE



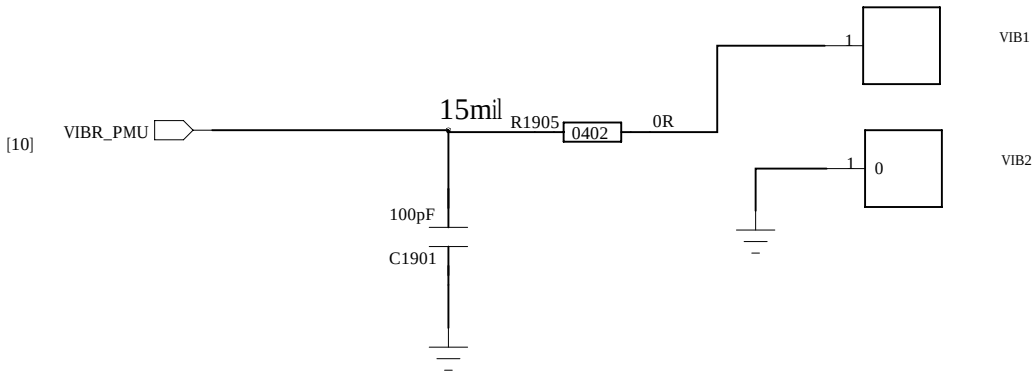
TEST POINT



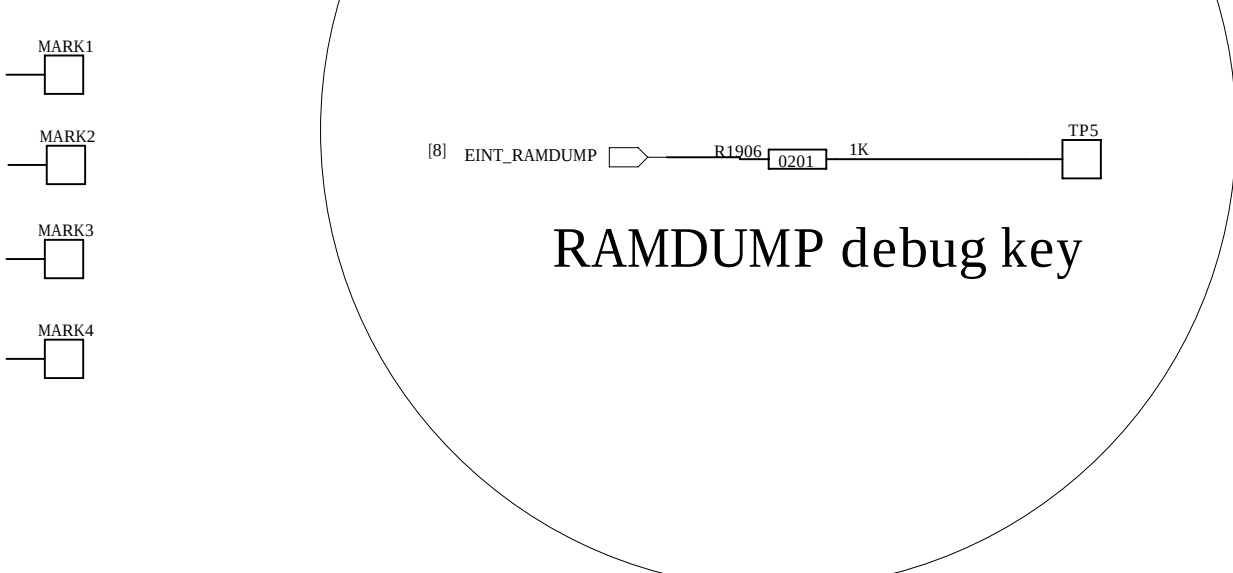
Shiedling Case



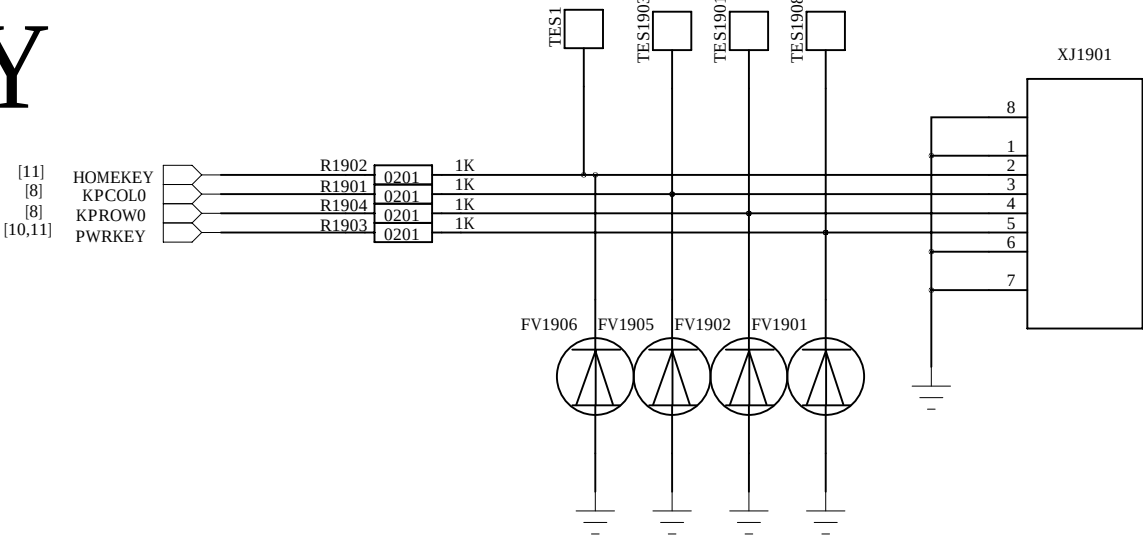
VIBRATOR



MARK POINT



KEY

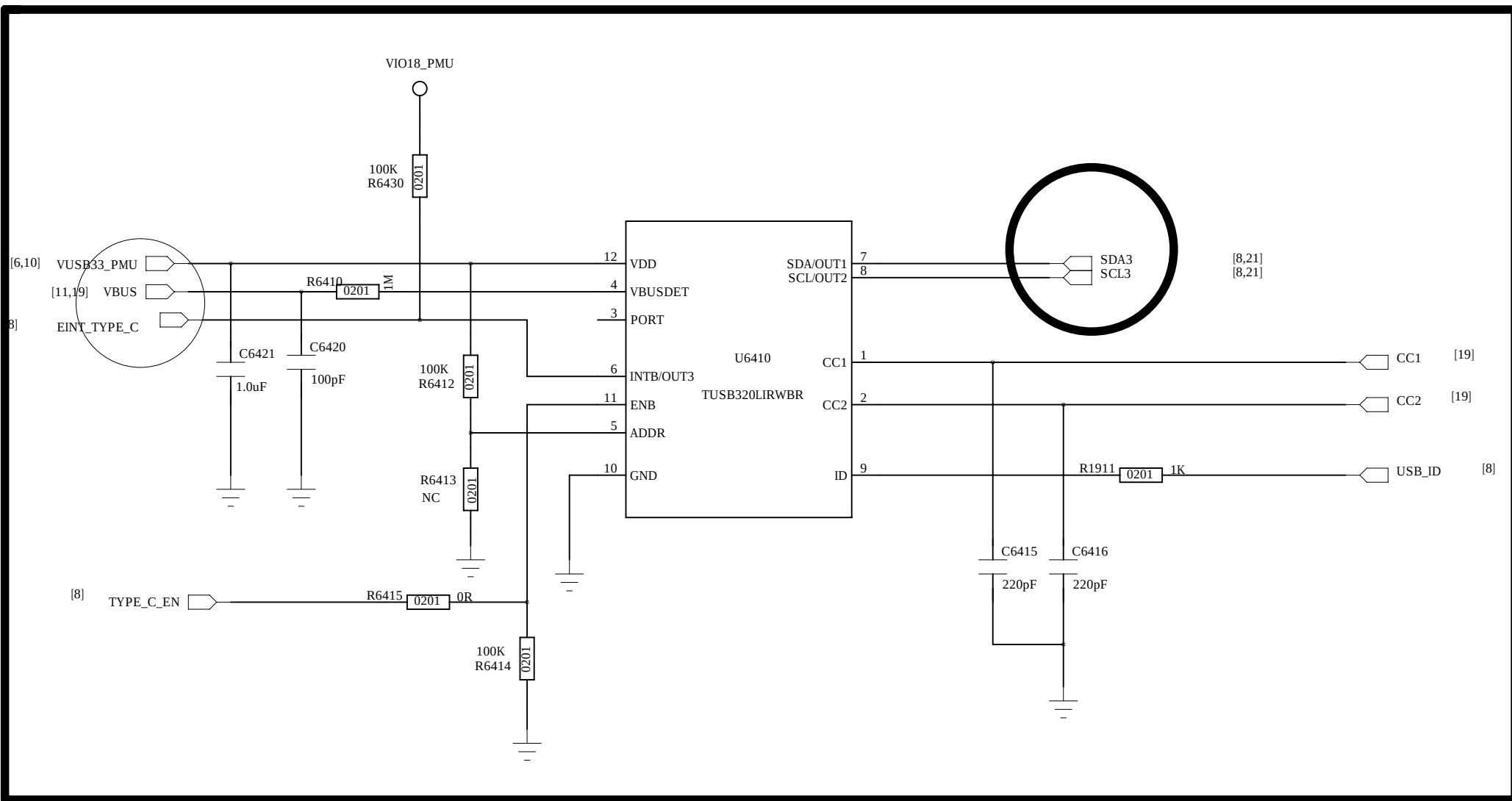


Schematic design notice of "65_PERI_Dual_SIM_ICUSB_KEYPAD" page.

Note 65-1: DO NOT put pull-up resistor on PWRKEY

Note 65-2: Volume Up : HOME Key / GND
Volume Down: KPROW/KPCOL0

Note 65-3: If do not need NFC SIM function, let Vpp pin floating



COMPANY: Longcheer			
TITLE: <15_AUDIO>			
DRAWN: <Drawn By>	DATE: <Drawn Date>	CODE: <Code>	SIZE: A0
CHECKED: <Checked By>	DATE: <Checked Date>	DRWING NO: CM212	REV: A
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REVISION RECORD			
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		COMPANY: Longcheer			
		TITLE: RF_TX			
DRAWN: <Drawn By>	DATED: <Drawn Date>	CODE	SIZE	DRAWING NO	REV.
CHECKED: <Checked By>	DATED: <Checked Date>	<Code>	A0	CM212	A
QUALITY CONTROL: <QC By>	DATED: <QC Date>				
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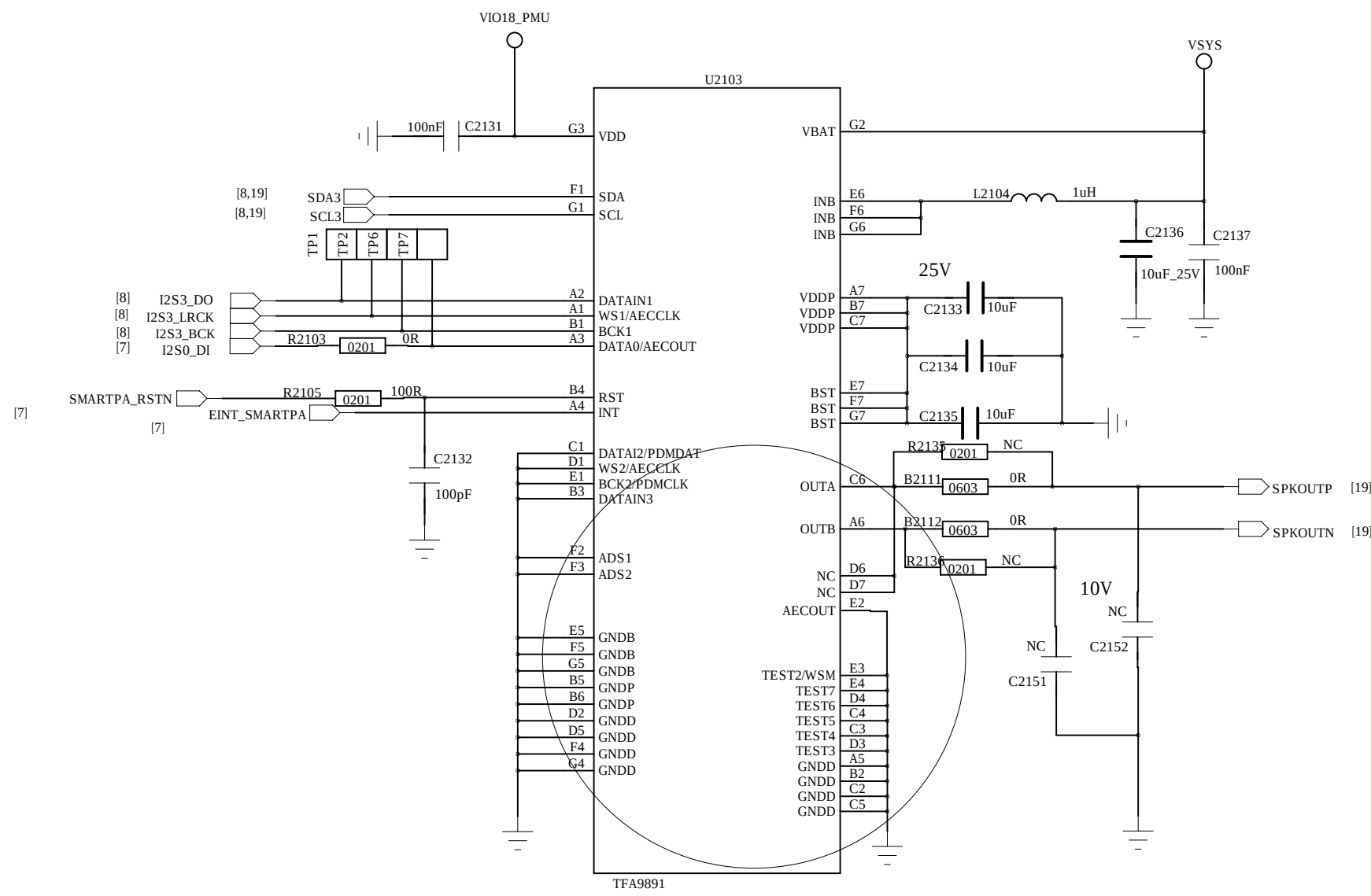
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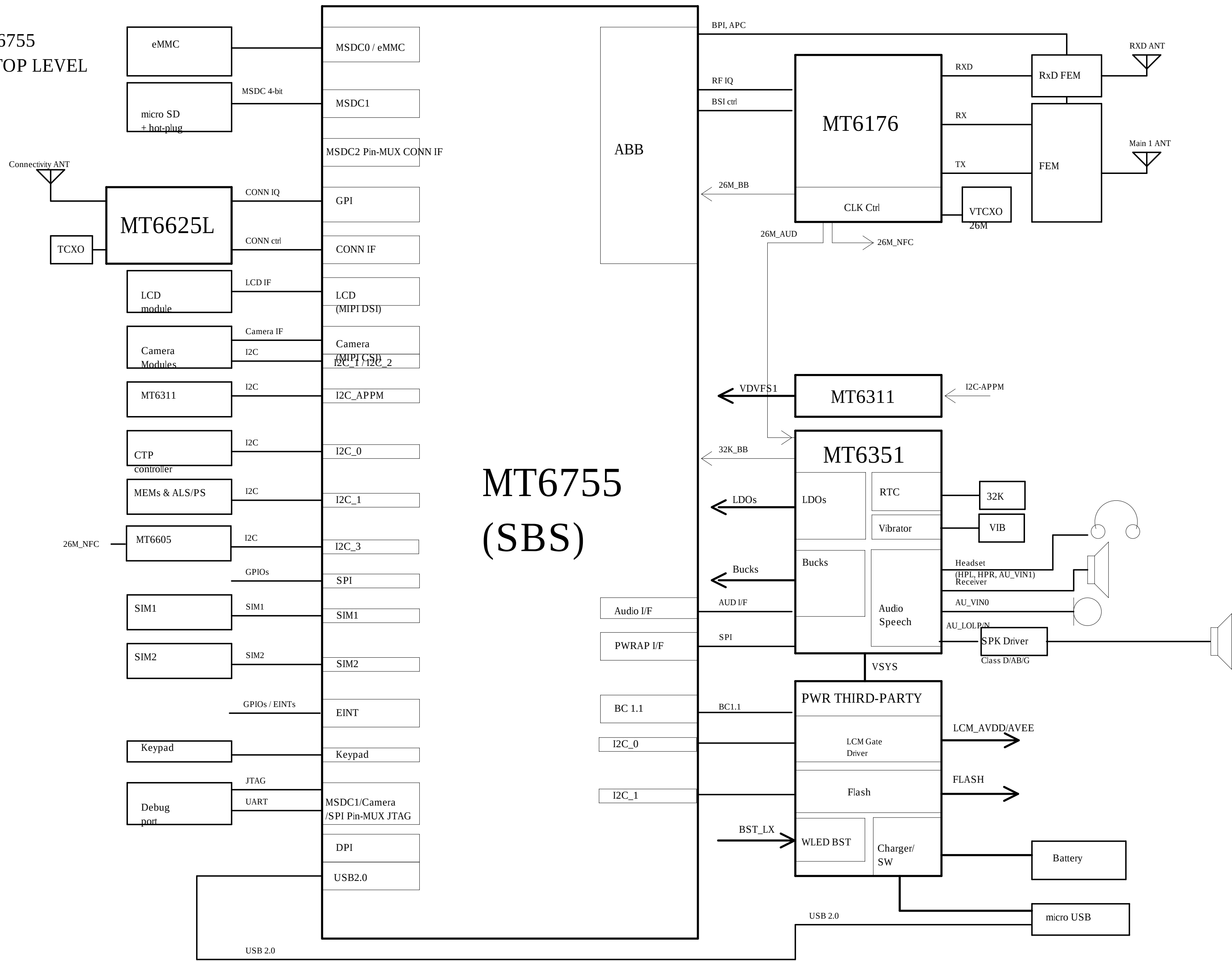
Use TFA9890



COMPANY: Longcheer			
TITLE: <15_AUDIO>			
DRAWN: <Drawn By>	DATED: <Drawn Date>	CODE: <Code>	SIZE: A0
CHECKED: <Checked By>	DATED: <Checked Date>	DRAWING NO: CM212	REV: A
QUALITY CONTROL: <QC By>	DATED: <QC Date>	SCALE: <Scale>	
RELEASED: <Released By>	DATED: <Release Date>	SHEET 23	

REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE

Project : MT6755
REF_SCH TOP LEVEL



COMPANY: Longcheer			
TITLE: <15_AUDIO>			
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CHECKED: <Checked By>	DATE: <Checked Date>	<Code>	A0
QUALITY CONTROL: <QC By>	DATE: <QC Date>	CM212	A
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CHECKED: <Checked By>		DATED: <Checked Date>		TITLE: <15_AUDIO>			
QUALITY CONTROL: <QC By>		DATED: <QC Date>		CODE: <Code>	SIZE: A0	DRAWING NO: CM212	REV: A
RELEASED: <Released By>		DATED: <Release Date>		SCALE: <Scale>		SHEET: 08 23	